GREEN CLIMATE FUND

Meeting of the Board
4 – 7 March 2024
Kigali, Rwanda
Provisional agenda item 10

GCF/B.38/02/Add.09

12 February 2024

Consideration of funding proposals – Addendum IX

Funding proposal package for FP229

Summary

This addendum contains the following six parts:

a) A funding proposal summary titled "Acumen Climate Action Pakistan Fund" submitted by Acumen Fund, Inc;
b) No-objection letter(s) issued by the national designated authority(ies) or focal point(s);
c) Environmental and Social report(s) disclosure;
d) Independent Technical Advisory Panel's assessment;
e) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
f) Gender documentation of the funding proposal.

These documents are presented as submitted by the accredited entity and the national designated authority(ies) or focal point(s), respectively. Pursuant to the Comprehensive Information Disclosure Policy of the Fund, the funding proposal titled "Acumen Climate Action Pakistan Fund" submitted by Acumen Fund, Inc is being circulated on a limited distribution basis only to Board Members and Alternate Board Members to ensure confidentiality of certain proprietary, legally privileged or commercially sensitive information of the entity.

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Funding Proposal

PUBLIC VERSION

Project/Programme title:	*Acumen Climate Action Pakistan Fund*
Country(ies):	*Pakistan*
Accredited Entity:	*Acumen Fund, Inc.*
Date of first submission:	*[2023/08/21]*
Date of current submission	*[2024/01/22]*
Version number	*[V.004]*

GREEN CLIMATE FUND

Contents

Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the GCF Information Disclosure Policy, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]"

A. PROJECT/PROGRAMME SUMMARY

A.1. Project or programme	Programme	**A.2. Public or private sector**	Private
A.3. Request for Proposals (RFP)	*If the funding proposal is being submitted in response to a specific GCF Request for Proposals, indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+.* Choose an item.		

A.4. Result area(s)

*Check the applicable GCF result area(s) that the overall proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of **GCF and Co-financers' contribution** devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.*

	GCF contribution	Co-financers' contribution[1]
Mitigation total	Enter number %	Enter number %
☐ Energy generation and access	Enter number %	Enter number %
☐ Low-emission transport	Enter number %	Enter number %
☐ Buildings, cities, industries and appliances	Enter number %	Enter number %
☐ Forestry and land use	Enter number %	Enter number %
Adaptation total	100 %	100 %
☒ Most vulnerable people and communities	75 %	75 %
☒ Health and well-being, and food and water security	25 %	25 %
☐ Infrastructure and built environment	Enter number %	Enter number %
☐ Ecosystems and ecosystem services	Enter number	Enter number

A.5. Expected mitigation outcome *(Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)*	*Indicate greenhouse gas (GHG) emission reductions or removals in tCO2eq over total lifespan of the project/programme[2]*	**A.6. Expected adaptation outcome** *(Core indicator 2: direct and indirect beneficiaries reached)*	*Total Beneficiaries: 13,055,893* *Male: 6,658,912; Female: 6,396,980*	
			1,993,266 direct beneficiaries	*11,062,627 indirect beneficiaries*
			0.8%	*4.6%*
A.7. Total financing (GCF + co-finance[3])	90M USD	**A.9. Project size**	Medium (Upto USD 250 million)	
A.8. Total GCF funding requested	28M USD			

[1] Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.
[2] The total lifespan of the project/programme is defined as the maximum number of years over which the outcomes of the investment are expected to be effective. This is different from the project/programme implementation period.
[3] Refer to the Policy of Co-financing of the GCF.

A.10. Financial instrument(s) requested for the GCF funding	*Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.* ☒ Grant 3M USD ☐ Loan Enter number ☐ Guarantee Enter number ☒ Equity 25M USD ☐ Results-based payment Enter number		
A.11. Implementation period	*10 years*	**A.12. Total lifespan**	*12 years (10 years + two 1-year extensions)*
A.13. Expected date of AE internal approval	*This is the date that the Accredited Entity obtained/will obtain its own approval to implement the project/ programme, if available.* Click or tap to enter a date.	**A.14. ESS category**	*Refer to the AE's safeguard policy and GCF ESS Standards to assess your FP category.* I-2
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	**A.16. Has Readiness or PPF support been used to prepare this FP?**	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	**A.18. Is this FP included in the country programme?**	Yes ☐ No ☒
A.19. Complementarity and coherence	*Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.* Yes ☐ No ☒		
A.20. Executing Entity information	*"ACAP Fund Manager" or "the Manager", a to-be-formed Dutch private limited company* *"ACAP" or "the Fund", a to-be-formed Dutch cooperative*		

A.21. Executive summary (max. 750 words, approximately 1.5 pages)

Climate Problem

1. **Pakistan is one of the most vulnerable countries to climate change.** It consistently ranks among the top 10 countries most affected by climate change and natural disasters globally[4]. Between 1992-2021, climate change induced disasters have resulted in economic losses equivalent to nearly USD 29.3B[5]. Recently, the 2022 floods

[4] Eckstein et al, Global Climate Risk Index 2021, 2021, https://www.germanwatch.org/sites/default/files/Global%20Climate%20Risk%20Index%202021_2.pdf

[5] World Bank, Pakistan Country Climate and Development Report, 2023, https://openknowledge.worldbank.org/server/api/core/bitstreams/2d1af64a-8d35-5946-a047-17dc143797ad/content

alone caused losses amounting to USD 15.2B[6]. These floods impacted nearly 33 million people, and were a 1 in 100 year event catalyzed by anthropogenic warming[7]. As global warming worsens the frequency and duration of climate disasters faced by Pakistan will be amplified leading to increasing vulnerability, economic losses and lives lost.

2. **By the end of the 21st century, under the worst GHG emissions scenario, Pakistan is projected to warm by 5.3°C which is twenty percent higher than the global average of 4.4°C[8].** The impact of this warming will be particularly pronounced for regions in the North of Pakistan and will increase the pace of glacial melting which will significantly impact the availability of water for the Indus River Basin System[9]. Additionally, extreme weather events such as heatwaves, droughts and floods will become more frequent under all future emissions scenarios. For example, for SSP1-1.9, SSP2-4.5 and SSP5-8.5 i.e. the low, medium and high emissions scenarios, the probability of heat waves occurring in Pakistan will increase by 18%-56%[10].

3. **The agriculture sector will be significantly impacted by climate change across multiple dimensions with significant socio-economic consequences for Pakistan.** Nearly 90% of the water supplied to the agriculture sector is dependent on water from the Indus River Basin[11]. Over time, climate change will decrease the water flows to this system by nearly 30%-40%, due to accelerated glacial melt[12]. Rising temperatures and extreme weather conditions will lower yields for key staple crops, alter growing seasons, worsen food waste and amplify crop losses. Extreme heat will further lower the productivity of agricultural workers and beyond directly affecting food production, extreme weather events will disrupt food distribution by slowing shipments, limiting market access by damaging already fragile infrastructure and further increasing the risks of damaged, spoiled or contaminated produce. Through its impact on the agriculture sector, climate change can have severe consequences for farmer livelihoods, rural economics and food security.

4. **The impact of climate change on the agriculture sector will likely cause Pakistan's GDP to contract by nearly 4.6%[13].** Given that nearly 67% of the rural population is engaged in agriculture, climate change will have a huge impact on rural livelihoods[14]. Smallholder farmers are the most vulnerable as they do not have the capacity to adapt to the effects of climate risks on yields and production. Approximately 90% of farmers in Pakistan are smallholders and any impact on their production will have significant consequences for the country's nutrition and food supply[15].

Proposed Interventions

5. **Acumen Climate Action Pakistan (ACAP) Fund will be Pakistan's first adaptation focused fund supporting venture, early growth, and growth stage agribusinesses building climate resilience.** The Fund is a USD 90M blended finance facility with USD 80M of catalytic investment capital and a USD 10M grant funded Technical Assistance Facility (TAF). The Fund will invest in 15-17 agribusinesses developing innovative adaptation solutions across three key themes to support smallholder farmers and build ecosystem level climate resilience. ACAP's blended capital fund structure with a singular sector focus will not only incentivise the growth

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[6] Ministry of Planning Development & Special Initiatives, Pakistan Floods 2022 Post Disaster Needs Assessment, https://thedocs.worldbank.org/en/doc/4a0114eb7d1cecbbbf2f65c5ce0789db-0310012022/original/Pakistan-Floods-2022-PDNA-Main-Report.pdf
[7] ibid
[8] World Bank Climate Change Portal. Accessed June 18, 2023
[9] The World Bank Group and the Asian Development Bank, Climate Risk Country Profile: Pakistan, 2021, https://climateknowledgeportal.worldbank.org/sites/default/files/2021-05/15078-WB_Pakistan%20Country%20Profile-WEB.pdf
[10] World Bank Climate Change Portal. Accessed June 18, 2023
[11] Talpur et al, Water consumption pattern and conservation measures in academic building: a case study of Jamshoro Pakistan, 2020, https://doi.org/10.1007/s42452-020-03588-z
[12] FAO, Water availability, use and challenges in Pakistan, 2021, https://www.fao.org/3/cb0718en/cb0718en.pdf
[13] State Bank of Pakistan, Monetary Policy Statement, 2023, https://www.sbp.org.pk/m_policy/2023/MPS-Apr-2023-Eng.pdf
[14] Arif et al, Climate, population and vulnerability in Pakistan: Exploring evidence of linkages for adaptation, 2019, https://knowledgecommons.popcouncil.org/cgi/viewcontent.cgi?article=1754&context=departments_sbsr-pgy
[15] Hazoor et al, Determinants of Small Farmers Poverty in the Central Punjab (Pakistan), 2006, https://www.fspublishers.org/published_papers/96458_..pdf

of strong companies building resilience, but also - by de-risking the market - catalyze further private sector investment in climate resilient agriculture.

6. **ACAP Fund will be investing in three investment themes that will specifically target building adaptation capacity across the value chain:**
    1. Farmer Platforms and Financial Solutions
    2. Smart Farming
    3. Post-Harvest Technologies

7. **ACAP Fund's TA facility will be used to accelerate impact by supporting capacity enhancement, reducing knowledge gaps & promoting ecosystem development.** TAF will provide support to portfolio companies, smallholder farmers, and vulnerable agri-dependent livelihoods across five areas:

    1. Climate Adaptation
    2. Gender Initiatives
    3. Business Development
    4. ESG Reporting
    5. Impact Evaluation

    The TA facility will also amplify ACAP's ecosystem level impact through developing and disseminating knowledge products/insights and convening stakeholders to create sectoral connections.

8. **ACAP's investment thesis is informed by local context, climate and market needs, and sector dynamics in Pakistan.** The Fund has a local team with almost two decades of local investing experience, which has spent over 24 months conducting targeted market research, engaging key stakeholders and building a robust pipeline of high climate impact, investable agri-businesses that are building critical solutions across the value chain. ACAP's investment and impact strategy was developed with climate adaptation of smallholder farmers and vulnerable agri-dependent populations at its core.

9. **Acumen has decades of experience investing at the nexus of poverty and climate change.** Through its early-stage pioneer portfolio, Acumen has invested approximately USD 154.4M in 167 companies, including 36 agribusinesses across eight regions. The organization has launched three climate-focused investment funds, with GCF as an anchor investor, that have raised a combined USD 222M to date. One of these funds, Acumen Resilient Africa Fund (ARAF) (FP078), was the world's first impact fund focused on building smallholder farmer climate resilience. ACAP Fund draws on learnings from ARAF, Acumen's sector expertise in impact investing in emerging markets, and its two decades of local experience investing in Pakistan.

Climate Results/Benefits

10. **ACAP's investments should help strengthen the resilience of approximately 13.1M people in agricultural communities, both directly and indirectly, against the effects of climate change.** This is expected to be accomplished by boosting farmer incomes, improving access to climate smart solutions, improving gender equity, and strengthening food systems.

11. **On an ecosystem level ACAP should help crowd in climate resilient investments in Pakistan's vulnerable agriculture sector by demonstrating the commercial viability of agribusiness models.** This should reduce risk perceptions amongst commercial investors while providing tangible evidence of profitability, growth potential and competent financial stewardship. Furthermore, ACAP will establish stakeholder platforms to disseminate insights, knowledge and research. Enhancing stakeholder awareness and providing opportunities to collaborate should create increased capacity and encourage the development of climate adaptive solutions for the agriculture sector of Pakistan.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Country Context

1. Pakistan is the fifth largest country in the world and remains one of the most vulnerable to climate change.[16] Pakistan has consistently ranked among the top 10 most vulnerable countries on the Climate Risk Index with financial losses amounting to about USD 29.3B from 199 extreme weather events between 1992-2001.[17] Beyond the devastating floods, Pakistan suffered at least three additional extreme weather events in 2022 alone, including the hottest March since 1901.[18] As the country simultaneously struggles with various macro-economic challenges and the burden of a growing population, UNESCAP estimates that Pakistan will lose up to 6% of its GDP under the current climate change scenario - with projected GDP losses surpassing to 9% under the worst-case climate change scenario. Despite accounting for less than 1% of global GHG emissions, Pakistan has already sustained climate related damage worth billions of dollars - and as is the case with most catastrophes, those who are most economically vulnerable suffer disproportionately.

Climate Context and Baseline

2. Over the past several decades, Pakistan has suffered repeated and increasingly worse climate shocks. These include:

(i) Floods and Glacial Lake Outburst Floods (GLOFs):
Since 1999, flooding has become the most common climate hazard faced by Pakistan (68% of all severe weather events have been floods) and has caused damages worth approximately USD 41.4B while affecting 80.5M lives cumulatively. Over the last two decades, the country has faced three major flood events which were ruinous for the country's economy, and most specifically its agriculture sector. In 2010, changing weather patterns led to unprecedented levels of rainfall which devastated the sector by destroying key agricultural inputs and infrastructure (i.e., fertilizers, seed stocks, tube wells, animal shelters, water channels, and houses). Two other major floods hit Pakistan in the subsequent five years, causing further loss of life and repeatedly hurting the already impacted farmers. For more details on the 2022 floods.

Pakistan is reliant on a single major river system, the Indus River Basin System. The basin's major water sources include glacier melt and seasonal rains (monsoon). As such, Pakistan is dependent on a delicate ecosystem to supply its water (the country has the most glacial ice outside the polar regions and is surrounded by more than 7,000 mountain glaciers melting at an increasingly quickening pace). The Himalayas and its glaciers, one of the largest snowy mountain ranges in the world, make up the principal source of the water that flows through Pakistan. However, the range and its glaciers are now melting at a rate ten times faster than in the previous century. The rapid and excess flow of water from the snowy mountains is also causing increasing, catastrophic glacial lake outbursts. 2022 saw more than a dozen glacial lake outbursts in the country, which is more than double the historic average of five.

Further, monsoon rainfall has been increasingly unpredictable. 2022 saw seven times more rain compared against the previous year, breaking rainfall records set in 1961. All these factors contribute to rivers overflowing, increasing the frequency and intensity of floods throughout the country.

(ii) Extreme temperatures:
In the last fifty years, Pakistan's temperatures have risen by 0.5°C, and could potentially rise an additional 1.3°C–4.9°C by the 2090s (measured against the 1986–2005 baseline). The country now faces rates of warming

[16] German Watch, Global Climate Risk Index, 2021, https://germanwatch.org/sites/default/files/Global%20Climate%20Risk%20Index%202021_1.pdf
[17] EM-DAT, CRED/UCLouvain (Brussels, Belgium), last accessed January 17, 2024, www.emdat.be
[18] World Weather Attribution, Climate change made devastating early heat in India and Pakistan 30 times more likely, 2022, https://www.worldweatherattribution.org/wp-content/uploads/India_Pak-Heatwave-scientific-report.pdf

considerably above the global average.[19] Despite the speed with which temperatures are rising, increases in both the annual maximum and minimum temperatures are projected to be larger than the rise in average temperature, amplifying the pressure on human health, livelihoods, and ecosystems. Extreme heat waves and prolonged periods of droughts are now a common occurrence in the summer months, wreaking havoc on agricultural yields and further worsening conditions for Pakistan's economy. The IPCC Sixth Assessment Report's findings support climate projections which point to an observable increase in the variance in temperature.[20] Projected changes in the variance of minimum and maximum temperatures are expected to increase by as much as 3°C in northern temperate regions.[21] In other words, this indicates that temperatures are becoming more extreme and fluctuating more often than they have historically. This also means that the frequency and intensity of heatwaves, cold spells, and other extreme climate events are likely to increase, leading to an array of challenges for the ecosystem, the country's infrastructure, and the agriculture sector and economy at large. When reviewing projections for Pakistan, maximum and minimum temperatures have risen throughout the country. This increasing trend of maximum temperature is more significant in the future (2021-2050) as compared to the present climate (1981-2010).[22] Additionally, extreme night temperatures have also decreased in the current climatic period. Mostly occurring in the northern parts of the country, cold wave episodes encourage these negative anomalies of extremely cold temperature.[23]

(ii a) Heatwaves:
According to the IPCC, global surface temperatures have been 1.09°C higher in recent years as compared to the last century. Furthermore, from 1850 to 2019, the human-mediated global surface temperature increase is around 1.07 °C. What is more alarming is how surface temperatures across Asian regions, like Pakistan have risen at a much higher rate than the global comparable.[24] For the last five decades, Pakistan has been facing increasingly brutal heat waves, with the country seeing temperatures frequently exceeding 50 degrees Celsius. The 2022 heat wave produced the hottest March in Pakistan since 1901; temperatures soared to over 50°C in parts of Pakistan, leading to water shortages, and a loss of life, crops, and livestock.[25]

(ii b) Droughts:
Approximately 25% of agricultural land in Pakistan is rain-fed ("*barani*") and has little to no irrigation infrastructure.[26] With rising temperatures and increasing variability in rainfall, droughts have become more frequent in the country. One of the worst droughts Pakistan experienced occured between 1998 to 2002 after the 1997-98 El Nino Southern Oscillation (ENSO).[27] Due to the close linkage between atmospheric circulation patterns and climatic variables, researchers have established that extreme weather events such as droughts can be attributed to variations in atmospheric circulation patterns.[28] In Pakistan, southern regions are more likely to be adversely impacted by droughts due to pre-existing water scarcity issues. Water shortages associated with periods of droughts are not only responsible for compromising food production but also health and ecosystem sustainability.[29] Over time, extremely high temperatures coupled with low rainfall have made droughts more frequent.

(iii) Water Scarcity:

[19] World Bank, Climate Risk Country Profile: Pakistan, 2021, https://climateknowledgeportal.worldbank.org/sites/default/files/2021-05/15078-WB_Pakistan%20Country%20Profile-WEB.pdf
[20] IPCC, Sixth Assessment Report, 2021, https://www.ipcc.ch/assessment-report/ar6/
[21] Vasseur, David A., et al. "Increased Temperature Variation Poses a Greater Risk to Species than Climate Warming." *The Royal Society Publishing* (2014), The Royal Society, https://royalsocietypublishing.org/doi/10.1098/rspb.2013.2612.
[22] Cheema, S. B. (2023), A Comparison of Present and Future Climate Anomalies over Pakistan. Pakistan Journal of Meteorology. 15 (29):85-94
[23] Cheema, S. B. (2023), A Comparison of Present and Future Climate Anomalies over Pakistan. Pakistan Journal of Meteorology. 15 (29):85-94
[24] Sharma, Ayushi, et al. (2022), "Heatwaves in South Asia: Characterization, Consequences on Human Health, and Adaptation Strategies." MDPI, Multidisciplinary Digital Publishing Institute, https://www.mdpi.com/2073-4433/13/5/734#B4-atmosphere-13-00734.
[25] Amnesty International, Pakistan: Extreme weather: Searing heatwaves and torrential rains in Pakistan, and their impact on Pakistan, 2022, https://www.amnesty.org/en/documents/asa33/5828/2022/en/
[26] Baig, M. B., Shahid, S. A., & Straquadine, G. S. (2013). Making rainfed agriculture sustainable through environmental friendly technologies in Pakistan: A review. International Soil and Water Conservation Research, 1(2), 36-52
[27] Hina, Saadia, et al. "Droughts over Pakistan: Possible Cycles, Precursors and Associated Mechanisms." Taylor & Francis, 2021, https://www.tandfonline.com/doi/full/10.1080/19475705.2021.1938703.
[28] Omidvar K, Fatemi M, Narangifard M, Hatami Bahman Beiglou K. 2016. A study of the circulation patterns affecting drought and wet years in central Iran. Adv Meteorol. 2016:1–14
[29] Hina, Saadia, et al. "Droughts over Pakistan: Possible Cycles, Precursors & Associated Mechanisms." Taylor & Francis, 2021, https://www.tandfonline.com/doi/full/10.1080/19475705.2021.1938703.

About 92% of Pakistan is classified as semi-arid to arid, and the vast majority of Pakistanis are dependent on surface and groundwater sources from a single source—the Indus River basin.[30] The country ranks 14 among the 17 'extremely high water-risk' countries of the world, a list that includes hot and dry countries like Saudi Arabia. Over 80% of the total population in the country faces 'severe water scarcity' for at least one month of the year. In addition to surface water, Pakistan's groundwater resources—the water supply of last resort—are severely overdrawn, mainly to supply water for irrigation. In a report published by the UN, Pakistan was among 23 countries ranked 'critically water insecure.[31] If the situation remains unchanged, the whole country may face 'water scarcity' by 2025 as water availability in Pakistan has plummeted from 5,229 cubic meters per inhabitant in 1962 to 1,017 in 2021.[32][33]

Climate Projections

3. For different SSP pathways there are various projections for temperature, precipitation and extreme weather events that demonstrate a worsening future climate for Pakistan. Under the worst emissions pathway Pakistan will experience 5.3°C of warming as compared to 4.4°C for the rest of the world.[34] Regions in the Northern areas of Pakistan are expected to face higher temperatures as compared to the South. Additionally, Pakistan will experience between 11.5%-26.4% higher precipitation, at the end of the 21st century, which is higher than any other country in South Asia[35].

4. Increased temperatures will accelerate the melting of glaciers which will result in a threefold increase in GLOF risk for the Northern Areas of Pakistan. Concurrently, climate change is projected to reduce the supply of water as accelerated glacial melt will decrease flows to the Indus River Basin System by 30%-40%[36]. With the agriculture sector utilizing 90%[37] of surface water in Pakistan, water shortages for this vital sector will negatively impact yields and cause GDP to contract by nearly 4.6% per year[38]. Higher temperatures will also result in a 75% increase in the severity and 65% increase in the duration of droughts. Similarly, under the highest emissions pathway, the probability of heatwaves occurring in Pakistan will increase by 62%-140% as a result of climate induced warming. Over the next 25 years the country will also face a doubling in high end flood risk due to an increase in extreme precipitation events and glacial melt[39]. The most affected regions will be the provinces of Sindh and Punjab with the projected affected population standing at between 9.2M-13.3M.

Pakistan does not have the resources to respond to a growing climate emergency

5. Pakistan faces the concurrent challenges of providing basic socio-economic support for its growing population while struggling with national debt and a devaluing currency. Unfortunately, the vulnerability of the agriculture sector has placed yet another burden on the economy.[40] Pakistan faces many economic pressures - the country has a large budget deficit, suffers from a circular debt problem and has been in 22 International Monetary Fund (IMF) programs, most recently in 2023. Challenging macroeconomic conditions resulted in the Pakistani Rupee (PKR) losing 25.9% of its value since Q1 2023 as policy rates rose to a record 23% in 2023 with conservative estimates putting inflation at 29%.[41] [42] [43] The mounting burden of billions of dollars in damages from climate change (USD 30B in the recent floods alone) further reduces fiscal space for investing in climate adaptation.

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[30] International Institute of Sustainable Development, 2016, https://www.iisd.org/articles/insight/making-every-drop-count-pakistans-growing-water-scarcity-challenge
[31] UN, Global Water Security Assessment, 2023, https://inweh.unu.edu/wp-content/uploads/2023/03/Global-Water-Security-Assessement-2023_F.pdf
[32] PIDE, Water Crisis in Pakistan: Manifestation, Causes and the Way Forward, 2022, https://file.pide.org.pk/uploads/kb-060-water-crisis-in-pakistan-manifestation-causes-and-the-way-forward.pdf
[33] Ahmad et al, Impact of water insecurity amidst endemic and pandemic in Pakistan: Two tales unsolved, 2022, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9464874/#:~:text=As%20per%20IMF%2C%20Pakistan's%20per,approximately%2083%20MAF%20%5B21%5D
[34] World Bank Climate Change Portal. Accessed June 18, 2023
[35] World Bank Climate Change Portal. Accessed June 18, 2023
[36] FAO, Water availability, use and challenges in Pakistan, 2021, https://www.fao.org/3/cb0718en/cb0718en.pdf
[37] Pakistan Academy of Sciences, Water Security Issues of Agriculture in Pakistan, 2019, https://www.paspk.org/wp-content/uploads/2019/06/PAS-Water-Security-Issues.pdf
[38] State Bank of Pakistan, https://www.sbp.org.pk/m_policy/2023/MPS-Apr-2023-Eng.pdf
[39] Willner et al, Adaptation required to preserve future high-end river flood risk at present levels, 2018, https://www.science.org/doi/10.1126/sciadv.aao1914
[40] Trading Economics, https://tradingeconomics.com/pakistan/currency
[41] Trading Economics, https://tradingeconomics.com/pakistan/currency
[42] State Bank of Pakistan, https://www.sbp.org.pk/m_policy/2023/MPS-Dec-2023-Eng.pdf
[43] Trading Economics, https://tradingeconomics.com/pakistan/inflation-cpi

6. Climate financing is critical for Pakistan and, even prior to the recent 2022 floods, its adaptation needs stood at an estimated USD 7B-14B annually.[44] As discussed, the floods (only one severe weather event in 2022) caused economic losses worth USD 30B leaving the country with even more limited fiscal space to invest in adapting to worsening climate impacts. Despite the crucial need, Pakistan's share of the available international climate finance has also been very low, relative to other climate-vulnerable countries (standing at USD 1B in 2022).[45] In comparison, during the same period, neighboring countries such as India and China which aren't as vulnerable to the effects of global warming raised approximately USD 3.7B and USD 2.6B in climate finance. For reference, India and China rank 116th and 39th on the ND-GAIN Index, which assesses a country's climate vulnerability and readiness to improve resilience, whereas Pakistan ranks 150th[46]. As the world further warms, and Pakistan becomes even more vulnerable, adaptation finance from both public and private sector sources is critical in reducing the impact of and increasing resilience to climate catastrophes like those seen recently.

The agriculture sector is critical for Pakistan's economy

7. Agriculture is widely regarded as the mainstay of Pakistan's economy and is fundamental to food security for a large and growing population. Presently, agriculture contributes 22.9% to the national GDP and provides employment for 37.4% of the country's labor force[47]. Demographically, slightly less than two-thirds of the population live in rural areas, and more than 65-70% of them are either directly or indirectly dependent on agriculture for their livelihoods.[48] To illustrate, out of 22 million rural households, livestock rearing provides 8 million households with 35%-40% of their income[49]. In addition, while agriculture serves as the main income source for 34% and 74% of economically active men and women respectively[50], in terms of equitable distribution of resources based on contribution, the sector is gender discriminatory. Nevertheless, it is vital to the economy as it generates over 77% of the country's USD 28B export revenue (FY23) through agriculture-based textiles and value-added food products.[51] The sector (which includes the country's textile industry) forms strong linkages with downstream industries as a primary supplier of raw materials.[52]

8. The various socioeconomic and ecological forces shaping the agricultural context remain complex.

Agriculture sector suffers catastrophic climate impact: Pakistan is an example

9. There is a significant body of research linking climate change to an outsized impact on agriculture.[53] The crop cycles for staple crops (i.e., wheat and rice) have been fixed for centuries. More recently, however, rapid temperature variations have produced changes that range from the duration of growing seasons to volatility in weather and rainfall patterns. IPCC's highest scenario of warming found a mean effect on the yields of four crop groups (coarse grains, oil seeds, wheat and rice, accounting for about 70% of global crop harvested area) of -17% globally by 2050[54]. Figure 1 depicts the ND-Gain Index score for countries with a large share of GDP accounted for by agricultural activities. For reference, a high score represents high exposure and sensitivity to climate change with a low ability to adapt to its consequences, and vice versa. Through the scatter plot, it is evident that agriculture-dependent economies are more vulnerable to the negative impacts of climate change.

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[44] ADB, Climate Change Profile of Pakistan, 2017, https://www.adb.org/sites/default/files/publication/357876/climate-change-profile-pakistan.pdf
[45] European Investment Bank, Joint Report On Multilateral Development Banks' Climate Finance, 2023, https://publications.iadb.org/en/2022-joint-report-multilateral-development-banks-climate-finance
[46] ND-GAIN Index, 2021, https://gain.nd.edu/our-work/country-index/rankings/
[47] Ministry of Finance, Pakistan Economic Survey 2022-2023, https://www.finance.gov.pk/survey/chapters_23/02_Agriculture.pdf
[48] Ministry of Finance, Pakistan Economic Survey 2020-2021, https://www.finance.gov.pk/survey/chapters_21/02-Agriculture.pdf
[49] Ministry of Finance, Pakistan Economic Survey 2022-2023, https://www.finance.gov.pk/survey/chapters_23/02_Agriculture.pdf
[50] FAO, Country Programming Framework for Pakistan 2018-2022, https://www.fao.org/fileadmin/user_upload/FAO-countries/Pakistan/docs/FAOPK_CPF-2018-2022.pdf
[51] State Bank of Pakistan, https://www.sbp.org.pk/ecodata/Export_Receipts_by_Commodities_and_Groups.pdf
[52] Ministry of Finance, Pakistan Economic Survey 2020-2021, https://www.finance.gov.pk/survey/chapters_21/02-Agriculture.pdf
[53] Gorst et al. (2018) *Crop productivity and adaptation to climate change in Pakistan*. Environ Dev Econ 23(6):679–701; Tariq et al. (2014) *Food security in the context of climate change in Pakistan*, PJCSS; Asif (2013) *Climatic Change, Irrigation Water Crisis, and Food Security in Pakistan*; Rasul et al. (2011) *Effect of temperature rise on crop growth & productivity*. Pak J Meterol 8:53–62; Siddiqui et al. (2012) *The Impact of Climate Change on Major Agricultural Crops: Evidence from Punjab, Pakistan*. Pak Dev Rev 51:2012;
[54] FAO, Climate change and food security: risks and responses, 2015, https://www.fao.org/3/i5188e/I5188E.pdf

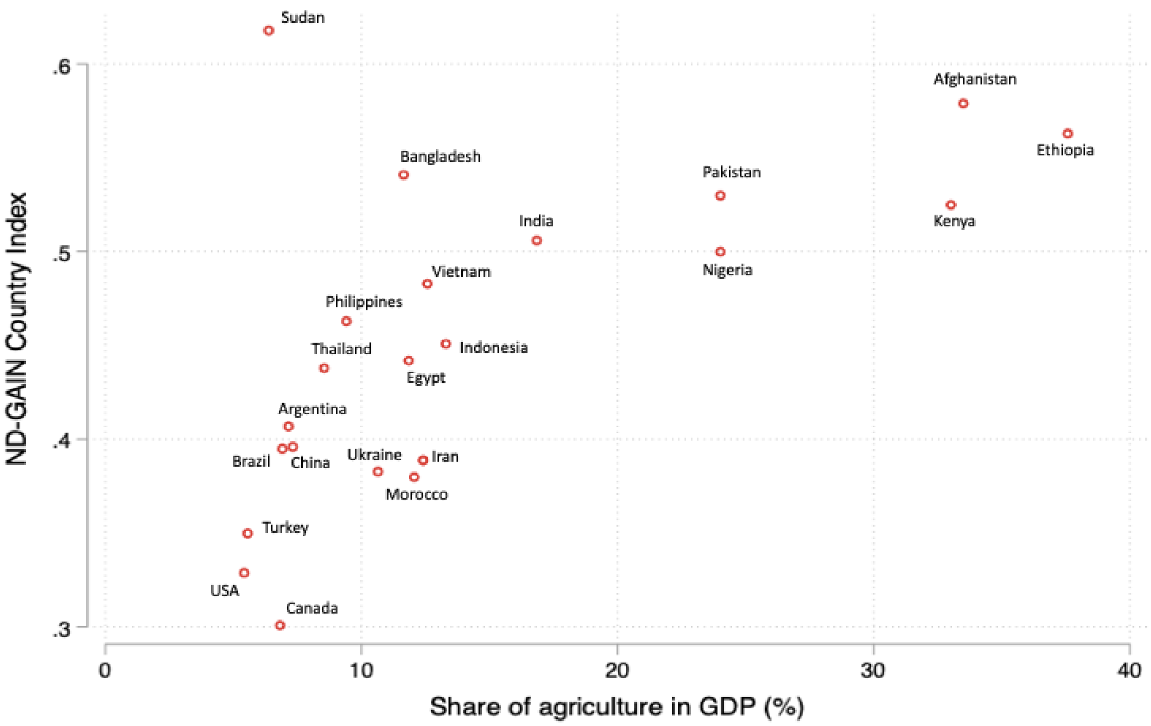

Figure 1. Climate vulnerability (according to ND-GAIN Index score) for 23 agrarian economies

10. Table 1 summarizes the impact of climate change on the agriculture sector, with a focus on Pakistan. Climate change impacts the agriculture sector through direct and indirect effects on infrastructure, supply chain, crop growth processes and labour. The direct effects include changes in temperature, precipitation, and carbon dioxide availability. The indirect effects include the impact of unexpected and extreme changes in water availability and seasonality, soil erosion and organic matter transformation, increased occurrences of pests and diseases, the arrival of invasive species, and decline in arable land due to desertification and coastal land submergence.[55] According to the latest climate projection models (CIMP6), at the end of the 21st century under RCP 4.5 (moderate climate scenario) and 8.5 (very high baseline emission scenario), the pathway where emissions peak by 2040 and the business as usual case persists, Pakistan faces median temperature increases of 1.92°C and 5.18°C, respectively.[56] For reference, temperature rises of 0.5°C to 2°C are projected to cause an 8-10% loss in agricultural productivity.[57] As an example, under the high emissions scenario, major crops such as sugarcane and rice will fare poorly, experiencing a projected 25% and 20% reduction in yield respectively.[58] The productivity of wheat, another staple, is also expected to fall by a substantial 50% by the end of this century when temperatures are predicted to exceed by 5-6°C.[59]

11. Temperature increases are also expected to impact the health and productivity of agricultural labourers. During peak months, productivity levels for labour have already declined by 10% and, under RCP 8.5, are expected to decline by 20%.[60] The frequency and intensity of extreme weather events, which are responsible for 60%-80% of agricultural loss in developing countries, are also projected to increase in Pakistan. Extreme heatwaves hit the agriculture sector hard, which, to-date, have occurred during the crucial final weeks of the wheat growing season, killing the crop shortly before harvest. Given that wheat is a staple crop in the country and Pakistan maintains one of

[55] World Bank, Climate Risk Country Profile: Pakistan, 2021, https://climateknowledgeportal.worldbank.org/sites/default/files/2021-05/15078-WB_Pakistan%20Country%20Profile-WEB.pdf
[56] ADB, Climate Change Profile of Pakistan, 2017, https://www.adb.org/sites/default/files/publication/357876/climate-change-profile-pakistan.pdf
[57] Syed, Areeja, et al. "Climate Impacts on the Agricultural Sector of Pakistan: Risks and Solutions." Environmental Challenges, Elsevier, 2 Jan. 2022, https://www.sciencedirect.com/science/article/pii/S2667010021004078
[58] World Bank, Climate Risk Country Profile: Pakistan, 2021, https://climateknowledgeportal.worldbank.org/sites/default/files/2021-05/15078-WB_Pakistan%20Country%20Profile-WEB.pdf
[59] Syed, Areeja, et al. "Climate Impacts on the Agricultural Sector of Pakistan: Risks and Solutions." Environmental Challenges, Elsevier, 2 Jan. 2022, https://www.sciencedirect.com/science/article/pii/S2667010021004078
[60] Dunne, J. P., Stouffer, R. J., & John, J. G. (2013). Reductions in labour capacity from heat stress under climate warming. Nature Climate Change, 3(6), 563–566. URL: http://www.precaution.org/lib/noaa_reductions_in_labour_capacity_2013.pdf

the highest per capita wheat consumption rates in the world (on average, Pakistanis consume 124 kg per year with wheat flour making up 72% of total caloric intake), losses resulting from extreme heat have been especially impactful. The 2022 heatwave gave way to drought, causing rainfall totals to fall between a quarter to a third of normal levels. Mango, the second largest fruit crop in Pakistan, also saw a 50% drop in yield.[61] As one of Pakistan's major fruit exports, this decline in yield hit vulnerable farmers disproportionately hard and contributed to a significant loss in export earnings for the country.

12. Over the past two decades, the country has been hit by two deadly floods that have cumulatively led to USD 14.3B worth of losses to the agricultural sector, killed approximately 2.5 million livestock, and inundated millions of acres of arable land[62][63]. Furthermore, climate change will reduce total water availability across the whole Indus Basin by 30%-40% over the second half of the 21st century. This can be attributed to a prolonged shortfall in precipitation and accelerated melting of glacial lakes which will eventually lead to this reduction. Another consequence of climate change is how it has aggravated Pakistan's shortfall in domestic food production. As an example, the 2022 heatwave wiped out 3M metric tons of the country's wheat production which resulted in a 15.46% increase in the country's food import bill.[64]

13. Due to the sensitivity of crops and livestock to temperature, water availability, and extreme weather, climate change threatens agricultural production and heightens the vulnerability of smallholder farmers as a result. Farmers need different adaptation tools to reduce climate vulnerability. These include access to drought-resistant varieties of crops; timely and reliable meteorological information; crop diversification; improved irrigation efficiency using micro/drip irrigation; adoption of green energy for on-farm energy needs; and efficient livestock management systems (e.g., increased herd disease surveillance). Currently, smallholder farmers don't have access to such adaptation tools. Scientists predict that agricultural productivity will reduce by 4.5% to 9% in the short-term and by 25% (in the long term, i.e., 2070-2099) if farmers do not adapt and make use of advancements in science and technology.[65]

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[61] Climate Tracker, Mango crops in Pakistan ravaged after record heat and drought, 2022, https://climatetracker.org/mango-crops-in-pakistan-ravaged-after-record-heat-and-drought/

[62] UNDP, Pakistan Floods 2022: Post-Disaster Needs Assessment (PDNA), 2022, https://www.undp.org/pakistan/publications/pakistan-floods-2022-post-disaster-needs-assessment-pdna

[63] ADB, Pakistan Floods 2010: Preliminary Damage and Needs Assessment, 2010, https://www.adb.org/sites/default/files/linked-documents/44372-01-pak-oth-02.pdf

[64] Pakistan Bureau of Statistics, Annual Analytical Report On External Trade Statistics Of Pakistan Fy 2020-21, https://www.pbs.gov.pk/sites/default/files/external_trade/annual_analytical_report_on_external_trade_statistics_of_pakistan_2020-21.pdf

[65] Gbetibouo, G. A., Ringler, C., and Hassan, R. (2010). Vulnerability of the South African farming sector to climate change and variability: an indicator approach. Nat. Resour. Forum 34, 175–187. doi: 10.1111/j.1477-8947.2010.01302.

Climate Change Hazards	Impact on Agriculture			
	Production and Harvest	Processing and Storage	Transportation and Retail	Pakistan Example
High Temperature and Heat Waves	**Reduced crop yields.** The yield for basmati rice, Pakistan's second largest export, falls by 7.2% for every 10% rise in temperature[66] **Reduced milk, meat quantity and animal fertility.** Heat stress can reduce milk production between 14%-35% **Increase in animal mortality.** The direct effects of high temperatures on livestock health include metabolic disruptions, oxidative stress and immune suppression leading to infections and death[67]	**Spread of pests and diseases.** Increasing temperatures can encourage the incidence of pests such as the khapra beetle. Infestations of khapra beetle cost Pakistan USD 1B in rice exports between 2011-2014[68] **Food spoilage.** A 2°C-3°C rise in temperature will likely reduce chilled storage life by half[69]	**Changes in consumer requirements.** The impact of climate change on food production can raise the market price for consumers. This reduces their purchasing power and dietary diversity. Changes in weather can also influence consumer behavior. For example, warmer temperatures tend to increase beverage sales	• High temperatures led to an outbreak of the cotton leaf curl virus (CLCuV) in 2010-2011 which destroyed approximately 40% of cotton crops[70] 2022 heatwave led to: • Wide-scale damages to wheat harvest leading to an estimated shortfall of 3M metric tons[71] • 50% reduction in mango yields[72] • 10-30% increase in water demand for crops • Death of up to 200 animals due to the extreme heat in Cholistan Desert, Punjab • 62M hectares, out of 79.6M hectares of land, presently vulnerable to desertification[73]

[66] World Bank Group, From Swimming in Sand to High and Sustainable Growth: A Roadmap to Reduce Distortions in the Allocation of Resources and Talent in the Pakistani Economy, 2022, http://hdl.handle.net/10986/38133
[67] Umberto Bernabucci, Climate change: impact on livestock and how can we adapt, 2019, https://doi.org/10.1093/af/vfy039
[68] Honey et al, Trogoderma Granarium (Everts) (Coleoptera: Dermestidae), An Alarming Threat To Rice Supply Chain In Pakistan, 2017, https://esciencepress.net/journals/index.php/IJER/article/view/2046
[69] Duchenne-Moutien, Climate Change and Emerging Food Safety Issues: A Review, 2021, https://doi.org/10.4315/JFP-21-141
[70] Hassan, Feehan, et al. "Cotton Leaf Curl Virus (Clcuv) Disease in Pakistan: A Critical Review." *Research Gate* , 2010, www.researchgate.net/publication/305994089_Cotton_Leaf_Curl_Virus_CLCuV_Disease_in_Pakistan_A_Critical_Review.
[71] Pakistan Bureau of Statistics, Annual Analytical Report On External Trade Statistics Of Pakistan Fy 2020-21, https://www.pbs.gov.pk/sites/default/files/external_trade/annual_analytical_report_on_external_trade_statistics_of_pakistan_2020-21.pdf
[72] Climate Tracker, Mango crops in Pakistan ravaged after record heat and drought, 2022, https://climatetracker.org/mango-crops-in-pakistan-ravaged-after-record-heat-and-drought/
[73] PMAS Arid Agriculture University, https://www.uaar.edu.pk/news-detail.php?news_id=403

	Heavy Rains and Flooding	**Increase in crop damage.** Floods in 2010-2014 floods damaged 4.3M ha of cropland **Harvest delays.** Flooding causes soil waterlogging which promotes nitrogen loss. This causes harvest delays and reduces crop productivity[74]	**Supply chain disruptions.** Floods can significantly increase the cost of doing business. Headline inflation increased by 2% following the 2010 floods in Pakistan[75] **Damage to buildings, machinery and finished products.** The 2022 floods resulted in a USD 5.3B economic loss for SMEs with nearly 40% operating in the agriculture sector[76]	**Infrastructure damage, unfavorable driving conditions and lack of access to markets.** The 2022 floods damaged 13,000 km of roads and 410 bridges causing significant disruption to transport and people's ability to access markets[77]	2022 Floods led to:[78] • ~USD 3.7B in losses to the agricultural sector (food, livestock, fisheries) • Loss of 13.3M tons in the production of major crops • Death of 1.2M livestock • Inundation of 2M acres of cultivated land	
	Droughts	**Reduced crop yields.** Between 2011-2014 farmers in Balochistan, affected by droughts, suffered yield losses of 50%-80%[79] **Drought stress on animals.** Average livestock losses due to drought in	**Reduced access to rainfed and groundwater resources.** The severe drought in 2001-2002 resulted in water shortages up to 51% of normal supplies	**Decrease in food supply.** For net food importing countries, such as Pakistan, a drought can cause a 10%-20% decline in food supply[82] **Increase in prices.** Regions in Pakistan, affected by drought, have experienced food price increases of up to 65%[83]	• Across Pakistan agriculture is facing water shortages of up to 50%[84] 2015-2017 drought led to:[85] • 53% reduced crop harvest • 48% decreased livestock output	

[74] Kaur et al, Impacts and management strategies for crop production in waterlogged or flooded soils: A review, 2019, https://doi.org/10.1002/agj2.20093
[75] Miles Parker, The impact of disasters on inflation, 2016, https://www.ecb.europa.eu/pub/pdf/scpwps/ecbwp1982.en.pdf
[76] SMEDA, Preliminary Assessment of Impact of Floods on SMEs, 2022, https://www.researchgate.net/publication/366635147_PRELIMINARY_ASSESSMENT_OF_IMPACT_OF_FLOODS_ON_SMEs
[77] OCHA, Revised Pakistan 2022 Floods Response Plan, 2022, https://reliefweb.int/report/pakistan/revised-pakistan-2022-floods-response-plan-01-sep-2022-31-may-2023-04-oct-2022
[78] UNDP, Pakistan Floods 2022: Post-Disaster Needs Assessment (PDNA), 2022, https://www.undp.org/pakistan/publications/pakistan-floods-2022-post-disaster-needs-assessment-pdna
[79] UNDP, Drought Risk Assessment in the Province of Balochistan, Pakistan, 2015, https://www.undp.org/pakistan/publications/drought-risk-assessment-balochistan-province-pakistan
[82] FAO, Managing Risks to Build Climate-Smart and Resilient Agrifood Value Chains: The Role of Climate Services, 2022, https://doi.org/10.4060/cb8297en
[83] Islamic Relief, Drought Assessment Report Chaghai, Noshki, Kharan, Washuk and Quetta, 2019, https://reliefweb.int/report/pakistan/drought-assessment-report-chaghai-noshki-kharan-washuk-and-quetta
[84] Abdullah Khan, Ameer. "Pakistan's Climate-Induced Water Scarcity and Its Impact on Agriculture." Hilal Publications, 2023, www.hilal.gov.pk/index.php/eng-article/pakistan%E2%80%99s-climate-induced-water-scarcity-and-its-impact-on-agriculture/NjM0Nw==.html
[85] Relief Web, Pakistan: Drought - 2014-2017, https://reliefweb.int/disaster/dr-2014-000035-pak

	Balochistan are 37%[80] **Soil degradation.** Topsoils, due to drought, can become highly vulnerable to erosion. 90% of the total soil loss, in a 20-30 year cycle, can be due to drought breaking rain[81]				

Table 1. Impact of climate change at each stage of the agriculture value chain[86]

Climate Rationale for Agri Sector Barriers

Ecosystem Level Barriers

14. With an already fragmented value chain, the physical and economic challenges created because of climate variability not only affect small farmer livelihoods and rural economics, but also worsen food security and health outcomes at a national level. Given the position of smallholder farmers within the value chain, any change in their output owing to climate shocks affects the revenue of all downstream actors through a reduction in quantity and/or quality of produce[87]. For example, one study found that cotton spinners in Pakistan faced increased costs in cleaning the cotton boll after heavy rainfall due to elevated moisture content in the boll.[88] Additionally, storage-processing-transportation facilities are generally concentrated within a geographic area and focus on specific commodities with little flexibility to shift to other activities.[89] Hence for foods/crops with few substitutes, long growing periods, and limited actors, the network interdependency significantly raises the impact of climate hazards throughout the chain.[90] Colon et al also demonstrated that fragmentation within a value chain results in firms holding less inventory, leaving them at risk of climate disaster related shortages.[91] Beyond simply impacting production, extreme weather events also disrupt food distribution and market access by slowing food shipments and damaging infrastructure, increasing the risks of damaged, spoiled or contaminated produce.[92]

15. The lack of engagement between adaptation focused stakeholders leads to decreased national capacity to leverage climate finance, resulting in a mounting gap between adaptation needs and adaptation finance. Pakistan's climate financing needs stand at approximately USD 348B, for the period 2023-2030.[93] Of this amount USD 152B is required for adaptation and resilience. However, attracting this capital will be difficult as Pakistan only received an average of USD 1.3B per year from multilateral development banks (the largest source of climate finance) between 2015-2021. This was primarily due to poor resource utilization, a policy and regulatory environment that discourages private sector investment and limited government capacity to access international finance. By enhancing coordination between key stakeholders and mainstreaming climate risk considerations into sectoral policies it is possible for

[80] ibid
[81] Jenkins et al, Soil management - drought recovery, 2020, https://www.snowymonaro.nsw.gov.au/files/assets/public/environment-and-waste/documents/prime-fact-soil-management-drought-recovery.pdf
[86] FAO, Managing Risks to Build Climate-Smart and Resilient Agrifood Value Chains: The Role of Climate Services, 2022, https://doi.org/10.4060/cb8297en
[87] ibid
[88] SDPI, Towards a climate resilient cotton value chain in Pakistan: Understanding key risks, vulnerabilities and adaptive capacity, 2018, https://sdpi.org/sdpiweb/publications/files/TowardsaclimateresilientcvcinPakUnderstandingkeyrisksvulnerabilities&adaptivecapacities.pdf
[89] Canevari-Luzardo, LM, Berkhout, F, Pelling, M. A relational view of climate adaptation in the private sector: How do value chain interactions shape business perceptions of climate risk and adaptive behaviours? *Bus Strat Env*. 2020; 29: 432– 444. https://doi.org/10.1002/bse.2375
[90] Davis, K.F., Downs, S. & Gephart, J.A. Towards food supply chain resilience to environmental shocks. Nat Food 2, 54–65 (2021). https://doi.org/10.1038/s43016-020-00196-3
[91] Colon, C., Brännström, Å., Rovenskaya, E., & Dieckmann, U. (2020). Fragmentation of production amplifies systemic risks from extreme events in supply-chain networks. Plos one, 15(12), e0244196.
[92] FAO, Managing Risks to Build Climate Smart and Resilient Agrifood Value Chains, 2022, https://www.fao.org/3/cb8297en/cb8297en.pdf
[93] Ministry of Climate Change and Economic Coordination, National Adaptation Plan, 2023, https://www.mocc.gov.pk/PublicationDetail/ZDg5NjgwMjMtMWVkZC00ODMyLTg1NzctNmQzODFhNTVmMWEz

Pakistan to strengthen its access to international climate finance and scale up green investment. ACAP will help supplement this by establishing platforms to catalyze engagement between stakeholders and publishing insights to spread awareness of climate adaptation.

ACAP Fund intends to address the following ecosystem level barrier:

- Barrier 1: Underinvested, fragmented and highly vulnerable agriculture sector
- Barrier 7: Lack of sector engagement in the financing, development, and ownership of climate resilience interventions

Business Level Barriers

16. For businesses operating in the agriculture sector, climate risks are significant and will impact the supply of raw material (causing further price volatility) and induce stresses on the local economic environment.[94] Climate hazards will likely make investment in the sector even more challenging given commercial lenders' practice of pricing 'increasing business risks' into their credit assessments. On the demand side, weather variability can have a significant impact on the purchase of agrifood products by limiting market access and impacting consumer behavior. For example, extreme weather events such as floods can damage infrastructure and block roads thereby preventing customers from reaching retailers. This can lead to significant food loss and wastage. It is critical for agribusinesses to adapt to climate change to limit future economic losses.

17. For Pakistan to address its adaptation challenges, it must build domestic solutions to climate change. To do so, it must incentivize the private sector to invest in agriculture businesses given their crucial role within the domestic supply chain (e.g., operating cold chains, providing storage/packaging, controlling lead times).[95] Further, agribusinesses that are solving the problem of climate change by developing smart-agriculture and food systems are absolutely critical for a climate vulnerable country like Pakistan. It is only by supporting the growth of companies developing scalable, local solutions that help farmers adapt to climate change while building resilient supply chains and ecosystems, that Pakistan can adapt to climate change in the long run.[96]

ACAP Fund intends to address the following business level barriers:

- Barrier 2: Lack of patient and flexible capital for agribusinesses
- Barrier 3: Agribusinesses are largely informal and lack the capacity to absorb private capital
- Barrier 6: Limited private sector capital flows due to high-risk perception by private sector investors

Farmer Level Barriers

18. Farmers suffer from a lack of information on topics ranging from seed quality to modern farm management techniques specific to location (geography, soil conditions, etc.) and crop type. This results in lower yields and higher losses at the harvest stage. For example, eight percent of the potatoes produced in Pakistan are lost at the farm level, the majority of which are lost because of mechanical damage that occurs during harvesting due to inadequate training.[97] Furthermore, limited farm-to-market connectivity and poor agriculture marketing systems necessitate farmers to sell their produce at lower prices to local agents, preventing them from accessing high value markets. There is also a limited availability of weather advisory services that enable farmers to prevent losses during the production or harvest stages due to extreme weather events. Increased weather variability arising because of climate change will negatively impact farmer yields and subsequent income, making it harder to invest in ex-ante adaptation solutions. Furthermore, global warming will change biophysical relationships which can alter the suitability of existing crop mixes. In the high latitude areas of Pakistan, a rise in temperature could enhance the growth of winter crops.[98] As climate change leads to greater weather variability it is critical for farmers to have access to information to prevent losses during the production and harvest stages.

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[94] ISF Advisors, The state of the Agri-SME sector – bridging the finance gap, 2022, https://www.casaprogramme.com/wp-content/uploads/2022/03/the-state-of-the-agri-sme-sector-bridging-the-finance-gap.pdf
[95] ibid
[96] ibid
[97] ADB, Building Horticulture Value Chains and Reducing Postharvest Losses in Pakistan, 2022, https://www.adb.org/sites/default/files/publication/843386/adb-brief-235-horticulture-value-chains-pakistan.pdf
[98] Saif Ullah, Climate Change Impact on Agriculture of Pakistan, 2017, https://juniperpublishers.com/ijesnr/pdf/IJESNR.MS.ID.555690.pdf

19. Nearly two-third of the workers in the agriculture sector are women. They are involved in farming, livestock rearing and cottage industries. Women also support male members of the household in crop production, harvesting and processing. However, much of their labor is unpaid and only 4% of women are landowners. Among the populations and communities most vulnerable to climate change women are disproportionately affected due to their limited capacity to withstand natural disasters. This is a result of limited access to capital and skills training, exclusion from decision making processes and difficulties in accessing markets. For example, when climate change impacts or natural disasters render a family's situation precarious, recovery is also more difficult for women, with agriculture and livestock rearing activities disrupted and limited livelihood opportunities compared to men, such as migration or seeking employment. ACAP will address this by providing training to portfolio companies that will enable them to grow in gender inclusive ways, tailor products towards females, and also run workshops that will train women working in the agriculture sector.

ACAP Fund intends to address the following farmer level barriers:

- Barrier 4: Limited farmer understanding of climate change impacts
- Barrier 5: Low gender equity across agriculture sector

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

Paradigm Shift

20. The ACAP Fund is based on the premise that, **IF** ACAP Fund provides climate focused agribusinesses with capital, TA, expertise, & partnerships **THEN** this will improve vulnerable farmer and agri-dependent livelihoods and resilience, catalyze private capital to scale agribusinesses, support stakeholder engagement & strengthen sector capacity to develop robust climate solutions **BECAUSE** the investments & partnerships will demonstrate a commercially viable model enhancing Pakistan's ability to contribute to national climate adaptation priorities in the vulnerable agriculture sector.

21. Through its operations and investments, the Fund will target long-term goals through six outcomes: i) improved farmer and vulnerable, agri-dependent livelihoods, ii) improved climate adaptive capacity of farmers, iii) improved farmer health and food security, iv) demonstrated commercial viability of business models attracting more private sector capital, v) scalable, replicable fund models developed for climate adaptation and vi) enabling environment for adaptation solutions supporting climate resilient agriculture. There are two expected co-benefits that will contribute towards the fund's long-term goal: i) enhanced gender inclusivity and ii) reduction in greenhouse gas emissions. However, there are a number of multifaceted assumptions that must be considered, and barriers that must be addressed, before this solution can be realized.

22. The Theory of Change (ToC) describes how ACAP aims to shift Pakistan's agricultural / agribusiness landscape towards climate resilience by illustrating causal linkages between and among the results hierarchy. This ToC forms the basis of ACAP's logical framework, which will help track performance of long-term and intermediate goals and objectives of the project, accomplished through core and supplementary indicators for adaptation result areas, enabling environment outcomes, and co-benefits prescribed in the IRMF and related guidance.

23. The results / outcomes in the ToC can broadly be defined at the following levels: farmer level, business level, and ecosystem level. The outcomes identified in ACAP's ToC schematic are grounded in evidence related to climate resilience and adaptation projects accomplished in similar country contexts.

24. Keeping the goal statement in mind and then working backwards, five pre-conditions or project outcomes necessary to bring about high-level change have been proposed. Three co-benefits have also been identified and depicted right below the outcome level.

25. Assumptions:

The following assumptions must hold true for ACAP Fund to achieve the intended results:

Assumption 1: *Private sector investors willing to participate if they can identify potentially profitable business opportunities building climate adaptation solutions* - without private sector investors willing to invest in projects deemed sufficiently profitable, ACAP will be unable to catalyze significant additional investment in target agribusinesses.

Assumption 2: *The creation and dissemination of sector insights along with stakeholder engagement will lead to increased sector capacity for developing scalable climate solutions, catalyzing sector collaboration, and increasing private capital investments into agribusinesses in Pakistan* – sufficient engagement and dissemination of relevant insights will transmit relevant knowledge of opportunities and risk to appropriate parties and can provide a basis to engage in potential investment opportunities in the agribusiness landscape.

Assumption 3: *Long term climate adaptation strategies at the national and provincial levels will continue to be implemented and provide a supportive framework for developing adaptive responses* – ACAP will be built off the supportive framework of national and provincial level strategies. These strategies and frameworks should remain in place for ACAP to be implemented effectively.

Assumption 4: *Increasing demand for local agriculture production and supply chains* – ACAP relies on market demand for agricultural products and supply chains, so they should remain stable or increase during program duration to achieve success.

Assumption 5: *Macroeconomic and political context remains sufficiently stable for agribusiness activities* – since macroeconomic conditions as well as political stability affect all agribusiness activities, including those supported by ACAP, these should remain stable enough for the program to be implemented effectively.

26. Barriers:

Barrier 1: *Underinvested, fragmented and highly vulnerable agriculture sector*

The agriculture sector in Pakistan, like many other developing countries, remains fragmented, inefficient, and highly climate vulnerable. The country's agriculture production depends largely on archaic farming methods that have left farmers increasingly vulnerable to factors outside their control (unpredictable weather patterns, disease outbreaks, pricing uncertainty, etc.). Owing to these structural inefficiencies, Pakistan's agricultural productivity is approximately 30%-50% lower than world averages for major commodities like wheat, cottonseed, rice, and maize. Agriculture productivity levels are a further 30% lower than local averages for small farmers, who account for 90% of the total farming population and about half the country's total cultivated land.[99] To compare, even though Pakistan is the 8th largest producer of wheat worldwide, its yield per hectare is 2.9 tonnes - 72% lower than the yield in Ireland, the 60th largest producer of wheat.[100] The low efficiency of the agriculture sector has hindered growth, as the sector grew at an average rate of 2.5%, between 2001 and 2021, despite Pakistan's GDP increasing by 4.1% during the same period.[101]

In addition to inefficient production, Pakistan's agriculture value chains are extremely fragmented with multiple middlemen (traders, commission agents, wholesalers, distributors, and retailers), each of which takes a sizable portion of the final retail price. To illustrate, a farmer's share of the retail price of rice in Pakistan is only 35%, whereas farmers in India and Bangladesh capture two thirds of the final price.[102][103] Value chain fragmentation has a negative impact on farmer income and productivity, increasing the cost of delivery for value added activities, raising the market price for consumers, and further contributing to food losses.[104]

Furthermore, wholesale agricultural produce markets in Pakistan are inefficient and exploitative in nature and thus hamper the production of higher value-added output. These markets dominate the agricultural landscape in the country; however, they are largely traditional with limited potential to grow on a national level. Thus, the sector has been characterized by a disparity in market power, decreased competitiveness and transparency in agriculture marketing and trade, and limited direct purchases and contracting between Agri-processors and farmers.

Barrier 2: *Lack of patient and flexible capital for agribusinesses*

Access to credit remains a significant challenge for private sector business growth in Pakistan. The banking sector, which is the largest player in the financial sector primarily lends to the government - and this factor is particularly strong during a high-rate environment where there is even greater incentive to park all liquidity in government securities in order to earn high, risk free returns. Private sector credit as a percentage of GDP is under 15% in

[99] IFAD, Pakistan Country Profile, https://www.ifad.org/en/web/operations/w/country/pakistan
[100] Food and Agriculture Organization of the United Nations, FAOSTAT Statistical Database, https://www.fao.org/faostat/en/#data/QCL
[101] World Bank, Pakistan Data, https://data.worldbank.org/country/pakistan
[102] ADB, The Transformation of Rice Value Chains in Bangladesh and India: Implications for Food Security, 2013, https://www.adb.org/sites/default/files/publication/30393/ewp-375.pdf
[103] State Bank of Pakistan, Report on Basmati Rice Value Chain in Pakistan, 2015, https://www.sbp.org.pk/publications/ChainReport/2015/Report%20on%20Basmati%20Rice%20Value%20Chain%20in%20Pakistan.pdf
[104] FAO, Managing Risks to Build Climate Smart and Resilient Agrifood Value Chains, 2022, https://www.fao.org/3/cb8297en/cb8297en.pdf

Pakistan - which is the second lowest in South Asia.[105] The situation is even worse for the agriculture sector, where there is a critical need for growth capital on flexible terms, sensitive to the cyclical nature of the sector. Credit to the agriculture sector as a percentage of total credit is only 4.94%, whereas for regional comparisons of other agrarian countries such as India and Sri Lanka, it stands at 11.92% and 7.59% respectively.[106] Furthermore, commercial lending to growth stage businesses in Pakistan has remained on average between six to nine percent of total private sector financing and is characterized by some of the highest interest rates charged in the region.[107] A key reason for this limited access to finance is that Pakistan's capital markets are extremely shallow with limited liquidity and a lack of venture capital and institutional impact investors. Illustrating this point, Pakistani startups only raised USD 355M in 2022, compared to USD 2.06B in Brazil, USD 23.5B in India and USD 530M in Egypt.[108] Businesses in Pakistan find it very difficult to access growth capital – and agribusinesses (due to the higher perceived risk associated with them) struggle the most.

Barrier 3: *Agri-SMEs are largely informal and lack the capacity to absorb private capital*

Due to the high cost of deploying credit to informally structured entities, formal banking institutions are often reluctant to provide much-needed credit from venture to growth stage businesses. This is a major factor impeding the growth of agribusinesses, as most agribusiness models operate in the informal economy, with limited incentives to formally structure, register, and maintain accurate financial records.[109] This makes it even more difficult for these entities to access formal credit. Even when agribusinesses are formally structured, the high costs of commercial credit, combined with stringent collateral requirements, dissuade businesses from accessing credit through formal financial institutions. This keeps agribusinesses small and underinvested in growth – particularly growth robust enough to navigate climate threats that are a core business risk.

Barrier 4: *Limited farmer understanding of climate change impacts*

Pakistan faces some of the highest disaster risk levels in the world, ranked 18 out of 191 countries by an index developed by INFORM, a multi-stakeholder forum for developing shared, quantitative analyses relevant to humanitarian crises and disasters. In addition, Pakistan has high exposure to flooding, including riverine, flash, and coastal, plus some exposure to tropical cyclones and their associated hazards. The country also deals with risks stemming from prolonged drought. Such extreme weather events driven by climate change will lead to severe crop damage, livestock loss, and disruption to transport infrastructure - all outcomes that negatively impact food security. Despite this dire food security situation and Pakistan's high climate vulnerability (it ranks 37 out of 191 per the World Bank), research suggests that the majority of the farmers do not consider climate change a potential threat to agriculture; therefore, they do not make any intentional efforts to change farming practices.[110] One study suggests that owing to a recent increase in extreme weather events, some Pakistani farmers have begun to perceive future climate change as a threat to agriculture, yet, they have no access to agricultural extension or farmer training that could build adaptive capacity.[111]

Barrier 5: *Low gender equity across agriculture sector*

In Pakistan, nearly 68% of the agricultural workforce is made up of female workers. However, according to the UNDP, in spite of women's contribution to the sector, women remain at a disadvantage as compared to male farmers in terms of (i) accessing agricultural extension services, (ii) lacking decision power related productive assets, (iii) achieving literacy rates (40.84% of women are literate vs. 67.15% of men) and (iv) land ownership (only 1.2% of women own land in Pakistan). Depriving women of access to resources results in lower farm productivity, as women managed farms are 20%-30% less productive than farms managed by men. It is estimated that closing this gender gap can boost agricultural production by 2.5%-4%, strengthen food security and improve the welfare of rural households.[112] Additionally, this low level of gender equity reduces the capacity of women to adapt to the impact of

[105] World Bank, Domestic credit to private sector (% of GDP) - Pakistan, Afghanistan, https://data.worldbank.org/indicator/FS.AST.PRVT.GD.ZS
[106] FAO, Credit to Agriculture, https://www.fao.org/faostat/en/#data/IC
[107] Trading Economics, Interest Rates, https://tradingeconomics.com/country-list/interest-rate
[108] KPMG, Venture Pulse Q4, 2022, https://assets.kpmg.com/content/dam/kpmg/xx/pdf/2023/01/venture-pulse-q4-2022.pdf
[109] Mekonnen, D. A., Termeer, E., Soma, K., van Berkum, S., & de Steenhuijsen Piters, B. (2022). How to engage informal midstream agribusiness in enhancing food system outcomes: what we know and what we need to know better (No. 2022-034). Wageningen Economic Research
[110] Salman et al, Farmers Adaptation to Climate Change in Pakistan: Perceptions, Options and Constraints, 2018, http://researcherslinks.com/current-issues/Farmers-Adaptation-to-Climate-Change-in-Pakistan-Perceptions-Options-and-Constraints/14/1/1833/html
[111] Ullah et al, Understanding climate change vulnerability, adaptation, and risk perceptions at household level in Khyber Pakhtunkhwa Pakistan, 2018, https://www.emerald.com/insight/content/doi/10.1108/IJCCSM-02-2017-0038/full/html#sec0012
[112] CABI, Gender and Rural Advisory Services Assessment in Pakistan, 2022, https://www.cabi.org/wp-content/uploads/Gender-and-Rural-Advisory-Services-Assessment-in-Pakistan.pdf

climate change. Evidence from across the globe demonstrates the importance of gender responsive actions for natural resource management, food security, investment in public goods, and many other activities aimed at building climate resilience.[113]

Barrier 6: *Limited private sector capital flows due to high-risk perception by private sector investors*

According to the 2012 Food and Agriculture Organization report titled,[114] "Priority areas for investment in the agricultural sector in Pakistan", commercial banks consider the disbursement of agricultural finance risky principally due to production risks stemming from a variety of interrelated factors. Outdated farming techniques and low productivity levels deter private investors looking for efficient and high-yield operations. This technological limitation restricts the scope of potential partnerships between private investors and farmers. The prices of agricultural products can be highly volatile due to the lag between the start of the food production cycle and when the product is finally delivered to the market, resulting in supply and demand mismatches. This volatility is exacerbated by extreme weather events which can significantly impact agribusiness revenues. Finally, the inconsistency in government policies and regulations adds to an unstable investment environment as changes in agricultural policies, subsidies, and tariffs can impact investors' returns and long-term plans.

Barrier 7: *Lack of sector engagement in the financing, development, and ownership of climate resilience interventions*

Pakistan's access to international climate finance is extremely limited due to a lack of coordination between stakeholders.[115] Many actors operate in silos limiting the development of climate resilient interventions. There are no multi stakeholder forums which can build relationships across value chains that improves the uptake and promotion of CSA (Climate Smart Agriculture) and RA (Resilient Agriculture) practices. For example, the adoption of globally accepted environmental and social risk management systems is very low in Pakistan.[116] Furthermore, institutional capacity at the regulatory, sectoral and corporate level is limited with weak adaptation finance and investment frameworks to attract private capital. Given Pakistan's relatively low share of climate finance, there are currently very few (if any) commercially viable examples of climate-focused vehicles deploying catalytic capital.

27. Activities:

Component 1: Investment Fund

Activity 1.1 *Establish ACAP Legal & Operational Structure*

Activity 1.1 aims to lead to the establishment of an USD 80M returnable, venture, early growth, and growth stage agriculture investment fund with a 10-year term (subject to two 1-year extensions) and a capital structure with USD 27M junior equity and USD 53M senior equity tranches. The investment approach would be to make equity, mezzanine, and debt investments in venture, early growth, and growth stage agribusinesses with a ticket size of USD 1M – 3M.

Activity 1.2 *Identify pipeline companies across prioritized themes: (i) Farmer Platforms and Financial Solutions, (ii) Smart Farming, and (iii) Post-Harvest Technologies*

Activity 1.3 *Deploy capital to investee companies*

Activities 1.2 and 1.3 will identify pipeline companies across prioritized themes for ACAP's investments in local agribusinesses which will i) develop solutions which include digital platforms and marketplaces, ii) enable smart farming techniques, and iii) provide access to modern post-harvest technologies which will ultimately benefit the farmer in terms of better access to end-user/market, resulting in better profits, improved productivity, and decreased losses at the post-harvest stage. This will improve farmer income and boost resilience to challenges related to climate change.

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[113] IUCN Pakistan, Climate Change Gender Action Plan of the Government and People of Pakistan, 2022, https://iucnhq-my.sharepoint.com/personal/saeedh_iucn_org/_layouts/15/onedrive.aspx?id=%2Fpersonal%2Fsaeedh%5Fiucn%5Forg%2FDocuments%2FccGAP%2FClimate%20Change%20Gender%20Action%20Plan%2Epdf&parent=%2Fpersonal%2Fsaeedh%5Fiucn%5Forg%2FDocuments%2FccGAP&ga=1

[114] FAO, Priority areas for investment in the agriculture sector: Pakistan, 2012, https://www.fao.org/3/i2879e/i2879e.pdf

[115] Mako et al, Recent developments in climate finance: Implications for Pakistan, 2022, https://www.theigc.org/sites/default/files/2022/09/Mako-et-al-2022-Working-paper.pdf

[116] World Bank Group, Pakistan Country Climate Development Report, 2022, http://hdl.handle.net/10986/38277

Activity 1.4 *Monitoring, reporting, and verification of portfolio company performance and impact*

ACAP's MEL team, along with an external, third-party firm, will provide regular, periodic performance monitoring of various KPIs in the log frame and verify the achievement and compliance of portfolio companies on key performance indicators. This is expected to include designing and implementing robust perception surveys, consumer voice surveys and conducting evaluations of adaptation focused KPIs. The team also intends to monitor and report on financial data, environmental and social activities, and gender activities.

Component 2: Technical Assistance Facility (TAF) to Support Capacity, Knowledge Gaps & Ecosystem Development

Activity 2.1 *Establish TA Facility*

Activity 2.1 entails establishing a technical assistance facility which will focus on improving the viability of the business models of portfolio companies while also building overall capacity and skill sets of smalllholder farmers and vulnerable agri-dependent livelihoods (through training workshops, seminars, and demonstrations) to develop resilience to climate change. The TA facility will also amplify ACAP's ecosystem level impact through developing and disseminating knowledge products/insights and convening stakeholders to create sectoral connections.

Activity 2.2 *Provide targeted TA across five components: (i) Climate Adaptation, (ii) Gender Initiatives, (iii) Business Development, (iv) ESG reporting, and (v) Impact Evaluation*

The USD 10M TA will be deployed in the above-mentioned thematic and functional areas to augment the impact of the investment fund for portfolio companies, smallholder farmers and vulnerable agri-dependent lives. Please see the table below for ACAP TA activities and their beneficiaries.

Component 3 TAF to Support Insights & Partnerships

Activity 3.1: Generating sector insights and knowledge products

Activity 3.1 aims to use TA funds to share insights and knowledge products from the first climate-focused fund in Pakistan in order to enhance the overall ecosystem level impact of ACAP Fund. As a part of this activity the Fund intends to support academic research and insights into the agriculture sector and climate resilience in Pakistan by having a dedicated insights expert to create climate focused insights reports that can be shared with sector players and other agri-dependent geographies.

Activity 3.2: Convening and engaging stakeholders and co-investors to amplify impact

Recognising the key role that the ACAP Fund seeks to play in demonstrating viable investments in climate resilient agriculture, the team aims to engage relevant stakeholders, experts and co-investors to build sector partnerships. Given the massive scale of the climate change challenge in Pakistan, ACAP aims to use its convening ability to create a platform that will enable partnerships for building effective solutions.

The table below summarises the activities and sub-activities for Components 2 and 3 and the linkages with outputs. In addition, please also find an overview of beneficiaries for each sub-activity.

Output	Activities and Sub-Activities	Sub-Activity Beneficiaries
	Component 2: TAF to Support Capacity, Knowledge Gaps & Ecosystem Development	
Output 1.3: Strengthened business performance of portfolio companies	**Activity 2.1: Establish TA Facility** • Sub-activity 2.1.1: Establishment & Management of ACAP TAF **Activity 2.2: Provide targeted TA across five components** • Sub-activity 2.2.3: Business Development • 2.2.3.1 Support companies with financial management and governance	Sub-activity 2.1.1: All TAF beneficiaries <u>2.2.3: Business Development</u> Sub-activity 2.2.3.1: Portfolio companies Sub-activity 2.2.3.2: Portfolio companies

	• 2.2.3.2 Support companies on improving investment readiness including marketing, data room management, and valuation support • 2.2.3.3 Support companies with the development of human resources processes and talent development and acquisition • 2.2.3.4 Support distressed companies with access to recovery best practices	Sub-activity 2.2.3.3: Portfolio companies Sub-activity 2.2.3.4: Portfolio companies

Output 2.1: TA deployed across strategic themes to maximize portfolio viability and fund impact	**Activity 2.1: Establish TA Facility**
	
	• Sub-activity 2.1.1: Establishment & Management of ACAP TAF
	
	Activity 2.2: Provide targeted TA across five components:
	
	• Sub-activity 2.2.1: Climate Adaptation
	  • 2.2.1.1 Support activities increasing farmer/rural communities awareness of climate resilient agriculture practices and products
	  • 2.2.1.2 Support activities increasing agribusiness awareness of climate resilient practices and risks
	  • 2.2.1.3 Execute programmes to accelerate early-stage agri businesses building climate resilience
	
	• Sub-activity 2.2.2: Gender Initiatives
	  • 2.2.2.1 Support activities to promote skills development and income enhancement of women in agriculture sector
	  • 2.2.2.2 Support initiatives to increase entrepreneurial participation of women in the agriculture sector
	  • 2.2.2.3 Support companies to set, implement, and evaluate gender action plans and provide company level gender trainings
	  • 2.2.2.4 Recruit a full-time gender expert to ensure that gender is embedded across ACAP investment processes and generate gender insights
		Sub-activity 2.1.1: All TAF beneficiaries
		
		2.2.1: Climate Adaptation
		
		Sub-activity 2.2.1.1: Farmers/rural communities
		
		Sub-activity 2.2.1.2: Portfolio companies
		
		Sub-activity 2.2.1.3: Early-stage agribusinesses building climate resilience
		
		2.2.2: Gender Initiatives
		
		Sub-activity 2.2.2.1: Rural women
		
		Sub-activity 2.2.2.2: Female agri entrepreneurs
		
		Sub-activity 2.2.2.3: Portfolio companies
		
		Sub-activity 2.2.2.4: Portfolio companies, rural women, female agri entrepreneurs, gender-focused climate adaptation stakeholders, ecosystem
		
		2.2.3: Business Development
		
		Sub-activity 2.2.3.1: Portfolio companies
		
		Sub-activity 2.2.3.2: Portfolio companies

	• Sub-activity 2.2.3: Business Development
	  ◦ 2.2.3.1 Support companies with financial management and governance
	  ◦ 2.2.3.2 Support companies on improving investment readiness including marketing, data room management, and valuation support
	  ◦ 2.2.3.3 Support companies with the development of human resources processes and talent development and acquisition
	  ◦ 2.2.3.4 Support distressed companies with access to recovery best practices
	
	• Sub-activity 2.2.4: ESG Reporting
	  ◦ 2.2.4.1 Support companies to set, implement, and monitor their climate and ESG action plans
	  ◦ 2.2.4.2 Support companies with Environmental and Social Impact Assessment when companies engage in higher risk activities
	  ◦ 2.2.4.3 Hire an ESG associate for monitoring ESG outcomes for portfolio companies
	
	• Sub-activity 2.2.5: Impact Evaluation
	  ◦ 2.2.5.1 Pre-investment impact assessment of pipeline companies
	  ◦ 2.2.5.2 Support portfolio companies with post-investment detailed customer surveys to enhance impact awareness and solutions
	  ◦ 2.2.5.3 Support broader industry with impact deep dives on selected subset of portfolio companies
	  ◦ 2.2.5.4 Support companies with development of impact evaluation processes and systems
		Sub-activity 2.2.3.3: Portfolio companies
		
		Sub-activity 2.2.3.4: Portfolio companies
		
		<u>2.2.4: ESG Reporting</u>
		
		Sub-activity 2.2.4.1: Portfolio companies
		
		Sub-activity 2.2.4.2: Portfolio companies
		
		Sub-activity 2.2.4.3: Portfolio companies
		
		<u>2.2.5: Impact Evaluation</u>
		
		Sub-activity 2.2.5.1: Portfolio companies
		
		Sub-activity 2.2.5.2: Portfolio companies
		
		Sub-activity 2.2.5.3: Portfolio companies, climate adaptation stakeholders/ecosystem
		
		Sub-activity 2.2.5.4: Portfolio companies

Output 2.2: Monitoring and evaluation of financial performance and climate impact of investments	**Activity 2.2: Provide targeted TA across five components:**
	
	• Sub-activity 2.2.3: Business Development
	  • 2.2.3.1 Support companies with financial management and governance
	  • 2.2.3.2 Support companies on improving investment readiness including marketing, data room management, and valuation support
	  • 2.2.3.3 Support companies with the development of human resources processes and talent development and acquisition
	  • 2.2.3.4 Support distressed companies with access to recovery best practices
	
	• Sub-activity 2.2.4: ESG Reporting
	  • 2.2.4.1 Support companies to set, implement, and monitor their climate and ESG action plans
	  • 2.2.4.2 Support companies with Environmental and Social Impact Assessment when companies engage in higher risk activities
	  • 2.2.4.3 Hire an ESG associate for monitoring ESG outcomes for portfolio companies
	
	• Sub-activity 2.2.5: Impact Evaluation
	  • 2.2.5.1 Pre-investment impact assessment of pipeline companies
	  • 2.2.5.2 Support portfolio companies with post-investment detailed customer surveys to enhance impact awareness and solutions
	  • 2.2.5.3 Support broader industry with impact deep dives on selected subset of portfolio companies
	  • 2.2.5.4 Support companies with development of impact evaluation processes and systems
		<u>2.2.3: Business Development</u>
		
		Sub-activity 2.2.3.1: Portfolio companies
		
		Sub-activity 2.2.3.2: Portfolio companies
		
		Sub-activity 2.2.3.3: Portfolio companies
		
		Sub-activity 2.2.3.4: Portfolio companies
		
		<u>2.2.4: ESG Reporting</u>
		
		Sub-activity 2.2.4.1: Portfolio companies
		
		Sub-activity 2.2.4.2: Portfolio companies
		
		Sub-activity 2.2.4.3: Portfolio companies
		
		<u>2.2.5: Impact Evaluation</u>
		
		Sub-activity 2.2.5.1: Portfolio companies
		
		Sub-activity 2.2.5.2: Portfolio companies
		
		Sub-activity 2.2.5.3: Portfolio companies, climate adaptation stakeholders, ecosystem
		
		Sub-activity 2.2.5.4: Portfolio companies

Output 2.3: Increase in inclusivity and climate impact of portfolio companies	**Activity 2.2: Provide targeted TA across five components:**
	
	• Sub-activity 2.2.1: Climate Adaptation
	  ◦ 2.2.1.1 Support activities increasing farmer/rural communities awareness of climate resilient agriculture practices and products
	  ◦ 2.2.1.2 Support activities increasing agribusiness awareness of climate resilient practices and risks
	  ◦ 2.2.1.3 Execute programmes to accelerate early-stage agri businesses building climate resilience
	
	• Sub-activity 2.2.2: Gender Initiatives
	  ◦ 2.2.2.1 Support activities to promote skills development and income enhancement of women in agriculture sector
	  ◦ 2.2.2.2 Support initiatives to increase entrepreneurial participation of women in the agriculture sector
	  ◦ 2.2.2.3 Support companies to set, implement, and evaluate gender action plans and provide company level gender trainings
	  ◦ 2.2.2.4 Recruit a full-time gender expert to ensure that gender is embedded across ACAP investment processes and generate gender insights
	
	• Sub-activity 2.2.4: ESG Reporting
	  ◦ 2.2.4.1 Support companies to set, implement, and monitor their climate and ESG action plans
	  ◦ 2.2.4.2 Support companies with Environmental and Social Impact Assessment when companies engage in higher risk activities
	  ◦ 2.2.4.3 Hire an ESG associate for monitoring ESG outcomes for portfolio companies
	
	• Sub-activity 2.2.5: Impact Evaluation
		<u>2.2.1: Climate Adaptation</u>
		
		Sub-activity 2.2.1.1: Farmers/rural communities
		
		Sub-activity 2.2.1.2: Portfolio companies
		
		Sub-activity 2.2.1.3: Early-stage agribusinesses
		
		<u>2.2.2: Gender Initiatives</u>
		
		Sub-activity 2.2.2.1: Rural women
		
		Sub-activity 2.2.2.2: Female agri entrepreneurs
		
		Sub-activity 2.2.2.3: Portfolio companies
		
		Sub-activity 2.2.2.4: Portfolio companies, rural women, female agri entrepreneurs, gender-focused climate adaptation stakeholders, ecosystem
		
		<u>2.2.4: ESG Reporting</u>
		
		Sub-activity 2.2.4.1: Portfolio companies
		
		Sub-activity 2.2.4.2: Portfolio companies
		
		Sub-activity 2.2.4.3: Portfolio companies
		
		<u>2.2.5: Impact Evaluation</u>
		
		Sub-activity 2.2.5.1: Portfolio companies
		
		Sub-activity 2.2.5.2: Portfolio companies
		
		Sub-activity 2.2.5.3: Portfolio companies, climate adaptation stakeholders, ecosystem
		
		Sub-activity 2.2.5.4: Portfolio companies

	• 2.2.5.1 Pre-investment impact assessment of pipeline companies • 2.2.5.2 Support portfolio companies with post-investment detailed customer surveys to enhance impact awareness and solutions • 2.2.5.3 Support broader industry with impact deep dives on selected subset of portfolio companies • 2.2.5.4 Support companies with development of impact evaluation processes and systems	
	Component 3: TAF to Support Insights & Partnerships	
Output 3: Established stakeholder platform to catalyze engagement and awareness of climate adaptation	**Activity 3.1 Generating sector insights & knowledge products** • Sub-activity 3.1.1: Support academic research and insights into climate adaptation in Pakistan • Sub-activity 3.1.2: Hire an insights expert to develop sector insights and knowledge products • Sub-activity 3.1.3: Generate comprehensive insights reports to disseminate fund level learnings from investing in climate resilience in Pakistan Sub-activity 3.1.4: Develop industry research on climate adaptation, environmental safeguarding, and specific impact areas to develop sector best practices for Pakistan	Sub-activity 3.1.1: Academic/research institutions, climate adaptation stakeholders/ecosystem Sub-activity 3.1.2: Portfolio companies, climate adaptation stakeholders/ecosystem Sub-activity 3.1.3: Climate adaptation stakeholders/ecosystem Sub-activity 3.1.4: Climate adaptation stakeholders/ecosystem
	Activity 3.2 Convening & engaging stakeholders & co-investors to amplify impact Sub-activity 3.2.1: Convene stakeholders including public and private sector leaders, agricultural leaders, academics, CSOs, and entrepreneurs to share best practices and develop knowledge networks	Sub-activity 3.2.1: Climate adaptation stakeholders/ecosystem

28. Outputs / Project Results

Output 1.1: *ACAP Fund Established*

The Acumen Climate Action Pakistan (ACAP) Fund aims to be Pakistan's first climate focused fund targeting climate adaptation in the critical agriculture sector. The Fund seeks to invest in venture, early growth, and growth stage agribusinesses focused on farmer and system level climate resilience to reconfigure the agriculture sector.

Output 1.2: *Fund capital disbursed to venture, early growth, and growth-stage agribusinesses*

Output 1.2 aims to build climate resilience at scale among farmers by investing in agribusinesses that can provide: (i) **Farmer Platforms and Financial Solutions** with the aim of removing informational asymmetries and providing access to finance / credit solutions and better end-users / markets, ii) **Smart Farming** to help improve on-farm productivity, yields, and climate resilient production practices utilizing renewable energy, precision agriculture, and modern farm management to improve on-farm net returns and reduce farmers' exposure to downside risks and crop failure and iii) **Post-Harvest Technologies** to support on-farm loss avoidance.

Output 1.3: *Strengthened business performance of portfolio companies*

ACAP Fund will be managed by an experienced team that will provide critical guidance and support to each portfolio company. This support shall enable portfolio companies to strengthen their business models and scale their operations, thereby enabling them to earn higher revenues, optimize costs and develop a more resilient supply chain. Additionally, Acumen's deep experience in impact investing, bolstered by a well defined operational strategy and a hands-on portfolio team, will enable the Fund to support investee agri-businesses in a way that will significantly strengthen their overall business performance.

Output 2.1: *TA deployed across strategic themes to maximize portfolio viability and fund impact*

ACAP's TA facility aims to accelerate impact within its portfolio by providing support throughout a target company's lifecycle. The specific support may vary depending on specific company needs (e.g., training farmers on product usage, measuring impact, access to investors etc.) Furthermore, the TA seeks to directly build the climate resilience of smallholder farmers and vulnerable agri-dependent livelihoods through training workshops, seminars, and demonstrations to develop resilience to climate change. Support will be provided across the following five themes to maximize viability of the businesses and overall fund impact:

i. Climate Adaptation
ii. Gender Initiatives
iii. Business Development
iv. ESG Reporting
v. Impact Evaluation

This TA support, focused on key areas related to climate, inclusivity, business performance, and evaluation, seeks to serve multiple, self-reinforcing ends: boosting financial success while fostering a culture of collaboration and inclusion; tackling climate challenges while proving market viability; and tracking business performance while quantifying impact.

Output 2.2: *Monitoring and evaluation of financial performance and climate impact of investments*

The TA fund aims to support agricultural companies on critical dimensions of financial performance, impact and commercial viability. This includes developing and implementing rigorous company level business performance and climate impact measurements, as well as the monitoring and evaluation routines of the project, and programs to ensure the contractual compliance of portfolio companies. The TA will also support monitoring and evaluation of the impact, effectiveness, relevance, efficiency, and sustainability of proposed interventions. Activities such as conducting field visits, interviewing senior management and developing a quarterly scorecard will help track these key performance indicators.

Output 2.3: *Increase in inclusivity and climate impact of portfolio companies*

The TAF activities will aim to support and amplify a portfolio company's impact and performance through bespoke support provided along five key areas as discussed in above. ACAP seeks to play a critical role in demonstrating successful, viable and impactful business models through improving gender representation within portfolio companies, enhancing their impact measurement capacity, enabling access to impact measurement data from consumers (through consumer voice surveys) that will allow them to capture end-user feedback and experience to improve overall adaptation impact of processes and operations, and improve financial performance through strengthening their operations. All Fund investments will be tracked and supported on inclusivity and climate impact indicators to ensure that in addition to maximizing commercial viability of investments, there is an increase in portfolio companies' performance across these critical dimensions, creating examples of best-in-class companies for the ecosystem.

Output 3: *Established stakeholder platform to catalyze engagement and awareness of climate adaptation*

Workshops, seminars, and multi-channel dissemination of insights, knowledge, and research should enhance stakeholder awareness of key developments in the sector and provide opportunities for collaboration with potential co-investors and other relevant eco-system stakeholders. Such awareness and knowledge dissemination sessions will nurture and develop networks of relevant adaptation-focused organizations and institutions, which will nurture the forming of strategic partnerships for collaboration, co-investments, and the development of climate adaptive solutions for the agriculture sector of Pakistan.

29. Outcomes / Co-benefits

Farmer Level:

Outcome 1: *Improved farmer and vulnerable agri-dependent livelihoods*

Boosting sustainable farmer and vulnerable agri-dependent income leads to improved climate resilience, decreasing vulnerability to climate shocks and uncertainties. In addition to land owners, there is a large proportion of the rural population that is landless and is therefore extremely vulnerable. Investments in innovative climate suitable technology, precision agriculture, and mechanization likely lead to more efficient use of increasingly scarce resources such as water, fertilizers, and pesticides in crop farming, reduces diseases and mortality in poultry and livestock, and increases soil health, further promoting ecosystem-level climate resilience and enhanced food security. Investments will maximize employment opportunities for the large proportion of the vulnerable agri-dependent population that doesn't own any land but is reliant on the agriculture sector for providing employment. ACAP investments will aim to generate steady employment in rural areas to improve their incomes and livelihoods.

Outcome 2: *Improved climate adaptive capacity of farmers*

ACAP will seek to enhance the ability of farmers to respond to climate challenges by improving their access to finance and markets, providing them with climate smart technologies and training them in modern farm management techniques. This will likely increase their incomes and enhance their productivity thereby reducing production and post-harvest losses due to extreme weather events. By investing in businesses providing adaptation solutions to farmers, ACAP seeks to build resilience within the agricultural sector through increasing production, decreasing waste, improving farmer ability to respond to climate shocks and streamlining value chains, thereby also strengthening overall food security in the sector.

Outcome 3: *Improved farmer health and food security*

An increase in production as a result of improved farming practices, improved access to climate resilient inputs and services, and a reduction of food waste, will generally lead to an increase in farmer incomes and increased supply of food to local markets. As an example, investment in cold chain technologies has the potential to improve food security through the reduction of fresh produce wastage resulting from spoilage and post-harvest losses.[117] Further, the employment of unemployed or under-employed individuals from vulnerable agriculture dependent populations will also have a significant impact as food insecurity is often high in such households, and any income earned from employment is likely to translate well into improved food security and health outcomes. The increase in disposable income for farmers and vulnerable households engaged with portfolio companies will likely lead to improved food security and enable access to better health services and an overall increase in wellbeing. Additionally, access to more climate resilient agriculture practices will lead to an overall increase in farmer level resilience leading to an improved ability to cope with and recover from climate shocks.

Business Level:

Outcome 4: *Demonstrated commercial viability of business models attracting more private sector capital*

The demonstration effect of the commercial viability of agribusiness models provides tangible evidence of profitability, growth potential, and competent financial stewardship. These factors reduce risk perception and increase investor confidence, making venture, early-growth, and growth-stage agribusiness more attractive to private capital. The success of ACAP's portfolio companies will demonstrate the market viability of innovative agribusiness models in

[117] FAO, Improving Food Security and Income Generation of Smallholder Farmers in Food Deficit Areas, 2021, https://www.fao.org/3/cb6487en/cb6487en.pdf

Pakistan, boost investor confidence, and lay the groundwork for further venture, early growth, and growth stage agribusinesses to obtain funding, scale up operations, and expand growth projections. Furthermore, commercially viable agribusinesses that align with sustainable development goals, such as promoting climate resilience and adaptation, food security, and environmental sustainability, will benefit from funding from impact investors who will be more likely to deploy capital in the sector due to demonstrated success and business model viability of ACAP's portfolio companies.[118]

Ecosystem Level:

Outcome 5: *Scalable, replicable fund models developed for climate adaptation*

ACAP Fund, as the first fund of its kind in the region, has been designed as a model with high potential for replicability within Pakistan, and in other climate vulnerable frontier markets.

ACAP's investments aim to help develop models for climate adaptation in the agriculture sector which can serve as a blueprint for similar ventures in other agrarian, frontier markets. At the fund level, the success of ACAP's investment model will provide a clear example for other climate adaptation focused investment funds and draw capital towards supporting solutions at scale for the critical issue of climate adaptation. This replication potential can attract investors looking for opportunities within Pakistan and in other agri dependent, climate vulnerable markets. ACAP's success in an underinvested sector will derisk the sector and encourage potential investors to develop similar funding vehicles within or ourside the region. The Fund seeks to continue sharing insights and learnings widely in order to amplify replication potential.

This outcome will be measured at the enabling environment level.

Outcome 6: *Enabling environment for adaptation solutions supporting climate resilient agriculture*

The establishment of a stakeholder platform will likely provide an opportunity for key actors to learn, collaborate, discuss, plan, and ultimately implement key climate change adaptation interventions. Disseminating key findings, building industry consensus, and promoting engagement and involvement from key stakeholders may result in discrete and quantifiable achievements in the sector. For instance, a series of knowledge sharing related activities may be undertaken by ACAP to disseminate evidence-based insights for potential stakeholders and various ecosystem partners which should positively impact mobilization / availability of climate adaptation learnings for local agribusinesses. ACAP Fund aims to help to create a platform for climate-resilient development pathways consistent with relevant national climate change adaptation strategies and plans. As another example, a forum held for portfolio companies and impact investors can enable co-investment and follow-on capital from a values-aligned group of investors, further strengthening portfolio companies' financial health, developing the robustness of the impact investing ecosystem, and boosting the visibility and maturity of the agribusiness industry.

This outcome will be measured at the enabling environment level.

Co-benefit 1: *Enhanced gender inclusivity*

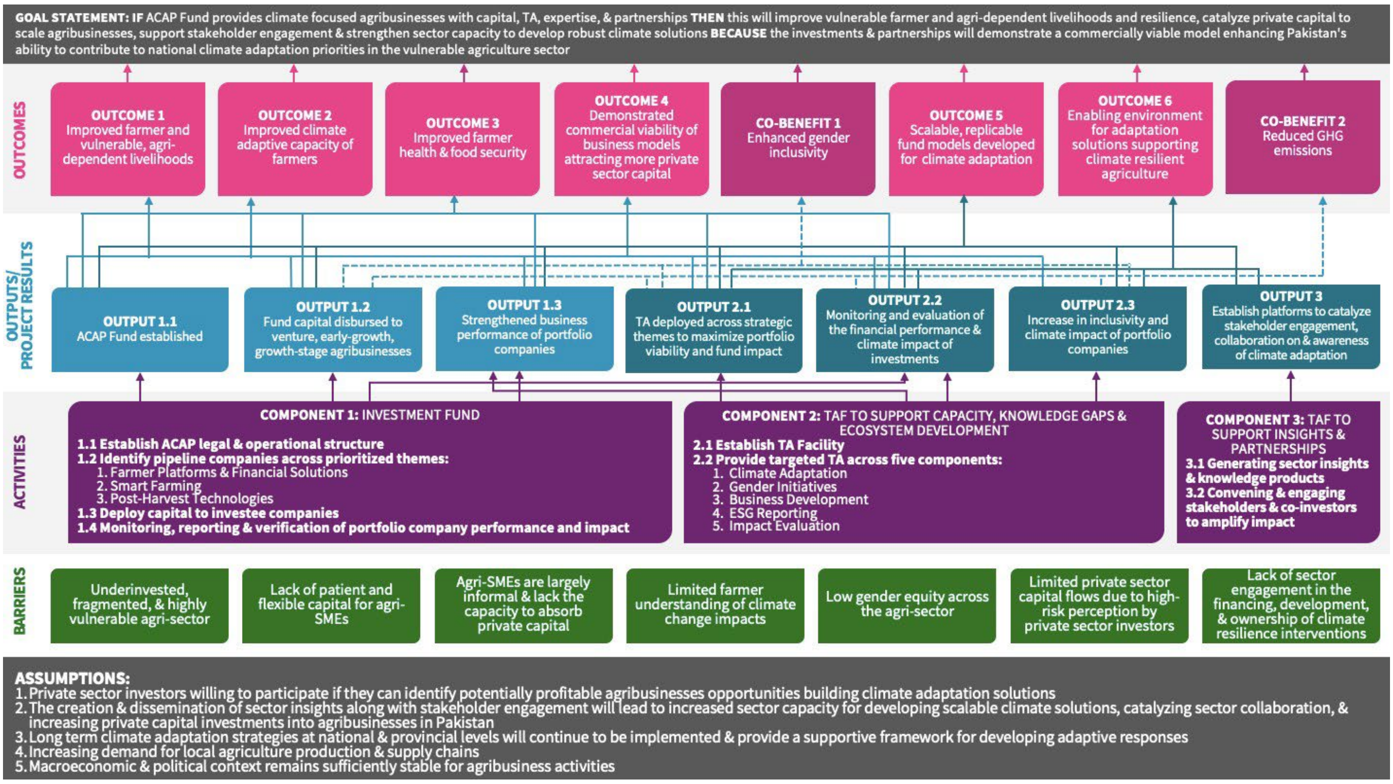
In view of the guidance provided in the IRMF, ACAP also aims to capture co-benefits wherever possible. Co-benefit 1 highlights the gender inclusiveness outcome from ACAP's support of portfolio companies as well as enhancing inclusivity in the ecosystem. With a triple bottom-line approach to impact, ACAP commits to measuring each portfolio company's top-line growth alongside good corporate governance practices, a gender action plan, and climate adaptation measures. ACAP intends to also provide trainings and grants to rural women focusing on income enhancement, support female run agribusinesses and engage a full time gender lead to disseminate gender focused insights and knowledge products.

Given the high percentage of women engaged in the agriculture sector, it is critical to improve inclusivity and gender focused activities in the sector. This is a key paradigm shift priority and ecosystem level impact for the Fund. ACAP will be using the 2X Challenge criteria to ensure that the Fund makes increasingly gender smart, inclusive investments, while helping move portfolio companies and the sector overall towards gender equity. The gains from gender smart investments made through ACAP can be captured through robust gender-disaggregated indicators in the logical framework.

Co-benefit 2: *Reduced GHG emissions*

[118] FAO, Understanding poverty and food insecurity at the household level, 2022, https://www.fao.org/3/cc2993en/cc2993en.pdf

The impact of post-harvest technologies provided by agribusinesses to the farmer can extend far beyond the impact at the farm-level and assist in reducing emissions from food systems due to food loss and food waste.[119] Thus, post-harvest technologies can increase climate resilience and reduce emissions footprint of the agricultural sector and the surrounding ecosystem.



Causal Linkage Pathways		
Outputs	**Outcomes**	**Co-benefits**
1.1	1, 2, 3, 4, 5	
1.2	1, 2, 3, 4, 5	1, 2
1.3	1, 2, 3, 4, 5	
2.1	1, 2, 3, 4, 5, 6	1, 2
2.2	1, 2, 3, 4, 5, 6	2
2.3	1, 2, 4, 5, 6	1, 2
3	5, 6	

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

[119] According to The African Postharvest Losses Information System (APHLIS), food loss accounts for 4.4 gigatonnes of greenhouse gas emissions each year (through on-farm emissions, and the energy used to produce, transport, and store food that is wasted).

Fill in the GCF results area table below to map each project/programme outcome identified in section B.2(a) to the contributing GCF results area(s) by referring to the description of eight results areas provided in the guidance note.

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	**MRA 2** Low-emission transport	**MRA 3** Building, citiies, industries, appliances	**MRA 4** Forestry and land use	**ARA 1** Most vulnerable people and communities	**ARA 2** Health, well-being, food and water security	**ARA 3** Infrastructure and built environment	**ARA 4** Ecosystems and ecosystem services
Outcome 1	☐	☐	☐	☐	☒	☒	☐	☐
Outcome 2	☐	☐	☐	☐	☒	☒	☐	☐
Outcome 3	☐	☐	☐	☐	☒	☒	☐	☐
Outcome 4	☐	☐	☐	☐	☒	☐	☐	☐
Outcome 5	☐	☐	☐	☐	☒	☐	☐	☐
Outcome 6	☐	☐	☐	☐	☒	☒	☐	☐

If any co-benefits have been identified in section B.2(a), fill in the Co-benefit table below to map each co-benefit to the corresponding category as defined in the FP guidance note.

Co-benefit number	Co-benefit					
	Environmental	**Social**	**Economic**	**Gender**	**Adaptation**	**Mitigation**
Co-benefit 1	☐	☐	☐	☒	☐	☐
Co-benefit 2	☐	☐	☐	☐	☐	☒

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

Define the project/programme. Describe the proposed set of components, outputs and activities that will be undertaken by the project/programme to attain the intended outcomes elaborated in Section B.2 (a). Please also elaborate how the project/programme activities will address the barriers described in Section B.2.

This section of the funding proposal should be consistent with sections C.2 (financing by component), E.5 (project/programme results level) and E.6 (project/programme activities and deliverables).

Referring to the feasibility study, describe why this set of interventions was selected instead of alternative solutions and how the project/programme can help unlock the needed support in a sustainable manner. Also identify trade-offs of the selected interventions, if applicable.

For Enhanced Direct Access (EDA) proposals and projects/programmes with financial intermediation (loans or on-granting), describe the selection criteria of the sub-project and types.

COMPONENT 1: INVESTMENT FUND

Output 1.1: ACAP Fund Established

Activity 1.1: Establish ACAP legal and operational structure.

ACAP Overview and Investment Strategy

30. Acumen intends to set up a USD 90M impact investment programme that will support 15-17 companies providing climate adaptation solutions aligned with Pakistan's adaptation objectives. This programme includes a USD 80M returnable, venture, early growth, and growth stage agriculture investment fund (with USD 25M requested from GCF) and a USD 10M Technical Assistance Facility (TAF) (with USD 3M requested from GCF) in grant funding. We believe that an investment vehicle like ACAP is uniquely positioned to start providing critical capital to high growth, catalytic agriculture sector companies that are building a more climate resilient ecosystem, increasing food security, building the rural economy, and benefiting the most vulnerable.

31. ACAP aims to be Pakistan's first climate focused investment fund, supporting venture, early growth, and growth stage agribusinesses. The Fund will make venture, early growth, and growth stage investments across the value chain in key priority sectors that have demonstrable impact on farmer and ecosystem level climate resilience (discussed below in Activity 2). Our blended capital fund structure with a singular sector focus implies that ACAP should not only incentivize the growth of strong agriculture sector companies, but also – by de-risking the market – catalyze further private sector investment in these areas. We expect that our initial investment capital of USD 80M will directly enable many multiples of this amount to be raised and invested in the agriculture sector. Such a catalytic impact would feed not only Pakistan's growing population, but also contribute to the provision of high quality, affordable food exports for the rest of the world.

32. ACAP's investment strategy intends to build the future of climate resilient agriculture in Pakistan by simultaneously focusing on a sustainable agricultural ecosystem and the increased climate resilience of vulnerable farmers. Given the magnitude of the problems being faced by the country's agriculture sector, Acumen realizes that it is critical to target its investments strategically. Not only does ACAP need to focus on the highest impact at a systemic and a farmer level, but it will also need to ensure that all investments are well positioned for growth, returns, and sustainability.

33. The ACAP Fund team has deep market knowledge and has spent the past few years building a robust pipeline of high impact, investable agri-businesses in Pakistan that are building critical solutions across the value chain. Given the team's knowledge of both the sector and the market, we believe that targeted investments will ensure maximum catalytic potential of the investment. In addition, we expect future ACAP investments to create sustainable change for millions of Pakistan's vulnerable farmers while, on an ecosystem level, leading to more sustainable investments by demonstrating the attractiveness and high impact of this sector to other investors. While these components are connected in many ways, we have broken down the Fund investment themes to differentiate direct value chain interventions from improvements in the ecosystem around it, allowing for a consideration of the pipeline and its associated impact in a more focused way. ACAP's compelling fund concept has already won the Luxembourg International Climate Finance Accelerator (ICFA) 2021, where it was selected as the Cohort Ambassador and presented at COP26, to investors and climate finance experts[120].

34. The Fund's investment and impact strategy has been formulated with climate adaptation of vulnerable farmers at the core and is guided by Acumen's +10 years of investment experience in the agriculture sector. Acumen's agriculture investments have already impacted 41M lives and strengthened various aspects of the agriculture value chains across eleven countries. These high-impact businesses are enabling critical interventions that include transparent supply chains, adoption of solar irrigation, index-based insurance, and solar warehouses. As a result, farmers have witnessed an increase in their incomes, productivity, and climate resilience. In the process of managing these investments, we have learned a great deal about how to simultaneously build for scale, impact, and sustainability. For example, given the many problems smallholder farmers face at every stage of production, Acumen knows that it is imperative to address multiple needs concurrently, instead of investing in piecemeal solutions. Furthermore, Acumen has learned that investing in bundled solutions is more financially viable than providing single input solutions. Through this investment lens, ACAP will prioritize businesses that provide bundled solutions to smallholder farmers over standalone, single-input business models.

Fund Structure

35. ACAP seeks to support innovative and scalable agribusinesses headquartered and operating in Pakistan, enhancing climate resilience throughout the agriculture value chain. In some cases, these investments may be made through international holding companies of Pakistani start-ups. All investments will be selected with the goal of

[120] ICFA, Ambassador of the 2021 Cohort, 2021, https://www.icfa.lu/cohort/acumen/

investing in best-in-class start-ups that have the potential to be regional success stories while building the climate resilience of smallholder farmers and a more climate resilient agriculture sector. As the first of its kind fund in Pakistan, ACAP aims to provide a blueprint for incentivising private investment in a critical sector for other climate vulnerable agrarian economies.

Activity 1.2: Identify pipeline companies across prioritized themes:

37. ACAP Fund intends to invest across three major themes – **Farmer Platforms and Financial Solutions; Smart Farming;** and **Post-Harvest Technologies.** These themes directly address some of the critical problems/barriers faced by the climate vulnerable agriculture sector that have been identified in Section B1 (as shown in the theory of change). There is also strong evidence that investing across these themes will have a significant impact on farmer level and ecosystem level climate adaptation as detailed below. The ACAP team has already compiled a list of over 200 pipeline agriculture companies in various stages of scaling, fundraising, and deploying climate action solutions within these themes. The ACAP Fund plans to invest in 15-17 venture, early growth, and growth stage agribusinesses, across the three investment verticals. These companies span across the value chain and have informed much of our investment thesis through extensive on-ground sector diligence, some examples which are shared below.

i. **Farmer Platforms and Financial Solutions**: Investments in business models that directly focus on farmers by providing access to finance/credit solutions, high quality inputs, and connectivity with stakeholders across the value chain as well as investments in scalable digital platforms that connect value chain components to increase the climate resilience and efficiency of the entire value chain. Investment archetypes include:

a. *High-yielding inputs*: Agricultural practices, technologies, and resources that are designed to increase crop or livestock production and improve the overall efficiency of farming operations. Enhancing farmer access to inputs that improve agricultural yields to improve farmer incomes, reduce food insecurity and poverty, and can help mitigate the impact of unpredictable factors, such as adverse weather conditions, pests, and diseases, by providing greater resilience and reducing production risks. Using technology to connect farmers with stakeholders across the value chain for greater farmer access to input- and end-markets, helping to build-up the ecosystem surrounding farmers.

b. *Information access and extension services:* Platforms that connect stakeholders across the value chain and ecosystem, helping to correct information asymmetries, create marketplaces, improve access to inputs, connect with retailers/final consumers, etc. These include farmer focused platforms that enhance farmer access to weather and market information, crop and soil advisory to improve farmer resilience.

c. *Access to financial services:* Access to farmer- and rural-friendly credit solutions is a massive problem faced by farmers when trying to scale. ACAP will invest in improved access to formal insurance and credit/financing options to improve farmer livelihoods and scaling capabilities, making them more climate resilient. Given our learnings in this space, ACAP will invest in credit access when it is bundled with efficient inputs (and extension services) to support the adoption of climate resilient practices and reduce financial intermediation risks.

Pipeline Examples: Platforms connecting retailers to end users, farmers to processors, and retailers to wholesale markets for inputs and outputs; farm to fork traceability in the meat industry. Digital farm input marketplace; spatial tagging of livestock to enable insurance; and access to credit; "Walmart" like models for bundled inputs.

Impact: Farmer focused investments directly impact the climate resilience of farmers by increasing their on-farm productivity (increasing yields and access to high value markets) and related incomes. Access to insurance will further reduce the volatility of their incomes. According to MicroSave Consulting, less than 22% of smallholder farmers in Asia have insurance protection for their crops.[121] A lack of insurance remains a significant hurdle in the effort to improve agricultural techniques and investment at the smallholder level. As One Acre Fund points out, insurance provides "a critical component in increasing farmers' confidence to invest in new, climate-smart agricultural approaches and crop diversification" both of which are critical to building smallholder climate resilience.[122] Thus, insurance instruments can help smallholders more quickly

[121] MicroSave Consulting, Enhancing resilience of smallholder farmers against climate change - can agricultural insurance make a difference?, 2023, https://www.microsave.net/2023/01/27/enhancing-resilience-of-smallholder-farmers-against-climate-change-can-parametric-agricultural-insurance-make-a-difference/

[122] One Acre Fund, The Case for Farming Insurance to Help Smallholders Build Climate Resilience, 2022, https://oneacrefund.org/articles/case-farming-insurance-help-smallholders-build-climate-resilience

recover from climate change related losses, while also increasing the appetite for risk (i.e., adoption of improved techniques, and investments), delivering value during years without a climate disaster. By specifically focusing on smallholder farmers and identifying investments that can increase their resilience to climate change-related risks, ACAP aims to help farmers raise productivity, reduce costs, realize better prices for their produce, and reduce post-harvest losses.

38. As discussed in farmer level barriers in Section B1, there is a large base of research that has shown how access to agriculture credit and insurance plays a pivotal role in uplifting the sector, building climate resilience and ensuring the economic development of an economy.[123] [124] The bundling of crop insurance with credit and input supplies has also been shown to provide a win-win for farmers, credit providers, and insurers.[125] Financial institutions are more willing to lend to small farmers when their loans are protected by crop insurance. Further, according to a research report published by the World Bank Group, extending agriculture credit to farmers is also critical for the widespread adoption of climate-smart agriculture activities. In Pakistan, agriculture credit is only available to less than 30% of farmers, almost all of whom are large farmers. Access to finance remains a critical issue that will require investment to increase the adaptive capacity of the sector.[126]

39. Given the highly fragmented and traditional nature of value chains in Pakistan, investments that improve linkages between value chain components should increase the overall efficiency of the sector and ensure that a higher percentage of farming revenues remains with the farmers, increasing their incomes and climate resilience. Digital solutions maximize market potential through five critical vectors of disruption: unlocking excess capacity of physical assets; creating liquid, transparent marketplaces; radical re-pricing of credit and risk; improving operational efficiency; and digitally integrating the value chain. [127] In a 2021 report analyzing the impact of digital agriculture solutions on climate resilience, the Consultative Group for International Agriculture Research (CGIAR) claims that in addition to bypassing intermediaries, digital marketplaces provide farmers with better access to information and the ability to obtain fairer prices. In doing so, CGIAR claims that digital markets “can help to reduce price market distortions caused by traditional intermediaries, thereby increasing the resilience of vulnerable farmers.”[128]

40. Commenting on Mercy Corps’ ongoing research on the impact of digital marketplaces in Africa, the Consulting Group to Assist the Poor (CGAP) points out that by further connecting farmers to buyers, smallholder incomes increased by 50% or more.[129] At scale, digital marketplace platforms can improve the efficiency and capacity of a country’s domestic market, reducing transaction costs related to the aggregation, transport, and payment of goods. Such an impact can improve country-wide resilience to climate related disasters. As Berquist and McIntosh (2021) explain in their paper analyzing the impact of digital marketplaces in Uganda, a platform can “increase the probability of trade, the number of traders, and the volume of crops traded between participating sub-counties.”[130]

41. While developed countries have led the innovation and adoption of digital agriculture, the potential impact in developing countries (such as the Middle East and North Africa (MENA) region) is massive.[131] Available evidence shows that improvements in primary production, supply chain and logistics performance, and optimized use of scarce natural resources (notably agricultural water) could be critical, if digital technologies can be implemented as envisioned.[132] Platforms that have leveraged information technology to create business models (as they aim at business opportunities resulting from market inefficiencies), often show how conventional companies have been

[123] Ghafoor, U.; Muhammad, S.; Chaudhary, K. Constraints in availability of inputs and information to citrus (kinnow) growers of tehsil Toba Tek Singh. Pak. J. Agric. Sci. 2008, 45, 520–522
[124] Usman, M.; Ashraf, I.; Chaudhary, K.M.; Talib, U. Factors impeding citrus supply chain in Central Punjab, Pakistan. Int. J. Agric. Ext. 2018, 6, 1–5
[125] Tsolakis, N.K.; Keramydas, C.A.; Toka, A.K.; Aidonis, D.A.; Iakovou, E.T. Agrifood supply chain management: A comprehensive hierarchical decision-making framework and a critical taxonomy. Biosyst. Eng. 2014, 120, 47–64.
[126] CIAT; World Bank. 2017. Climate-Smart Agriculture in Pakistan. CSA Country Profiles for Asia Series. International Center for Tropical Agriculture (CIAT); The World Bank. Washington, D.C. 28 p
[127] Brody, P. and Pureswaran, V. (2015), "The next digital gold rush: how the internet of things will create liquid, transparent markets", Strategy & Leadership, Vol. 43 No. 1, pp. 36-41. https://doi.org/10.1108/SL-11-2014-0094
[128] CGIAR Research Program on Climate Change, Agriculture and Food Security (CFAS), “Digital agriculture to enable adaptation: a supplement to the UNFCCC NAP Technical Guidelines,” Working Paper No. 372, Stephenson, Jim, et al, 2021.
[129] Shrader et al, CGAP, Super Platforms: Connecting Farmers to Markets in Africa, 2018, https://www.cgap.org/blog/super-platforms-connecting-farmers-to-markets-in-africa
[130] “Search Cost, Intermediation, and Trade: Experimental Evidence from Ugandan Agricultural Markets,” Berquist, Lauren Falcao and Craig McIntosh, July 29, 2021.
[131] Bahn, R. A., Yehya, A. A. K., & Zurayk, R. (2021). Digitalization for sustainable agri-food systems: potential, status, and risks for the MENA region. Sustainability, 13(6), 3223.
[132] ibid.

unable to access various drivers for scale. These platforms impact supply chain actors and consumers as well as economic development.[133] Digitizing markets is therefore a crucial lever for building climate resilience in agriculture value chains.

ii. **Smart Farming:** Investments in companies working on improving on-farm productivity, yields, and climate resilient production practices. Some of these solutions may utilize renewable energy to power the solutions (i.e., solar water pumps). These include**:**

a. *Small farm focused mechanization:* Hardware equipment that improves processes, production, and distribution through utilization of a variety of tools; machinery for the development of agricultural land, (i.e., tilling, planting, harvesting and primary processing). ACAP will be targeting models that prioritize small farmers

b. *Farm Optimisation:* On-farm technologies that enable better monitoring and management of crops and livestock to further improve yields

c. *Alternatives to traditional farming:* Alternatives to traditional farming methods that can optimize land usage in areas where soil degradation, salinity and climate change (among other factors) have caused land to become less arable and lose productivity. Additionally, it can provide farmers with an alternative, climate resilient income stream.This includes aquaculture, agroforestry and tissue culture.

d. *Regenerative agriculture:* Promoting practices like minimal tillage, cover cropping, crop rotation, and the use of organic matter to build soil fertility in order to sequester carbon, improve water retention, and increase long-term sustainability.

Pipeline Examples: Micro-propagation of high-yielding value chains (bananas, other high-value fruit and vegetable varieties) through tissue culture technology; locally manufactured, affordable farm mechanization equipment for small farms; high efficiency irrigation systems.

Impact: As farmers face increasing climate volatility, shifting to more smart agriculture practices becomes imperative. Traditional climate-smart agriculture strategies include changing input mix, changing cropping calendar, diversifying seed variety, and soil and water conservation measures. However, technology driven ‘smart agriculture’ can equally impact the sustainability of smallholder production. Farm-level data from various agroecological zones in Pakistan shows that the adoption of different smart agriculture practices is influenced by average rainfall, previous experience of climate-related shocks, and access to climate change information.[134] The findings further reveal that adoption of CSA practices positively and significantly improves farm net returns and reduces farmers’ exposure to downside risks and crop failure.[135] Results also show that unless smart agriculture interventions are specific, tailored to agroecological conditions and climate impacts, on farm net returns don’t significantly improve. Given that Pakistan has 17 unique agroecological zones, investments will be stringently screened on their climate resilience potential to maximize impact. Additionally, with low penetration of formal financial solutions in the agriculture sector, scaling up the adoption of smart agriculture practices can also serve as an ex-ante measure against crop failures (a kind of crop insurance), particularly in areas where formal insurance institutions are not effective or non-existent. Implementing smart farming practices could therefore contribute to improving rural farm household welfare.[136]

42. Given climate change impacts yields and productivity every year, smart farming technology tailored to smaller farmers and specific climatic challenges in impacted agroecological zones is key in aiding farmers in this transition. Climate impacts have caused increasing weather unpredictability and precision agriculture interventions are becoming more essential as they are intrinsically linked with predictive data analytics. While IoT and smart sensor technologies provide a wealth of relevant real-time data, data analytics can assist farmers in making sense of the data being collected, making critical predictions (i.e., crop harvesting time, disease and pest threats, yield volume, etc.). Frequently, smallholder farmers find weather data and better predictive models easier to access given that a large proportion of rural farmers currently use weather reports to improve the management and predictability of farming.[137] Climate services platforms can help farmers make decisions vis a vis crop selection, harvest timings,

[133] Schroder, A., Prockl, G., & Constantiou, I. (2021). How Digital Platforms with a Social Purpose Trigger Change towards Sustainable Supply Chains.
[134] Shahzad, Muhammad Faisal, Awudu Abdulai, and Gazali Issahaku. 2021. "Adaptation Implications of Climate-Smart Agriculture in Rural Pakistan" Sustainability 13, no. 21: 11702. https://doi.org/10.3390/su132111702
[135] Shahzad, Muhammad Faisal, Awudu Abdulai, and Gazali Issahaku. 2021. "Adaptation Implications of Climate-Smart Agriculture in Rural Pakistan" Sustainability 13, no. 21: 11702. https://doi.org/10.3390/su132111702
[136] ibid.
[137] Mohd Javaid, Abid Haleem, Ravi Pratap Singh, Rajiv Suman, Enhancing smart farming through the applications of Agriculture 4.0 technologies, International Journal of Intelligent Networks, Volume 3, 2022, Pages 150-164, ISSN 2666-6030, https://doi.org/10.1016/j.ijin.2022.09.004

investing in power storage infrastructure, etc.[138] Farmers with greater access to extension services and agricultural information tend to demonstrate higher efficiency in production and therefore profit.[139] For example, farmers in Ghana were able to increase their yam yields by 17% when given access to weather information.[140]

43. According to the 2022-2023 Pakistan Economic Survey, the country's agricultural sector contributes to 22.9% of GDP, while using as much as 90% of the country's water supply.[141] The report further claims that roughly 50% of the water used for agricultural purposes is wasted due to mismanagement and outdated techniques (i.e., flood irrigation). Overuse of such a precious resource not only negatively impacts yields, but can also waterlog soils, degrading their productivity. Investments in climate suitable technology, precision agriculture, and mechanization will lead to efficient use of increasingly scarce resources such as water, fertilizers, and pesticides in crop farming, reduce diseases and mortality in poultry and livestock, and increase soil health, leading to increased ecosystem-level climate resilience and enhanced food security. In fact, the Syngenta Foundation postulates that through smart-farming techniques (including minimum- or no-till technology and water use efficiency), smallholder farmers can simultaneously increase soil health (increasing yields and subsequent incomes) while also increasing soil's carbon dioxide sequestration (given soils ability to hold twice as much carbon as is present in the atmosphere).[142]

44. One of the main criticisms towards smart farming is the high cost and complexity of the technologies involved. Critics claim that to utilize smart farming techniques, farmers need to invest in equipment, software, and training (which may not be affordable or accessible for many smallholder farmers). Critics further contend that poor connectivity in rural areas could impact the accessibility of smart farming technologies for geographically remote farmers. For those able to utilize the burgeoning technology, a lack of standardization among different devices and platforms can create compatibility and data quality issues. While some smart farming technologies would be expensive, more crucial still is an overall shift required in farmer behavior. Farmer behavioral change (in terms of adopting new technology) requires an initial demonstration component to showcase the efficacy of new technology/methods to encourage farmers to switch.[143] As such, ACAP aims to strategically focus on smart farming models that are accessible to small and progressive farmers and intend to screen against models that cater exclusively to large farmers and will remain mindful of cost concerns for small farmers. As an example, the ACAP team will focus on mechanization specifically designed for small farm sizes (which is less expensive than standard size equipment), and hydroponic equipment for the same. The team seeks to opportunistically invest in smart farming technologies that have the potential to increase system-level resilience as well as the demonstration effect. It is important to note that smart farming technology models will shift employment patterns in rural areas. While some of these models may impact the need for human labor, research posits it is not enough to cause mass unemployment in the sector, especially in developing countries where factors such as education levels, lack of enabling policy and infrastructure, access to finance and poor connectivity can limit adoption at scale.[144] Lastly, the positive impact of smart farming technology on yields and productivity outweighs any potential negative impacts on employment.

iii. **Post-Harvest Technologies:** These critical investments intend to focus on innovative midstream technologies to reduce food loss and wastage, while enhancing food quality and safety, traceability, storage, logistics, food processing, and distribution. Some of these solutions may use renewable energy such as solar drying units. This includes:

a. *End to end value chains:* Models which enable smallholder farmer access to high-value markets (domestic and export) while upskilling rural communities and providing improved employment opportunities

b. *Farm gate storage and logistics:* Cold storage and warehousing receipts models that allow farmers to cut out middlemen, reduce food wastage, and increase incomes while also improving the transportation and connectivity between farmers and end markets.

[138] FAO, Managing Risks to Build Climate Smart and Resilient Agrifood Value Chains, 2022, https://www.fao.org/3/cb8297en/cb8297en.pdf
[139] Mubarik Ali, Profit Efficiency Among Basmati Rice Producers in Pakistan Punjab, 1989, https://www.researchgate.net/publication/237930142_Profit_Efficiency_Among_Basmati_Rice_Producers_in_Pakistan_Punjab
[140] Anuga et al, Adoption of climate-smart weather practices among smallholder food crop farmers in the Techiman municipal, 2016, https://www.researchgate.net/publication/311201979_Adoption_of_climate-smart_weather_practices_among_smallholder_food_crop_farmers_in_the_Techiman_municipal_Implication_for_crop_yield
[141] Finance Division, Government of Pakistan, Pakistan Economic Survey, 2022-2023, https://www.finance.gov.pk/survey/chapters_23/02_Agriculture.pdf
[142] Syngenta Foundation, Moving smallholder farmers higher up the climate change agenda, 2022, https://www.syngentafoundation.org/blog/policy/moving-smallholder-farmers-higher-climate-change-agenda
[143] OECD, Farmer Behaviour, Agricultural Management and Climate Change, 2012, https://www.oecd.org/greengrowth/sustainable-agriculture/farmerbehaviouragriculturalmanagementandclimatechange.htm
[144] Choi, Sung-Wook, and Yong Jae Shin. 2023. "Role of Smart Farm as a Tool for Sustainable Economic Growth of Korean Agriculture: Using Input–Output Analysis" Sustainability 15, no. 4: 3450. https://doi.org/10.3390/su15043450

c. *Post-harvest value addition:* Reducing food wastage and improving farmer incomes via innovative processing and value addition methods which aid in export readiness and alignment with quality standards.

Pipeline Examples: Solar drying units to preserve quality and reduce food waste; Controlled Atmosphere (CA) storage, cold storage; high value food processing units, efficient supply chain logistics, agriculture value chains in export focused sectors, such as end-to-end fisheries and aquaculture.
Impact: Post harvest losses can occur at every point along the value chain, whether during storage at the farm-level or at the market; during processing (whether threshing or drying); or during transport. Such losses not only decrease the income available to the producer, but negatively impact the availability of produce, increasing a country's vulnerability to a climate change related natural disaster. A 2017 study on the impact of solar milk refrigeration solutions in Kenya details the benefits of a project piloting off-grid solar milk chillers, meant to reduce the daily losses associated with evening milk (which previously either spoiled, was forcibly consumed, or sold at a loss). The study found that the solar-milk chillers, capable of storing 40-liters of evening milk, "allowed farmers to sell five to 40 extra liters per day, resulting in additional income of USD 60 to USD 500."[145]

45. Post-harvest technologies are critical to overcome significant food losses during various stages including harvesting, post-harvest handling, storing, packaging and transportation, and increase the shelf life of the products. In their research study, Vigneault et al., (2009) have shown adequate temperature control systems and air circulation systems are the most important means to ensure quality preservation of perishables.[146] Research finds that a cold supply chain provides ideal conditions to perishable goods from the point of source to the point of consumption (via thermal and refrigerated packaging methods which protect the quality and increase the shelf-life of fruits and vegetables).[147] [148] While the absence of refrigerated logistics impacts the longevity of fresh produce, so too does the lack of basic storage facilities throughout smallholder production. Controlled atmosphere storages could help alleviate the significant food losses.[149] [150] According to FAO, an estimated 25% of the world's food crops are affected by mycotoxins (post-harvest mold, common under poor storage conditions) that result in billions of dollars in losses.[151] According to James & Zikankuba (2017), use of post-harvest technologies such as application of ethylene and 1-MCP to enhance and delay ripening, respectively can significantly reduce post-harvest losses throughout the value chain in fruits and vegetables.[152]

46. Efficient processing of the agriculture produce is another important component of the post-harvest technologies. Spielman et al, 2016 have shown that the limited output growth in the agriculture sector in Pakistan has been driven by increased use of factor inputs, rather than technical change, a more efficient product mix, or upgrade processing of crops.[153] Additionally, post-harvest food losses in Pakistan also occur because of inefficient logistics and transportation. Various research highlights how logistics and supply chain management play a crucial role in the food industry.[154] [155]

47. The African Postharvest Losses Information System (APHLIS) suggests that without intervention, post-harvest losses will only continue due to climate change and the subsequent irregularities within weather

[145] Robert Foster, et al., "Direct drive photovoltaic milk chilling experience in Kenya," IEEE Photovoltaic Specialists Conference 44, June 2017.
[146] Vigneault, C., Thompson, J., Wu, S., Hui, K.C., & LeBlanc, D.I. (2009). Transportation of fresh horticultural produce. Postharvest technologies for horticultural crops, 2, 1-24
[147] Negi, S., & Anand, N. (2014). Supply chain efficiency: An insight from fruits and vegetables sector in india. Journal of Operations and Supply Chain Management, 7(2), 154-167
[148] Nyaoga, R.; Magutu, P. Constraints management and value chain performance for sustainable development. Manag. Sci. Lett. 2016, 6, 427–442.
[149] Siddique, M.I. Factors Affecting Marketing Channel Choice Decisions in Citrus Supply Chain Massey University; Massey University: Palmerston North, New Zealand, 2015
[150] Ezin, V.; Quenum, F.; Bodjrenou, R.H.; Kpanougo, C.M.; Kochoni, E.M.; Chabi, B.I.; Ahanchede, A. Assessment of production and marketing constraints and value chain of sweet potato in the municipalities of Dangbo and Bonou. Agric. Food Secur. 2018, 7, 15
[151] APHLIS, Climate change and postharvest loss, 2019, https://www.aphlis.net/en/news/29.
[152] Armachius James & Vumilia Zikankuba | (2017) Postharvest management of fruits and vegetable: A potential for reducing poverty, hidden hunger and malnutrition in sub-Saharan Africa, Cogent Food & Agriculture, 3:1, 1312052, DOI: 10.1080/23311932.2017.1312052
[153] Spielman, David J., ed.; Malik, Sohail Jehangir, ed.; Dorosh, Paul A., ed.; and Ahmad, Nuzhat, ed. 2016. Agriculture and the rural economy in Pakistan: Issues, outlooks, and policy priorities. Philadelphia, PA: University of Pennsylvania Press on behalf of the International Food Policy Research Institute (IFPRI)
[154] Manzini, Riccardo, and Riccardo Accorsi. "The new conceptual framework for food supply chain assessment." Journal of food engineering 115, no. 2 (2013): 251-263.
[155] Paciarotti, Claudia, and Francesco Torregiani. "The logistics of the short food supply chain: A literature review." Sustainable Production and Consumption 26 (2021): 428-442.

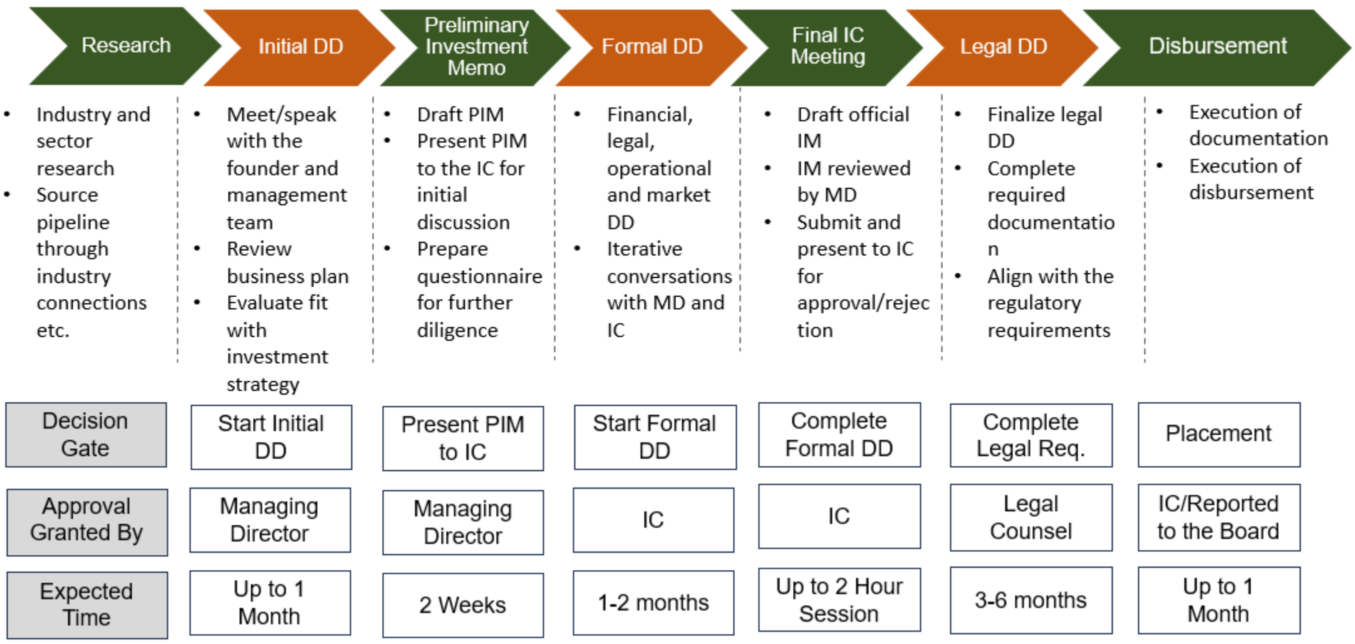
patterns.[156] Investments in post-harvest technologies can lead to decreased losses as well as higher value-additions, resulting in increased incomes and the creation of higher value markets (including food exports), such that farmers are more resilient and better able to withstand climate shocks. That said, the impact of post-harvest technology can extend far beyond the impact at the farm-level. According to the United Nations Environment Programme (UNEP) food loss and waste accounted for nearly 8%-10% of global GHG emissions.[157] Thus, post-harvest technologies can increase climate resilience of the entire agricultural sector and the surrounding ecosystem.

Output 1.2: Fund capital disbursed to venture, early-growth, growth-stage agribusinesses

Activity 1.3: Deploy capital to investee companies

Selection Processes

48. The team has sourced 200+ pipeline companies during 24+ months of market diligence. On first close, the Fund team will begin due diligence on high-priority targets identified by the team through this process.Figure 2 presents an overview of the selection process for ACAP's investment themes. The team is also developing customized selection criteria for each theme that will be informed by sectoral dynamics.



Note: "DD" = Due Diligence, "PIM" = Preliminary Investment Memo, "IM" = Investment Memo, "IC" = Investment Committee

Figure 2. ACAP Investment Process

Output 1.3: Strengthened business performance of portfolio companies

Activity 1.4: Monitoring, reporting & verification of portfolio company performance and impact

49. All ACAP investments will be rigorously pre-screened for impact. To that end, Acumen, in partnership with Winrock International, has developed a specialized climate resilience screening tool, the **Agriculture Resilience Investment Screen (ARIS)**. This tool provides a consistent, rapid, and robust methodology for investment managers to assess the climate resilience potential of a given agri-business opportunity. The tool is already being used by the

[156] ibid.

[157] United Nations Environment Programme, Food Waste Index Report, 2021, https://www.unep.org/resources/report/unep-food-waste-index-report-2021

Acumen Resilient Agriculture Fund (ARAF) (FP078 – a fund that GCF has invested in previously) and may be customized and adapted for ACAP based on the Pakistan context and the ACAP pipeline as needed. As such, the tool will be applied to all potential investment opportunities to assess the climate-adaptation impact, facilitate comparison, and aid decision making.

50. In addition to the climate resilience screening, we may use a stage-gate process to rule out deal breakers prior to carrying out any detailed due diligence. We intend to screen for environment, social, investment, and financial deal breakers, as well as impact potential of the company.

51. Post investment, we expect to track key impact metrics for all portfolio companies and use the TA grant facility to amplify specific impact - particularly those related to gender equity and smallholder farmers. Post-investment, ACAP intends to measure impact using a bespoke version of the **Climate Resilience Toolkit** developed by Acumen and tech-enabled impact measurement company 60 Decibels. The Toolkit allows companies to measure their contribution to building vulnerable farmers' capacity to absorb a shock, prepare/adapt to it, and access enablers that help with both absorption and adaptability, as well as company impact on agriculture sector resilience.

52. Given the significance of women in agriculture and the gender inequity within the sector, ACAP seeks to apply a gender lens across all investments. We aim to use the 2X Challenge criteria to ensure that we can make increasingly gender smart, inclusive investments and move portfolio companies towards gender equity, thereby creating exemplary businesses throughout the sector. The 2X Challenge was launched at the G7 Summit 2018 as a bold commitment to inspire DFIs/IFIs and the broader private sector to invest in the world's women and to measure value created by women's contributions. Investments that use 2X criteria reflect the multiplier effect of investing in women. Please see Annex 8 for further details on our gender assessment.

COMPONENT 2: TAF to Support Capacity, Knowledge Gaps & Ecosystem Development
Output 2.1: TA deployed across strategic themes to maximize portfolio viability and fund impact
Output 2.2: Monitoring and evaluation of the financial performance & climate impact of investments
Output 2.3: Increase in inclusivity and climate impact of portfolio companies

Activity 2.1: Establish TA facility

53. In addition to investment capital, ACAP aims to provide a USD 10M grant-funded Technical Assistance Facility (TAF) to further support the profitable growth of portfolio companies, amplify the impact on smallholder farmers, women and vulnerable rural populations, while strengthening the overall agri ecosystem in Pakistan. The TA Facility is split into two components (both managed as part of the ACAP TA Facility) as they will be targeting different beneficiaries and carrying out different activities. These components are *Component 2*: TAF to Support Capacity, Knowledge Gaps & Ecosystem Development and *Component 3*: TAF to Support Insights & Partnerships. The type of support under Component 2 will vary depending on the needs of the recipient (e.g., training farmers on product usage, supporting businsesses in measuring impact, access to investors etc.). Component 3 will target creating knowledge products/insights and stakeholder engagement to amplify ACAP's paradigm shift potential as detailed below. Additionally, please note that TAF grants are non-refundable.

Activity 2.2: Provide targeted TA across five components

54. The TAF is expected to be a key support for the Fund investing strategy to enhance the agricultural ecosystem level fund impact in Pakistan and build resilience in farmers, equitable and sustainable businesses, and a more impactful and viable sector overall. To that end, ACAP Fund seeks to use the TAF to engage businesses, networks, and stakeholders in TAF funded activities to provide bespoke support across five key areas: (i) climate adaptation, (ii) gender initiatives, (iii) business development, (iv) ESG reporting, and (v) impact evaluation.

COMPONENT 3: TAF TO SUPPORT INSIGHTS AND PARTNERSHIPS
Output 3: Establish platforms to catalyze stakeholder engagement, collaboration on & awareness of climate adaptation

Activity 3.1: Generating sector insights & knowledge products

55. Stakeholder engagement is a core part of ACAP Fund development and has consisted of broad outreach to diverse sector players. As Pakistan's first fund investing in agribusinesses targeting climate resilience, it was

essential to engage other investors within and outside Pakistan to ensure that ACAPs investment thesis was designed to incorporate all existing investment related learnings while building on a foundation of core sector knowledge and experience. Further, given the acute shortage of investment capital in the sector, it remains critical for ACAP to continue partnering with other investors to ensure that portfolio companies can scale by attracting co-investment and follow-on capital from a values-aligned group of investors. This would not only enable portfolio companies to grow faster and access valuable networks but would also de-risk the investments from a portfolio management perspective.

56. Our insights work will also inform monitoring mechanisms as the Fund team aims to refine adaptation metrics and measurement of food security and health impact through our investment through studies and insights projects. Component 3 will be funded through the TA facility (and managed as part of the ACAP TA Facility) and has been identified as a separate component to emphasize our focus on the paradigm shifting insights and knowledge generation that we aim to create through the Fund. We intend to synthesize insights from deploying the-first-of-its-kind ACAP Fund and, in turn, these will inform monitoring across the portfolio and allow us to share learnings with ACAP investees to amplify the impact of their work.

Activity 3.2: Convening & engaging stakeholders & co-investors to amplify impact

57. Given ACAP's position as the first of its kind impact fund focused on adaptation in Pakistan straddling the climate, agriculture, social impact, and private investment spaces, the Fund seeks to use its convening ability to ensure that we enable meaningful dialogue with relevant stakeholders to build further adaptation solutions for Pakistan. The Fund aims to convene annual meetings of relevant experts and stakeholders to enhance Pakistan's ability to create solutions for climate-resilient development and build adaptive capacity.

58. Key ACAP focus areas (i.e., adaptation finance, focused on the agriculture value chain) are already in line with Pakistan's revised NDC priority areas of agriculture and water management; Pakistan's Board of Investment (BOI) priority areas of investment such as food processing, transport, and logistics; and a focus on agri-tech enabled solutions such as cold-chain storage, warehousing, aquaculture, etc.

59. ACAP is also well aligned with National Climate Change Policy, Pakistan's UNFCCC Nationally Determined Contribution (NDC), National Adaptation Plans, and other appropriate national and international organizations and frameworks to ensure our investments meet the adaptation and mitigations goals of the Pakistan government as well as global adaptation goals and best practice standards**.** To validate our hypothesis, and to develop our investment strategy, we will be engaging with agriculture and climate change sector experts to help us identify early pioneer and growth-stage companies, carry out sector mapping, and conduct risk and feasibility assessments. These processes should enable the ACAP team to identify key intervention areas, key towards the team's goal to make climate resilience-focused and sustainable investments that have outsized impact for affected populations.

Impact Assessment: A focus on impact is embedded into every stage and process of the ACAP Fund

60. ACAP investments will be designed to increase adaptation capacity and will have a multi-level impact on Pakistan's agriculture sector on three levels: Ecosystem, Business and Farmer levels as shown below. Please see Section B.2 (a) for a detailed Theory of Change schematic and narrative.

61. ACAP seeks to build critical climate resilience at multiple levels: ecosystem, business, and farmer.

(i) The Fund is designed to have a critical *ecosystem level impact* that intends to increase the productivity of the overall value chain by catalyzing investments in paradigm shifting agri-business innovations. In addition to their direct impact, ACAP Fund investments intend to also incentivise the development of resilient food inputs, practices, and technologies. This investment strategy aims to further impact food security, health/nutrition outcomes, and rural economics across the country. Beyond Pakistan, as rising temperatures impact food systems and rural outcomes across the world, ACAP Fund can further demonstrate how even a limited amount of private capital, once structured effectively, can have outsized impact at a national level - and in doing so, provide a powerful blueprint for other agrarian countries across the world.

(ii) *Businesses* are expected to benefit directly from ACAP Fund. Not only will the Fund provide critical, missing capital for growth, along with robust technical assistance resources, but it will also leverage Acumen's strong network of global and local stakeholders to work with companies to grow more profitably and sustainably into regional success models. Access to finance from ACAP Fund should also further de-risk the companies and enable them to raise additional financing from public and private markets. In addition, we expect that the existence of a sector focused fund like ACAP will incentivise other entrepreneurs to develop more robust agriculture focused companies, which will in turn lead to a virtuous cycle of better companies, higher investment and growth, and better sector outcomes.

(iii) *Vulnerable farmers* are expected to directly benefit from investments in farmer platforms whereby they can access high quality inputs, finance, and high value markets more efficiently. Further, through investing in a more efficient value chain, farmers can increase returns, reduce the volatility of their incomes, and improve their resilience to climate shocks.

ACAP alignment with country level climate priorities

62. As stated above, key ACAP focus areas are already in line with:

- Pakistan's revised NDC 2021 priority areas of agriculture and water management; and
- Pakistan's Board of Investment (BOI) priority areas of investment

63. As stated above, to ensure ACAP investments meet the adaptation and mitigation goals of the Pakistan government as well as global adaptation goals and best practice standards, ACAP is well aligned with:

- National Climate Change Policy;
- Pakistan's UNFCCC Nationally Determined Contribution (NDC); and
- National Adaptation Plans and other appropriate national and international organizations and frameworks

Further, to validate ACAP's hypothesis, and to develop its investment strategy, ACAP will be engaging with agriculture and climate change sector experts to identify venture, early growth, and growth-stage companies, carry out sector mapping, and conduct risk and feasibility assessments to identify key intervention areas, allowing ACAP to make climate resilience-focused and sustainable investments that have outsized impact for affected populations.

64. ACAP is well-aligned with the UN's Sustainable Development Goals (SDGs) due to their interlinked nature and the Fund's ecosystem scope. The Fund's investments aim to create direct income improvement for vulnerable farmers and build resilience in the agriculture sector in Pakistan. ACAP directly aligns with the following SDGs:

1. **Goal 13:** Take urgent action to combat climate change and its impacts;
2. **Goal 12:** Ensure sustainable consumption and production patterns; and
3. **Goal 2:** End hunger, achieve food security and improved nutrition, and promote sustainable agriculture.

65. Additionally, as the Fund focuses on strengthening the overall agriculture ecosystem and improving vulnerable farmer livelihoods in Pakistan, it aims to also indirectly impact the following SDGs:

- **Goal 5:** Achieve gender equality and empower women and girls;
- **Goal 6:** Ensure availability and sustainable management of water and sanitation;
- **Goal 7:** Ensure access to affordable, reliable, sustainable, modern energy;
- **Goal 8:** Decent work and economic growth; and
- **Goal 11:** Sustainable cities and communities.

Acumen as Best-Placed Executor

66. Acumen is one of the world's leading impact investors, with extensive global experience of investing in companies building solutions for the most vulnerable across the world. Over the past 20 years, Acumen has already invested USD 154M in 167 companies in 13 countries, including Pakistan. The Executing Entity (EE) for this program will be the to be formed ACAP Fund entity. The EE will be responsible for the implementation of this program. The ACAP team will consist of investment professionals that are currently a part of the Acumen Pakistan team. Acumen will serve as the Accredited Entity (AE). In its capacity as the AE, Acumen will seek to provide oversight and ensure compliance with GCF requirements.

67. Over the past five years, Acumen has successfully raised three funds with anchor support from GCF totaling USD 230M through targeted climate.

68. Acumen has also successfully raised GCF's first private sector adaptation focused fund, ARAF (FP078) – a USD 58M fund focused on Africa, that closed above its target. ARAF is also the world's first climate resilience-focused agribusiness investment fund and has been heralded as an innovative and exciting proposition in climate adaptation financing. ARAFs model has attracted significant interest from private investors and as a result similar funds have been raised in other regions such as FP176 "Hydro-agricultural development with smart agriculture practices resilient to climate change in Niger". In the context of ACAP, we are mindful that Pakistan shares many of the agriculture sector challenges being addressed by ARAF. As such, not only has ACAP been developed using many of the learnings from ARAF, but we plan to develop deep linkages between the two funds through shared knowledge, insights, and portfolio connections that can further support the growth of scalable agri-business models and sector knowledge. Additionally, Acumen Pakistan has run accelerator programs focused respectively on climate change and agriculture sectors, engaging with early-stage entrepreneurs to better understand the context of these spaces in Pakistan.

69. Acumen has successfully mobilized capital from a broad range of climate investors, private funders, philanthropic organizations, family offices, and corporations. Over the past two decades, we have developed long term, trust-based relationships with a wide range of investors/funders and will be able to use these relationships to raise the capital required for ACAP Fund. The ACAP Fund team has made significant progress on this front and has already engaged with various potential investors interested in investing in ACAP.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

Explain why the project/programme requires GCF funding to address mitigation or adaptation measures, i.e. Why is the project/programme not currently being financed by public and/or private sector? Which market failure is being addressed with GCF funding? Are there any other domestic or international sources of financing?

Explain why the proposed financial instruments were selected in light of the proposed activities and the overall financing package. i.e. What is the coherence between activities financed by grants and those financed by reimbursable funds? How were co-financing amounts and prices determined? How does the concessionality of the GCF financing compare to that of the co-financing? If applicable, provide a short market read on the prevailing of the pricing and/or financial markets for similar projects/programmes.

Justify why the level of concessionality of the GCF financial instrument(s) is the minimum required to make the investment viable. Additionally, how does the financial structure and the proposed pricing fit with the concept of minimum concessionality? Who benefits from concessionality?

70. The following information is presented as an indicative summary of certain of only the principal terms of Acumen Climate Action Fund Pakistan ("ACAP") and its manager (the "Manager"). This Funding Proposal does not constitute an offer to sell or a solicitation of an offer to purchase any security of ACAP. Any such offer or solicitation will only be made pursuant to the final Confidential Private Placement Memorandum issued with respect to ACAP, which qualifies in its entirety the information set forth herein (subject to the requirement in the Term Sheet that all of the Programme documents are to be consistent in all material respects with the terms set out in the funded activity agreement and the accreditation master agreement) and which should be read carefully prior to any investment in ACAP for a description of the terms, merits and risks of such an investment. Further, the parties shall be bound only upon execution of definitive fund documentation, including the subscription agreement, limited partnership agreement and funded activity agreement. To the extent that the terms set forth below are inconsistent with those of the subscription agreement, limited partnership agreement or the funded activity agreement, such agreements, as the case may be, will control. ACAP and the Manager the executing entities.

1. ACAP is intended to bring together senior and junior equity capital to incentivize investors with different risk profiles to provide support for climate adaptation for vulnerable farmers. ACAP is expected to be formed as a Dutch Coöperatief U.A. which would allow ACAP to serve as a tax blocker and for investors to be able to utilize double taxation treaties between the Netherlands and Pakistan.

2. The to-be-formed ACAP Fund Manager is expected to be formed as a Dutch private limited company that would be indirectly wholly owned by the Accredited Entity. A management agreement would be entered into between the Manager and the Fund.

3. The TAF sits alongside but exists independently from ACAP, with separate management and governance arrangements. The Manager shall be under obligation to implement the TAF in line with the GCF requirements included in the FAA and AMA, as per a Subsidiary Agreement between the Manager and the AE. The Manager shall develop and present for approval by the Technical Assistance Committee ("TAC") a TA manual governing the operations and grant-making processes of the TAF ("TA Manual").

4. The Manager will adhere to all the relevant policies of Acumen that GCF reviewed in the re-accreditation of Acumen in 2022. It is possible that there will be some policies that differ between Acumen and the Manager; unlike Acumen, the Manager is not a tax-exempt entity.

5. Investment decisions for ACAP will be made by an investment committee designated by the Accredited Entity. The Investment Committee (IC) is the final decision-making body in the investment process. IC will assess the business model of each potential investment for its financial viability, social impact and climate contribution and make a final decision to approve each transaction. The IC will initially consist of up to 5 members. The IC members will comprise experts with a combination of (i) investment experience, (ii) Southeast Asian regional experience, and (iii) climate expertise. The Manager will only make investments that have been approved by the investment committee.

6. ACAP will have an advisory committee (the "Advisory Committee") to serve as an adviser to the Manager and to consult with the Manager concerning the ACAP's activities and operations, including, in particular, with respect to the identification and resolution of conflicts of interest. The Advisory Committee will consist of at least two but no more than seven voting members.

In addition, for over 20 years, Acumen has been directly investing in small and medium-sized enterprises that serve low-income communities in developing countries across South Asia, Sub-Saharan Africa, and Latin America, with the goal of changing the way the world tackles poverty. As of the end of 2022, Acumen had deployed approximately USD 154.4M into 167 companies, impacting approximately 501 million lives.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

Explain why the project/programme requires GCF funding to address mitigation or adaptation measures, i.e. Why is the project/programme not currently being financed by public and/or private sector? Which market failure is being addressed with GCF funding? Are there any other domestic or international sources of financing?

Explain why the proposed financial instruments were selected in light of the proposed activities and the overall financing package. i.e. What is the coherence between activities financed by grants and those financed by reimbursable funds? How were co-financing amounts and prices determined? How does the concessionality of the GCF financing compare to that of the co-financing? If applicable, provide a short market read on the prevailing of the pricing and/or financial markets for similar projects/programmes.

Justify why the level of concessionality of the GCF financial instrument(s) is the minimum required to make the investment viable. Additionally, how does the financial structure and the proposed pricing fit with the concept of minimum concessionality? Who benefits from concessionality?

In your answer, please consider the risk sharing structure between the public and private sectors, the barriers to investment and the indebtedness of the recipient. Please reference relevant annexes, such as the feasibility study, economic analysis or financial analysis when appropriate.

71. GCF's support of this project will be critical to ACAP's success. Capital markets are underdeveloped, and the private equity landscape is still at a nascent stage in Pakistan.[158] This is evidenced by the fact that there are only a handful of Private Equity/Venture Capital funds currently operating in Pakistan and only two funds registered locally. Despite the country's growing economy, large population and enormous potential for impact investing, many investors still perceive too much risk related to the security and political instability to make Pakistan a viable, investable market. Investment as a percentage of GDP in Pakistan was 13.6% compared with other Asian countries (for example, Bangladesh stands at 33%, Indonesia at 31%, Nigeria at 33%).[159] GCF capital acting as first-loss equity will, therefore, critically de-risk the Fund and attract private investment into a crucial sector that supports the entire economy. Please see Section F for further details on fund risks.

72. ACAP is targeting a blended structure with up to a 1:2 junior equity (first loss) to senior equity layer. Given that Pakistan remains a challenging frontier market, where it is difficult to attract commercial capital due to high perceived risk, a significant first loss layer is critical to mobilize meaningful funding - even from development finance focused funders and impact investors. This is even more important as ACAP is the first of its kind fund in a sector that continues to be perceived as higher risk. A blended finance fund structure is essential for the Fund to address the risk/return objectives of different types of investors. At a minimum, the fund structure would require a 1:2 first loss cover along with a relatively conservative investment strategy that focuses on export oriented/import substituting businesses to further reduce investment risk.

73. ACAP, as an adaptation fund focused on agriculture, is well-aligned with GCF adaptation result areas ARA1: Most vulnerable people and communities and ARA2: Health, Food and Water Security. According to sectoral guides on Agriculture and Ecosystems and Ecosystem Services, ACAP fits within paradigm shifting pathways through engaging private sector finance in the sector to invest in regenerative businesses across agriculture and food systems value chain, inclusion of women, and mobilizing finance at scale. Improving food loss and food security is a key aim for ACAP and building a sustainable food ecosystem is key in adapting to climate change for countries like Pakistan. ACAP as a single country fund will create a blueprint for adaptation finance for a population of 240M people in a climate vulnerable region. It is key to demonstrate success for mobilizing blended finance in an emerging market heavily dependent on agriculture. In addition, adaptation and agriculture ranked high on the priorities for the recent COP28 GCF's climate adaptation expertise will again be a strong guide to help our fund achieve meaningful climate resilience impacts for smallholder farmers in Pakistan.

74. The GCF will enable ACAP to channel much needed financing and technical assistance to agricultural companies to support their equitable, sustainable growth while supporting high poverty customers. GCF's participation in the TAF, much like with ARAF (FP078), will spur other investors into supporting the TAF. We want our investments to create scalable and sustainable agricultural businesses generating climate resilience for smallholder farmers and vulnerable agri-dependent populations. We also want these businesses to scale in environmental, social, and equitable ways. That is why our TAF is supporting gender initiatives for businesses, ESG initiatives to mitigate risks to their businesses, and enterprise support to help them strengthen their operations. Additionally, we believe that our TAF aligns with GCF's paradigm shift efforts, enabling environment, gender, and environmental and social goals so GCF is a fitting partner for these activities. Our impact and financial goals are enhanced by our TAF, and we believe that GCF's participation in the TAF is critical to ACAP's success.

75. In addition to the government, Acumen has engaged with a wide array of stakeholders, including farmers, rural communities, academics, and SMEs to understand key pain points and opportunities and incorporate those in the Fund strategy. As the world's only climate finance multi-development bank, GCF's expertise would help Acumen to create a community of practice in Pakistan around catalytic, climate finance and best practices for strengthening the climate resiliency of at-risk, vulnerable communities.

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[158] GIIN, The Landscape For Impact Investing In South Asia, 2015, https://thegiin.org/assets/documents/pub/South%20Asia%20Landscape%20Study%202015/Pakistan_GIIN_southasia.pdf

[159] CEIC Data, Pakistan Investment: % of GDP, Accesses on August 3, 2023, https://www.ceicdata.com/en/indicator/pakistan/investment--nominal-gdp#:~:text=Pakistan%20Investment%20accounted%20for%2013.6,an%20average%20ratio%20of%2017.0%20%25

76. Acumen has been the first impact investor on the ground in Pakistan and continues to remain one of a handful of institutional investors in the country. As seen from Acumen's other funded projects, participation by GCF as an anchor investor, especially in the form of a USD 25M first loss layer, would send a strong signal to the Pakistani capital market. Fundraising from potential investors will become significantly less challenging. As this would be the first climate change focused fund in Pakistan, adding a potential additional level of risk for traditional private investors, GCF's support would reassure other institutional investors and catalyze additional private sector participation.

77. GCF will also be filling a void in the commercial investment market in terms of funding for climate change-related investments, as there are currently no large investors working towards this goal in Pakistan. As mentioned above there is a shortfall of USD 7B - 14B in climate adaptation financing for Pakistan, in addition to the damage caused by unprecedented severe weather events, such as the heatwave and floods in 2022. We are confident that ACAP can mobilize GCF's capital in the most effective manner, catalyzing private sector funds and opening markets to new investments in one of the most vulnerable regions to climate change in the world.

B.6. Exit strategy (max. 500 words, approximately 1 page)

Explain how the project/programme will successfully exit once implementation is completed, including how results and benefits will continue beyond the project/programme period and how the contribution to paradigm shift will be maintained.

Include information pertaining to the longer-term ownership, project/programme exit strategy, operations and maintenance of investments (e.g. key infrastructure, assets, contractual arrangements). In case of private sector, please describe the GCF's financial exit strategy through Initial Public Offerings, trade sales, etc.

Provide information on additional actions to be undertaken by public and private sector or civil society as part of the project/programme to ensure sustainability of the results attained.

78. Investing in high impact, commercially viable companies and preparing for successful exits is a core element of ACAP's overall investment strategy. ACAP's investment professionals expend significant effort understanding the strategies and operational components of each portfolio company to support the growth of a financially sustainable enterprise that will attract a later-stage capital investment. A focus on sustainability will be present throughout the life of the Fund and multiple exit implications/assumptions are considered in our financial model. A robust pipeline of potential investments has been analyzed by the Fund investment team and the on-the-ground diligence of these climate focused agribusinesses underpins ACAPs investment strategy.

79. ACAP Fund's investment strategy was developed from the ground up and will deploy capital under the guidance of a world class investment committee to build scalable companies that respond to local market demand (end customer and downstream industry), as well as supply to export markets.

80. The Fund strategy is also determined by the availability of follow-on capital, exit options, and longer-term sustainability. Acumen has been investing in Pakistan for the past two decades. During this time, we have seen multiple investment cycles and identified the key gaps in building successful companies. We have seen that even as the market has become more attractive for investors over time, growth capital remains in short supply and is often the reason that otherwise successful business models fail or do not reach their potential. As such, we designed ACAP in a way that would allow us to support the company's growth journey through successive investment rounds (i.e. we will have enough investable capital to support our best companies as they grow) and be able to support the growth of the company until it reaches a maturity level that makes it an attractive investment option for later stage, more commercial investors, and allows us to exit the investment successfully.

81. The Fund team has consulted with various later stage (Series B/Series C) investors and DFIs focused on climate adaptation, agriculture and food systems in frontier markets like Pakistan to understand their investment strategy. Our fund investment strategy will be guided by a clear line of sight on developing the types of companies that serve a key market gap and have the capacity to attract later stage investment to create long term success and a regional blueprint for climate resilient agriculture. We are also actively engaging with local investors (both PE/VC funds and corporates) to develop a coalition of co-investors for each deal. This will likely not only reduce our investment risk (as we are able to reduce exposure to any single investment) but will also open further exit options via share buyouts.

83. In creating an attractive model for private capital deployment towards climate adaptation, ACAP Fund expects to create a powerful blueprint for other countries and fund managers. We are confident that not only will ACAP Fund unlock private sector capital towards adaptation and agriculture sector resilience, but this fund’s success will inspire new agribusinesses to emerge, as well as new funds to be developed with a focus on climate adaptation, particularly in agrarian economies that are vulnerable to climate change and food insecurity.

C. FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount	Currency
	USD 28M	USD

GCF financial instrument	Amount	Tenor	Grace period	Pricing
(i) Senior loans	Enter amount	Enter years	Enter years	
(ii) Subordinated loans	Enter amount	Enter years	Enter years	
(iii) Equity	USD 25M			
(iv) Guarantees	Enter amount	Enter years		
(v) Reimbursable grants	Enter amount			
(vi) Grants	USD 3M			
(vii) Results-based payments	Enter amount			

(b) Co-financing information	Total amount	Currency
	62M	million USD ($)

Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
Discussions ongoing (DFIs, private institutions, foundations, family offices)	Equity	USD 53M	million USD ($)			senior
Discussions ongoing (DFIs, private institutions, foundations, family offices)	Equity	USD 2M	million USD ($)		Enter	junior
TAF Component 2: TAF to Support Capacity, Knowledge Gaps & Ecosystem Development						
Discussions ongoing (DFIs, foundations, family offices)	Grants	USD 5.71M	million USD ($)	Enter years Enter years		Options
TAF Component 3: TAF to Support Insights & Partnerships						
Discussions ongoing (DFIs, foundations, family offices)	Grants	USD 1.29M	million USD ($)	Enter years Enter years		Options

(c) Total financing (c) = (a)+(b)	Amount	Currency
	90M	USD

(d) Other financing	

arrangements and contributions (max. 250 words, approximately 0.5 page)	

C.2. Financing by component

Please provide an estimate of the total cost per component and output as outlined in section B.3. above and disaggregate by source of financing. More than one co-financing institution can fund a single component or output.

Component	**Output**	**Indicative cost** Options	**GCF financing**		**Co-financing**		
			Amount Options	**Financial Instrument**	**Amount** Options	**Financial Instrument**	**Name of Institutions**
Component 1: Investment Fund	1.1 ACAP Fund Established	800,000	250,000	Equity	550,000		
	1.2 Fund capital disbursed to venture, early-growth, growth-stage agribusinesses	67,131,703	20,978,657	Equity	46,153,046	Equity	
	1.3 (a) Strengthened business performance of portfolio companies	8,424,653	2,632,704	Equity	5,791,949	Equity	
	2.2 (b) Monitoring & evaluation of the financial performance & climate impact of investments	3,643,644	1,138,639	Equity	2,505,005	Equity	
Component 2: TAF to Support Capacity, Knowledge Gaps & Ecosystem Development	1.3 (b) Strengthened business performance of portfolio companies	1,630,000	182,500	Grants	1,447,500	Grants	
	2.1 TA deployed across	1,682,281	504,684	Grants	1,177,597	Grants	Click here to

	strategic themes to maximize portfolio viability and fund impact						enter text.
	2.2 (a) Monitoring & evaluation of the financial performance & climate impact of investments	2,377,719	789,316	Grants	1,548,403	Grants	
	2.3 Increase in inclusivity and climate impact of portfolio companies	2,364,000	827,400	Grants	1,536,600	Grants	
Component 3: TAF to Support Insights & Partnerships	3. Establish platforms to catalyze stakeholder engagement,collaboration on and awareness of climate adaptation	1,946,000	656,100	Grants	1,289,900	Grants	
Indicative total cost (USD)		90,000,000	28,000,000		62,000,000		

This table should match the one presented in the term sheet and be consistent with information presented in other annexes including the detailed budget plan and implementation timetable.

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?	Yes ☒ No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☐ No ☐

If the project/programme is expected to support capacity building and technology development/transfer, please provide a brief description of these activities and quantify the total requested GCF funding amount for these activities, to the extent possible.

Through the proposed USD 10M Technical Assistance Facility (TAF), ACAP Fund will finance capacity development activities. Subcomponents and key activities, and associated GCF funding are further highlighted below.

Activity	Sub-Activities	Total Budget	GCF Financing (%)	Co-Financing (%)
2.1 Establish TA Facility	2.1.1: Establishment and Management of ACAP TAF	USD 1,682,281	30.0%	70.0%
Activity 2.2.1. Climate Adaptation	2.2.1.1 Support activities increasing farmer/rural communities awareness of climate resilient agriculture practices and products 2.2.1.2 Support activities increasing agribusiness awareness of climate resilient practices and risks 2.2.1.3 Execute programmes to accelerate early-stage agri businesses building climate resilience	USD 710,000	35.0%	65.0%
2.2.2. Gender Initiatives	2.2.2.1 Support activities to promote skills development and income enhancement of women in agriculture sector 2.2.2.2 Support initiatives to increase entrepreneurial participation of women in the agriculture sector 2.2.2.3 Support companies to set, implement, and evaluate gender action plans and provide company level gender trainings 2.2.2.4 Recruit a full-time gender expert to ensure that gender is embedded across ACAP investment processes and generate gender insights	USD 1,654,000	35.0%	65.0%
2.2.3. Business Development	2.2.3.1 Support companies with financial management and governance 2.2.3.2 Support companies on improving investment readiness including marketing, data room management, and valuation support 2.2.3.3 Support companies with the development of human resources processes and talent development and acquisition	USD 1,630,000	11.2%	88.8%

	2.2.3.4 Support distressed companies with access to recovery best practices			
2.2.4. ESG Reporting	2.2.4.1 Support companies to set, implement, and monitor their climate and ESG action plans	USD 1,217,719	30.0%	70.0%
	2.2.4.2 Support companies with Environmental and Social Impact Assessment when companies engage in higher risk activities			
	2.2.4.3 Hire ESG associate for monitoring ESG outcomes for portfolio companies			
2.2.5. Impact Evaluation	2.2.5.1 Pre-investment impact assessment of pipeline companies	USD 1,160,000	40.0%	60.0%
	2.2.5.2 Support portfolio companies with post-investment customer surveys to enhance impact awareness and solutions			
	2.2.5.3 Support portfolio companies with post-investment detailed customer surveys to enhance impact awareness and solutions			
	2.2.5.4 Support companies with development of impact evaluation processes and systems			
Activity 3.1. Generating sector insights & knowledge products	3.1.1 Support academic research and insights into climate adaptation in Pakistan	USD 1,446,000	35%	65%
	3.1.2 Hire an insights expert to develop sector insights and knowledge products			
	3.1.3 Generate comprehensive insights reports to disseminate fund level learnings from investing in climate resilience in Pakistan			
	3.1.4 Develop industry research on climate adaptation, environmental safeguarding, and specific impact areas to develop sector best practices for Pakistan			
Activity 3.2. Convening and engaging	3.2.1 Convene stakeholders including public and private sector	USD 500,000	30%	70%

stakeholders and co-investors to amplify impact	leaders, agricultural leaders, academics, CSOs, and entrepreneurs to share best practices and develop knowledge networks			
Total		USD 10M	30%	70%

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's Initial Investment Framework.

D.1. Impact potential (max. 500 words, approximately 1 page)

Describe the potential of the project/programme to contribute to the achievement of the Fund's objectives and result areas. As applicable, describe the envisaged project/programme benefits for mitigation and/or adaptation. Provide the intended outcomes for mitigation by elaborating on how the project/programme contributes to low-emission sustainable development pathways. Provide the intended outcomes for adaptation by elaborating on how the project/programme contributes to increased climate-resilient sustainable development. Calculations should be provided as an annex. This should be consistent with section E.3 reporting GCF's core indicators.

i. Climate Impact Potential

84. With a triple bottom line and a multi-layered approach to impact, ACAP has climate impacts on three levels - farmer, business level and ecosystem. We aim to invest in 15-17 growth companies and estimate that 13.1M lives will be impacted (directly and indirectly). ACAP aims to target 2M smallholder farmers through our investments. Our diligence in this regard remains ongoing.

85. We seek to make investments that are intended to generate the following impacts:

- Increased income for farmers;
- Higher yields and increased productivity;
- More food secure households;
- Increased awareness of climate information;
- Increased use of climate resilient tools;
- Agribusinesses that will be more climate aware and have better climate resilient tools; and
- In addition, ACAP will invest in some business models that use renewable energy/clean technology which will contribute to mitigation as well as climate resilience.

86. Like many other developing countries, the agriculture sector in Pakistan is largely female, with women making up ~70% of the agriculture workforce.[160] Pakistan is also the second most inequitable country in the world for women, making it even more imperative that we focus on increasing gender equity within our investment portfolio.[161] ACAP will use the 2X challenge criteria to apply a gender lens to our investments and to understand the value created by women in the sector.[162] Annex 8 discusses our gender assessment in more detail.

87. ACAP aims to improve adaptation capacity and build resilience in the Pakistani agriculture ecosystem through improving farmer incomes, increasing uptake of climate resilient solutions (such as improved access to high quality inputs and markets), reduction in on-farm food waste, and improved yields.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

Paradigm shift potential is defined as 'degree to which the proposed activity can catalyse impact beyond a one-off project or programme investment'. In this section, elaborate on the contribution to paradigm shift and how the proposed project/programme aims to contribute towards it based on the theory of change described in section B2(a). Also describe how and to what extent the project/programme will be able to promote or contribute to paradigm shift through the below.

- *Potential for scaling up and replication*
- *Potential for knowledge sharing and learning*
- *Contribution to the creation of an enabling environment*
- *Contribution to the regulatory framework and policies*
- *Overall contribution to climate-resilient development pathways consistent with relevant national climate change adaptation strategies and plans*

[160] FAO, Women in Agriculture in Pakistan, 2015, https://www.fao.org/3/i4330e/i4330e.pdf
[161] WEF, Global Gender Gap Report, 2022, https://www3.weforum.org/docs/WEF_GGGR_2022.pdf
[162] CDC, How to Measure the Gender Impact of Investments: Using the 2X Challenge Indicators in Alignment with IRIS+, 2020, https://assets.cdcgroup.com/wp-content/uploads/2020/03/16111901/How-to-measure-the-gender-impact-of-investments.pdf

88. ACAP will be Pakistan's first private investment vehicle focused on climate adaptation. Our investments will help increase incomes for vulnerable farmers and make them more resilient to climate change. Furthermore, ACAP's success aims to attract more private capital to this critical sector. ACAP is replicating the ARAF model in the Pakistan context and hopes to create a regional blueprint for adaptation finance in agrarian economies. ACAP also intends to incentivize the growth of climate focused agri-businesses.

89. ACAP's future portfolio hopes to demonstrate successful adaptation focused investments which would be catalytic for one of the most climate vulnerable regions in the world that is still largely dependent on agriculture. Pakistan's current adaptation needs are between USD 7B – 14B annually and will only increase as temperature increases continue across the region. ACAP will fulfill a vital need for private capital programmes focused on building climate resilient models. Aside from regional replication potential, the ACAP Fund model could also be replicated within Pakistan with a focus on building resilient infrastructure – which is also crucial for Pakistan and the wider region.

90. Agriculture contributes ~20% to Pakistan's GDP, and is linked to an additional ~60-70% of exports due to its backward and forward linkages with high-value sectors. However, Pakistan only invests 0.2% of GDP on research and development for the sector, the lowest percentage in South Asia.[163] This is despite the fact that the annual rate of return on agriculture investments is estimated to be 43%.[164] Investing in this sector (through ACAP and other funds) will be crucial to ensure that the 63% of all Pakistani livelihoods which are rural, are more climate resilient and have greater adaptation capacity in terms of resilient incomes, access to climate smart agriculture practices, and are part of a more productive and profitable agriculture ecosystem. [165]

91. The programme's paradigm shift and impact potential are embedded in its design. Pakistan's first climate fund aims to have an impact on multiple levels:

- **Vulnerable lives:** ACAP aims to impact 13.1M lives (directly and indirectly). This includes a broad spectrum of smallholder farmers and vulnerable agri-dependent populations as ACAP understands the need to anchor impact for those who are most affected by climate change. While the Fund has a mandate to invest in venture, early-growth, growth-stage businesses, the ACAP team will commit itself to measuring the following farmer-level KPIs: percentage reporting; income improvement; increase in yields; increased access to credit, improved inputs, weather information or extension services, and more equipped to absorb a climatic shock due to the company's product/service.
- **National level:** ACAP expects to demonstrate the catalytic impact of a partnership between public and private sector players towards the achievement of critical national goals that also tie in with global priorities. Key stakeholders are the government, DFIs, local banks, etc. Given ACAP's unique understanding of Pakistan's agriculture sector through both the upstream (supplier) and downstream (market) lens, the Fund is poised to play a unique role as an intermediary, connecting producers with the products needed for continued growth while also connecting those same producers with consistent buyers and high-value markets.
- **Business level:** By investing in and, actively building, 15-17 climate impact focused companies, we seek to provide a blueprint for attracting private capital towards climate investment in the country. To demonstrate the fund's impact, ACAP commits to measuring each portfolio company's top-line growth, as well as evaluating each company's good governance; gender action plan; and climate adaptation measures. Through our investments, we seek to prove that there are viable venture, early-growth, and growth-stage business models within Pakistan.
- **Sector level:** The success of ACAP should demonstrate the strong impact and commercial viability of Pakistan's agricuture sector. In turn, this should further attract much-needed private sector capital to Pakistan in sectors where it is needed the most.
- **Regional blueprint:** Through the creation of the climate vulnerable country's first climate change private capital fund, ACAP expects to demonstrate a blueprint for adaptation finance as the first-of-its-kind single

[163] World Bank, World Development Report:Agriculture for Development, 2008, https://openknowledge.worldbank.org/server/api/core/bitstreams/8d8ad2dd-5c98-5042-8aad-744fdd7b034f/content
[164] Acharya et al, Current Scenario and Future Perspective for Scientific Research in Nepal, 2020, https://www.researchgate.net/publication/346968145_Current_Scenario_of_and_Future_Perspective_for_Scientific_Research_in_Nepal
[165] Finance Division, Government of Pakistan, Pakistan Economic Survey, 2022-2023, https://www.finance.gov.pk/survey/chapters_23/Highlights.pdf

country fund targeting an emerging market. Furthermore, ACAP and Acumen plan on holding several convenings centered around the same idea.

92. ACAP's capacity to influence drivers for social and economic development is an important focal area for the Fund at an ecosystem level. Component 3 of the Fund Activities includes engaging various stakeholders (public, private) and developing/publishing insights that would help build institutional capacity and reduce information asymmetries that limit private investments in agribusinesses. Regarding gender, as part of the TA, we will not only focus on increasing gender equity at our portfolio companies but also build and share on-ground gender insights, along with examples of best practices and their impact on the agriculture ecosystem. ACAP will also have a full-time gender expert on the team to lead on this work, as well as a full-time insights lead to ensure that we keep sharing insights and learnings across different platforms - so that we are able to influence deeper social barriers that lead to underinvestment in agriculture and gender equity. Activity 3.2 directly connects with the focus on engaging stakeholders.

D.3. Sustainable development (max. 500 words, approximately 1 page)

Describe the wider benefits and priorities of the project/programme in relation to the Sustainable Development Goals and provide the potential in terms of:

- *Environmental co-benefits*
- *Social co-benefits including health co-benefit*
- *Economic co-benefits*
- *Gender-sensitive development benefit*

93. Environmental co-benefits: Climate-smart and climate resilient agricultural practices, technology, and services enable smallholder farmers to be better stewards of the environment. Excessive and increasing use of agrichemicals have harmed biodiversity and arable land in Pakistan. Climate smart services and practices including digital farmer platforms, drip irrigation, and efficient use of fertilizers can help limit nitrogen and other harmful chemical runoff, thereby better preserving farmland and critical habitats in Pakistan. Illustratively, while agricultural water use accounts for 90% of water withdrawals in Pakistan, 60-75% of water withdrawn is lost to either seepage into saline groundwater or surface water evaporation. Water is a limited resource and agricultural practices must change to maintain farmer livelihoods and access to water. ACAP seeks to invest in agribusinesses promoting the efficient usage of water including solar irrigation companies and companies providing other water management services and approaches. Additionally, ACAP seeks to fund farmer trainings on climate resilient agricultural practices that will protect the environment, sustain land use, and limit post-harvest loss.

94. Social co-benefits: Through it's investing activities, ACAP seeks to provide a number of social co-benefits to the Program's beneficiaries, primarily smallholder farmers and vulnerable agri-dependent populations. ACAP's investments intend to help increase crop productivity, decrease post-harvest loss, and improve incomes. These projected outcomes intend to generate further social impacts including improved quality of life, time saving, community improvements, decreased hunger, and increased climate awareness. Our investment pillars should increase time savings for famers and vulnerable agri-dependent populations through efficient processing, improved transportation, and access to information. The potential increase in incomes resulting from the adoption of climate-smart agricultural practices should also provide farming families with access to larger quantities of food and improved quality, likely making it healthier. Farmers and their families in Pakistan are both unaware of and vulnerable to climate hazards that could impact both their livelihoods and their homes. Our team expects that our investments will improve farmer awareness of climate hazards and climate resilient farming practices. Also, the ACAP TAF will fund trainings for farmers and vulnerable agri-dependent populations to help them adapt to and mitigate climate hazards impact on their lives.

95. Economic co-benefits: ACAP seeks to invests in businesses with products and services that will improve the incomes of smallholder farmers in Pakistan. We expect farmers to improve their incomes through the diversification of their crops, adoption of technologies that improve efficiencies, yields, and reduce post-harvest loss, and by accessing formal local and international markets. Short term impacts may include more regular cash infusion and stabilized incomes improving the management of daily expenses. Long term, ACAP expects that investments in agribusinesses will improve livelihoods and expand entrepreneurial opportunities for smallholder farmers.

96. Gender-sensitive development benefit: ACAP seeks to invest and support agribusinesses in Pakistan with a gender lens. Women account for about 23% of Pakistan's labor force. However, the average female worker makes about 16.3% of a man's salary. 81% of the 5.26 million people working in the informal sector in Pakistan are women.

Working in the informal economy exposes women to multiple vulnerabilities including limited income security, occupational health issues, and high economic vulnerability. ACAP seeks to both invest in companies that have gender smart business practices and to work with future portfolio companies to improve their operations and customer outreach through gender expertise, technical assistance, and binding commitments to safe and equitable workplaces. ACAP also seeks to fund trainings for women that will impart skills and knowledge to improve a). their employability in agribusinesses and b). climate awareness and climate smart agricultural practices, in an effort to improve their farming capacity. Improving the safety and accessibility of agribusinesses in Pakistan creates opportunities for women to enter the formal market. Climate awareness trainings can help female farmers improve family farming practices and improve incomes. Increased income, new skills, and access to technologies and services improves the safety and well-being of female farmers. Finally, by building the gender expertise of our fund, generating sex-disaggregated data across the ACAP portfolio, and sharing insights across the agricultural ecosystem, we aspire to contribute to a more equitable and sustainable agricultural sector in Pakistan.

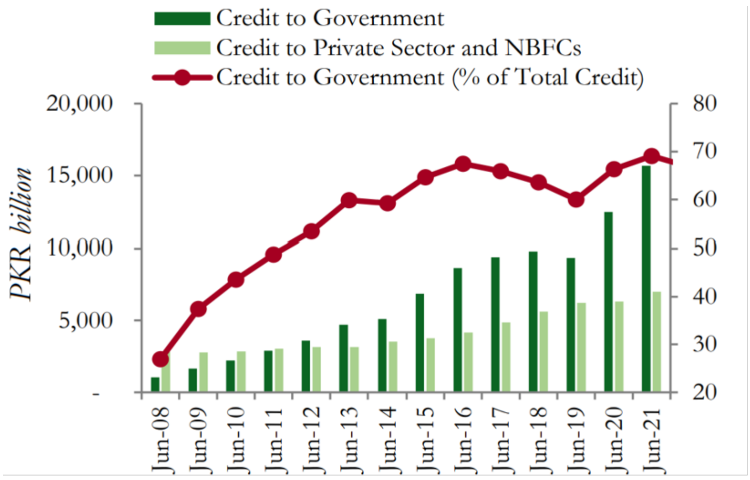
D.4. Needs of recipient (max. 500 words, approximately 1 page)

Describe the scale and intensity of vulnerability of the country and beneficiary groups and elaborate how the project/programme addresses the issue (e.g. the level of exposure to climate risks for beneficiary country and groups, overall income level, etc.). Describe how the project/programme addresses the following needs:

- *Vulnerability of the country and/or specific vulnerable groups, including gender aspects (for adaptation only)*
- *Economic and social development level of the country and the affected population*
- *Absence of alternative sources of financing (e.g. fiscal or balance of payments gap that prevents government from addressing the needs of the country; and lack of depth and history in the local capital market)*
- *Need for strengthening institutions and implementation capacity*

Pakistan suffers from a significant lack of capital

97. The country's vulnerability to climate change is well established, however it also suffers from a chronic lack of investment required to help build its resilience. The country's large and persistent fiscal deficits have led to public debt accumulation, crowding out private sector borrowing and posing macroeconomic risks. Fiscal deficits in Pakistan have been persistently large; in FY23 it stood at 7.8% of GDP as compared to the South Asian regional average of 6.3%.[166] As a result, there has been a rapid accumulation of public debt reaching 82.3% of GDP in FY23.[167]



[166] World Bank Group, Pakistan Development Update: Restoring Fiscal Sustainability, October 2023, https://thedocs.worldbank.org/en/doc/cfd9113f24c548efdc86dba482a5e2cf-0310062023/original/Pakistan-Development-Update-October-2023.pdf
[167] ibid.

Figure 3. Trend in Government of Pakistan Borrowing from Banks FY7-21[168]

98. Credit extended by the banking sector to the Government has risen by more than 400% over FY11–21, while private sector credit increased by 131% over the same period.[169] As of 2022, private sector credit stood at 14.8% of GDP, one of the lowest among emerging economies.[170] The reduced access to credit is one of the main contributing factors for the low levels of private investment and hence low productivity growth in Pakistan. Extensive government borrowing from the financial sector has kept capital markets shallow and underdeveloped. In the presence of a dominant borrower with no to low risk, the financial system has few incentives to design innovative financial products and institute robust risk management practices to extend finance to underserved vulnerable sectors like agriculture.

Absence of alternative sources of financing

99. There are supply-side challenges associated with financing bankable climate opportunities in Pakistan which affect both international investors and domestic investors. International private investors have low interest in climate financing opportunities due to the high return on investment barriers in Pakistan. Also, annual depreciation of the Pakistani Rupee estimated at ~30% over the next 2 years limits the potential climate investment options given high currency risk. Some international investors also price in perceived political instability in Pakistan, further increasing minimum viable return for projects. These factors mean most of the bankable opportunities in Pakistan are unable to stand on their own relative to other countries. For local investors, the most significant challenges are credit risk and long-term tenure of climate projects. Given the relatively high return today on money market debt instruments, local banks perceive climate projects as riskier given all the preparatory and logistical work that goes into getting projects off the ground. Also, the long-term nature of these projects does not fit the business model of most banks, thereby limiting participation by the domestic private sector.

100. According to the World Bank, Pakistan's banking sector is not effectively delivering on its role as an intermediary of capital and facilitator of climate finance and green growth. The banking sector has pivoted an ever-greater percentage of its asset base towards the government in recent years, crowding out credit to the private sector. Credit to the public sector accounts for nearly 70% of all the credit extended by the banking sector[171]. For the remaining comparatively small portion lent to the private sector, banks take a conservative, highly risk averse approach. Consequently, only 20 business groups in the country account for 30% of the banking sector's private sector lending portfolio. This high level of risk aversion is also reflected in the banking sectors loan-to-deposit ratio of 46%, an extraordinarily low figure compared to economies such as Bangladesh (73%) or Turkey (92%)[172].

101. Furthermore, Pakistan's capital markets - both debt and equity - are relatively underdeveloped. The debt market is largely dominated by government securities while the corporate bond market is mainly used by a few large state-owned enterprises (SOEs), private corporations, and banks to meet their capital adequacy requirements. Pakistan's equity market is also relatively shallow, concentrated, and volatile. The market capitalization to GDP has decreased considerably from a high of 34% in FY08 to 17 percent in FY21, much below the level of Pakistan's peers[173]. Moreover, the Pakistan Stock Exchange (PSX) is dominated by a few sectors, mainly financial, energy, and fertilizer, which altogether account for 48 percent of PSX's market capitalization[174]. There have been promising developments recently with several new listings and the operationalization of the Growth Enterprise Market Board. However, the market is generally subdued due to overall low listings, tax driven distortions between asset classes, and preference of raising funds through banks instead of capital markets.

There is a need to strengthen institutions and implementation capacity to address climate change issues in Pakistan.

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[168] World Bank Group, Pakistan Development Update: Financing The Real Economy, April 2022, https://thedocs.worldbank.org/en/doc/410d0506bba8afc6fd9d9541148bfe4d-0310062022/original/PDU-April-2022-April18-ForWEB-Final.pdf
[169] ibid.
[170] ibid.
[171] ibid
[172] ibid
[173] World Bank Group, Pakistan Development Update: Financing the Real Economy, April 2022, https://thedocs.worldbank.org/en/doc/410d0506bba8afc6fd9d9541148bfe4d-0310062022/original/PDU-April-2022-April18-ForWEB-Final.pdf
[174] ibid

102. Despite the presence of several policy documents, such as the National Adaptation Plan and National Climate Change Policy, there is no clear articulation of climate priorities at national and provincial levels[175]. Pakistan today does not have well-defined sectoral action plans especially for adaptation and resilience. The National Adaptation Plan has high level adaptation priorities that are not broken down into regional and detailed sectoral priorities. There is no scientific basis to determine the top five adaptation and resilience measures required in Pakistan, for instance, based on detailed modelling of climate risk and impact across sectors. Similarly, high level sectoral targets and objectives exist in the NDCs for mitigation, however, these targets are not time bound and do not have detailed activities and roadmaps to ensure success.

103. In Pakistan, there is no centralized platform to access data on climate issues. The information that exists is scattered across several ministries, departments and agencies making it difficult to get a comprehensive view on interventions within the climate space. Without adequate data, potential financiers of climate projects are unable to participate in funding opportunities. In instances where there may be an appetite, the unavailability of data makes funding difficult.

104. There are different climate finance units with overlapping responsibilities at the federal level. Also, provinces are largely working in silos, with no systematic coordination with relevant ministries, and no alignment of activities with national strategies and climate roadmaps.

Reduced productivity in the agriculture sector has kept incomes low, leaving farmers vulnerable to external shocks

105. Despite accounting for 37.4% of employment in Pakistan the agriculture sector's value-added share of the economy has steadily declined from 25.6% in 2000 to 22.3% in 2022.[176] The multi-decade decline is due to low productivity growth within the sector: annual growth in agriculture value added per worker amounted to 0.8%, which is much lower than the average rate of 3.8% among Pakistan's structural and aspirational peers such as India, Egypt, Indonesia etc.[177] As a result of this low productivity, Pakistan persistently lags behind its regional neighbors in terms of crop yields, negatively impacting the country's food security. Currently, Pakistan is ranked 84 out of 113 countries in the Global Food Security Index.[178]

106. According to the Labor Force Survey, in 2020-2021 median annual income in Pakistan was USD 755. In contrast, the median annual income of agriculture workers in Pakistan was USD 629.[179] Disaggregating by gender, male agriculture workers earned nearly 3.5 times more than their female counterparts (Male ~ USD 734; Female ~ USD 210) despite women making up nearly 70% of the agricultural workforce.[180] Low wages for agriculture workers (and particularly for female workers) make them susceptible to falling into poverty due to external shocks.

107. As described in section B.1, Pakistan's agriculture sector is extremely vulnerable to the impacts of climate change events, such as heatwaves, droughts and floods. The combination of existing structural deficiencies and climate change induced shocks will significantly impact the livelihoods and food security for rural workers (two thirds of whom are dependent on the agriculture sector). The 2022 floods stand as an unfortunate example: the floods are estimated to have pushed 9.1M people into poverty, having a disproporationate impact on rural populations (poverty rates increased by 5.4% in rural areas versus 1.1% in urban areas).[181]

108. In accordance with the GCF's goals, ACAP's investments intend to bring direct income improvement to vulnerable farmers. Such benefits include increased income, safer and healthier household environments, improved food and water security, increased access to energy, and adoption of environmentally sustainable farming practices. The Fund goals connect directly to SDG *Goal 13* (Climate Action) and SDG *Goal 2* (Zero Hunger) and are particularly reflected in the sub-goals of *Goal 2.3* (Increased agriculture productivity and incomes for vulnerable producers), and *Goal 2.4* (sustainable food production and resilient agricultural practices). In addition, ACAP also intends to align its

[175] UK International Development, Accelerating Green and Climate Resilient Financing in Pakistan, November 2023, https://growthgateway.campaign.gov.uk/wp-content/uploads/sites/138/2023/11/231120_Accelerating_Green_Climate_Financing_Report_vFinal-003.pdf
[176] World Bank Group, World Development Indicators, 2023, https://data.worldbank.org/indicator/NV.AGR.TOTL.ZS?locations=PK
[177] World Bank Group, From Swimming in Sand to High and Sustainable Growth: A Roadmap to Reduce Distortions in the Allocation of Resources and Talent in the Pakistani Economy, 2022, http://hdl.handle.net/10986/38133
[178] The Economist Group, Global Food Security Index, 2022, https://impact.economist.com/sustainability/project/food-security-index/reports/Economist_Impact_GFSI_2022_Global_Report_Sep_2022.pdf
[179] PBS, Pakistan Labor Force Survey 2020-2021, https://www.pbs.gov.pk/sites/default/files/labour_force/publications/lfs2020_21/LFS_2020-21_Report.pdf
[180] ibid.
[181] Knippenberg et al, Quantifying the poverty impact of the 2022 floods in Pakistan, 2023, https://blogs.worldbank.org/developmenttalk/quantifying-poverty-impact-2022-floods-pakistan

outcomes with SDGs on Goal 1 (No Poverty); Goal 5 (Gender Equality); Goal 8 (Decent Work and Economic Growth); and Goal 11 (Sustainable Cities and Communities).

109. With its catalytic focus on agriculture, ACAP's investments will aim to create more resilient food systems which, in turn, will have a knock-on effect throughout the economy on livelihoods. Food security is a rising concern for Pakistan, given the 2022 catastrophic weather events and their impact on the yields of staple crops such as wheat (up to 30% of the total crop lost in the heatwave in spring 2022) and rice (over 600,000 acres lost in the most recent floods).

110. Moreover, ACAP's plan to invest in businesses increasing the climate resilience of farmers will greatly aid a country with a significant food insecurity issue. A 2018 national nutrition survey found that 36.9% of the Pakistani population faces food insecurity.[182] The average Pakistani household spends over 50% of their monthly income on food.[183] Extreme weather events, climate shocks, and climbing food prices will have devastating consequences for rural communities throughout Pakistan. ACAP seeks to serve as a vital lifeline for farmers, rural communities, and the nation.

111. ACAP will demonstrate how to develop effective private sector solutions for climate adaptation in a vulnerable frontier market. The Fund will be developing and deploying climate finance solutions for agribusinesses delivering adaptation impact in a highly vulnerable agrarian context while catalyzing significant follow-on private capital. The process of successfully developing the blended finance fund model, identifying and supporting best-in-class investments with a gender lens, attracting private capital and focusing on generating insights and stakeholder engagement will be a powerful example for the wider sector to learn from and replicate within Pakistan and beyond in similarly challenging environments.

112. By investing in the agriculture value chain and enabling ecosystem across ACAP's three investment themes, the Program intends to plug existing gaps in the fragmented agriculture value chain and invest in more climate resilient solutions. ACAP aims to improve food security as well as create stronger economic outcomes for farmers and businesses. With 90M people food-insecure and climate disasters impacting the agriculture sector with increasing severity every year, ACAP will target its investments towards sustainable food systems which cut across all sustainable goals.

D.5. Country ownership (max. 500 words, approximately 1 page)

Please describe how the beneficiary country takes ownership of and implements the funded project/programme. Describe the following:

- *Existing national climate strategy*
- *Existing GCF country programme*
- *Relevance to and alignment with existing policies such as Nationally Determined Contributions (NDCs), Nationally Appropriate Mitigation Actions (NAMAs), and National Adaptation Plans (NAPs)*
- *NDCs, NAMAs, and NAPs*
- *Capacity of Accredited Entities or Executing Entities to deliver*
- *Role of National Designated Authority*
- *Engagement with civil society organizations and other relevant stakeholders, including indigenous peoples, women and other vulnerable groups*

113. With almost two decades of experience in Pakistan, Acumen has engaged with a wide variety of stakeholders and is uniquely positioned to understand the complex challenges that farmers and the Pakistani public face. Acumen is encouraged by the Pakistani government's commitment to fighting climate change (through prioritizing agriculture as a key adaptation sector in addition to creating a National Adaptation Plan to mainstream adaptation across the economy). Pakistan, as a low-emitting, highly vulnerable country, has made climate change a critical priority. To sustainably and impactfully structure ACAP and ensure the Program remains in alignment with national policy priorities, the ACAP team actively engaged with a wide variety of stakeholders such as the Pakistan NDA for GCF,

[182] State Bank of Pakistan, The State of Pakistan's Economy, Third Quarterly Report, 2019, https://www.sbp.org.pk/reports/quarterly/fy19/Third/Special-Section-2.pdf
[183] World Food Programme, Country Brief: Pakistan, https://www.wfp.org/countries/pakistan#:~:text=An%20average%20Pakistani%20household%20spends%2050.8%20percent%20of%20monthly%20income%20on%20food.

and the Ministry of Climate Change, to ensure the project is developed in accordance with GoP's domestic and international policies and priorities. Acumen has been operating in Pakistan for over 16 years, and is deeply embedded in the local ecosystem, making it easier to work with relevant local governments, NGOs, and private sector partners to execute on investment and impact goals in partnership with key stakeholders.

114. Please find below a partial overview of stakeholder engagement:
- **Development Finance Institutions**
- **Government bodies**
- **Academic and research institutions**
- **Private sector**

115. ACAP is well aligned with respect to government policies. The Fund is aligned with key national strategic priorities of increasing agricultural yields, improving food security and water management, and is therefore aligned with the adaptation focus of improving smallholder farmer livelihoods. Pakistan's National Climate Change Policy, Climate Change Gender Action Plan, and National Agriculture & Food Security Policy all identify and establish risks posed by climate change to food, water and energy security and identify extreme vulnerabilities to climate change in the water resources, agriculture, forestry, and biodiversity sectors.

116. Over two years, the ACAP team has reached out to a diverse set of stakeholders to effectively develop the Fund and deeply embed it in the local context. Our team learned from government entities, investors, donors, corporates, academia, pipeline companies, industry associations, affiliated nonprofits, civil society organizations, and prospective beneficiaries. For example, speaking with government entities allowed us to understand the regulatory frameworks in place and also facilitated participation with pipeline companies, sector players, policy makers, and the local ecosystem. Engaging with private sector entities (large corporates, prospective pipeline companies) gave us line-of-sight into their day-to-day functioning and their respective roles in the agri sector. Consultants supported our understanding of local climate risks and the investment landscape.

117. In addition, the ACAP team sought the guidance and expertise of groups engaging with and representing vulnerable communities, women and indigenous peoples. Meetings with representatives from indigenous communities provided valuable insights into sustainable practices, cultural considerations, and the preservation of traditional knowledge. Engaging with women's groups facilitated a deeper understanding of gender dynamics, empowering women economically, and promoting gender equality in business ventures. Moreover, discussions with representatives of vulnerable groups shed light on the unique challenges faced by marginalized communities and allowed the Fund to tailor its investment strategy to address these issues.

Overview of Pakistan's Key Climate Policies:
Pre-2005

118. Preceding 2005, Pakistan had multiple discrete policies and laws in place which addressed different aspects of environmental protection and conservation. The country's approach at the time can be attributed to an array of historical factors, i.e., Pakistan's overriding commitment to advancing economic development, a lack of institutional capacity and resources committed to environmental protection and conservation, and political instability. Also, during this timeframe, Pakistan ratified numerous multilateral environmental agreements (MEAs) including the **Ramsar Convention** in 1975, **Montreal Protocol** in 1992, **UNFCCC** in 1994, **Basel Convention** in 1994[184], amongst others, that encouraged Pakistan to enact environmental policies and laws. Some of the documents introduced included the **National Forest Policy** (NFP) **1955[185], Pakistan Environmental Protection Ordinance** (PEPO) 1983[186], **National Environmental Quality Standards** (NEQS) 1992[187], **National Conservation Strategy** (NCS) in 1992[188]**, Pakistan Environmental Protection Act** (PEPA) in 1997[189], and so forth.

2005

119. Prior to 2005, several developments led to a need for introducing an all-encompassing and coherent environmental policy. First, rapid industrialization and urbanization resulted in acute environmental problems, such as air and water pollution, that required immediate addressing. Second, there was an increased recognition of

[184] MEAs
[185] NFP 1955
[186] PEPO 1983
[187] NEQS 1992
[188] NCS 1992
[189] PEPA 1997

environmental conservation and sustainability at the international level, with the UN labeling the 1990s as the "Decade of Environmental Awareness and Sustainability", substantially influencing policy and decision-making within the country. Third, Pakistan experienced multiple natural disasters leading up to 2005, such as the 1999 drought and 2001 earthquake, spurring a need to develop a framework for natural disaster preparedness and mitigation. As a result, the Ministry of Environment introduced the **National Environment Policy** (NEP) in 2005. It aimed to address the environmental issues faced by Pakistan – notably pollution of freshwater bodies, air pollution, natural disasters, deforestation, and climate change. The overarching goals of the policy included:

- Integrating environmental considerations in policy making and planning
- Capacity building of all stakeholders for improved environmental management
- Merging international obligations in line with national goals
- Integrating and implementing international/multilateral/regional agreements in line with MDGs
- Disaggregating the impact of the environment on women
- Public-private-civil society partnerships[190]

2012

120. In 2012, Pakistan devised its first **National Climate Change Policy** (NCCP), focusing on adaptation and mitigation efforts through a multisectoral approach. The preliminary goals of the policy were:

- Maintaining appropriate economic goals while addressing the challenges of climate change
- Integrating climate change policies with other interconnected national policies
- Engaging in pro-poor, gender sensitive adaptation and advancing cost-effective mitigation efforts
- Guaranteeing water, food, and energy security
- Curtailing the risks associated with a rise in the intensity and frequency of climate extreme events
- Reinforcing inter-ministerial decision making and coordination mechanisms on climate change[191]

2016

121. On becoming a party to the Paris Agreement, Pakistan developed its first climate action plan, the **Nationally Determined Contribution** (NDC). The NDC presents a strategy to cap the country's GHG emissions in line with Paris Agreement's objective to limit global temperature increase between 1.5-2°C. Furthermore, the document provides a roadmap for implementing nature-based solutions (NbS), developing land-use change and forestry initiatives, as well as building resilient community infrastructure.[192] Some of the NDC's guiding principles included:

- NbS green livelihood opportunities
- Improved cross-referencing to climate change in national and provincial policies and action plans on climate adaptation and mitigation
- Climate-informed preparatory and approval systems dealing with the life-cycle of projects and schemes
- Exploration of the market and non-market based approaches in diversifying the funding sources for commissioning capital intensive projects
- Promotion of opportunities for youth groups to engage in, and benefit from, Pakistan's adaptation and mitigation objectives and targets

2021

122. In 2021, the Federal Ministry of Climate Change updated its **National Climate Change Policy** (NCCP) and **Nationally Determined Contribution** (NDC). The revisions to the documents were made to address the rapidly changing climate context and, in the case of the NDC, to incorporate additional sectors such as blue carbon ecosystems, health, waste, gender, and youth, etc. Key parties involved in the execution of the policies and plans include the Ministry of Climate Change, Federal and Provincial EPAs, the National and Provincial Disaster Management Authorities, and the National Disaster Risk Management Fund. To ensure effective policy

[190] National Environment Policy
[191] National Climate Change Policy 2012
[192] Intended Nationally Determined Contribution

implementation and monitoring of all progress, an NCCP Implementation Committee was established in 2021 at the Federal and Provincial levels.[193]

123. While the NCCP focuses on both mitigation and adaptation, many of the 238 adaptation-related policy measures are either crosscutting, unclear, or piecemeal, and resultantly incorrectly identified as pure adaptation efforts or undermine the overall efficacy of the policy document.

124. For instance, policy measure 4.2 II h states, “Promote biotechnology in terms of more carbon responsive crops, improved breeds and production of livestock using genetic engineering.” The aforementioned policy measure is cross-cutting rather than pure adaptation as carbon-responsive crops will have enhanced carbon sequestration abilities and genetic engineering of livestock will also produce breeds that release less methane gas. Both these actions will reduce GHG emissions as well as develop varieties of crops and livestock that are more resilient to the impacts of climate change.

125. Furthermore, policy measure 4.2 II d states, “Promote appropriate technologies for small-scale irrigation, water reuse (waste/water recycling), and rain water harvesting, etc.” The policy measure is relatively vague in specifying how the Government aims to promote the adoption of such technologies. The methods for doing so are manifold and include financial incentives such as tax breaks, subsidies, or low-interest loans for farmers, technical assistance, demonstration farms to highlight potential benefits, and regulations on water use, etc.

2022

126. In 2022, the Ministry of Climate Change in collaboration with IUCN and with financial support from the GCF launched the **Climate Change Gender Action Plan** (ccGAP). Despite being highly susceptible to the adverse impacts of Climate Change and wielding profound knowledge of natural resources, women were left out of decision-making on climate change. The Climate Change Gender Action Plan was developed to fill this void by supporting policy measures and bolstering institutional processes that enhance women’s participation in climate decision-making and implementation. Measures taken by the ccGAP include the support of gender-inclusive policy dialogue, capacity-building mechanisms, and pilot projects for women.[194]

2023

127. In late July 2023, the Ministry of Climate Change, with the support of GCF and UNEP, launched Pakistan’s first **National Adaptation Plan (NAP)**. The plan provides a framework to mainstream climate change adaptation at the systemic, institutional, and individual levels and integrates it within policies, legislation, strategies, and regulations. The NAP will share guidance, lessons, and experiences at the national and sub-national levels while creating a strategy to implement, monitor, and convey adaptation benefits at these levels. Furthermore, it outlines actions to strengthen resilience and adaptive capacity in eight vulnerable sectors, namely Water Resources, Agriculture & Livestock, Forestry, Human Health, Biodiversity and other Living Ecosystems, Disaster Preparedness, Urban Resilience and Gender.

128. The NAP prioritizes the agriculture sector in its six adaptation priorities, highlighting the critical nature of the agriculture-water nexus (in a water scarce country like Pakistan) and the need for climate-smart agriculture practices. The NAP identifies the need for aligning the agriculture sector with priority interventions (such as Pakistan’s NDCs). Given the pockets of small-scale successes across the sector, the challenge faced by the country is the scaling of these solutions and encouraging adoption of potentially transformational strategies, especially among smallholder farmers. Therefore, the NAP stresses coordination mechanisms aimed at minimizing redundancy and harnessing synergies among diverse public and private institutions engaged in climate smart agriculture.

Policy Gaps

A comprehensive analysis of Pakistan’s climate commitments exposes the gaps that must be addressed, particularly on the adaptation side. These are summarized below:

129. As Pakistan is highly susceptible to the adverse impacts of climate change, it is paramount to prioritize a clear and coherent adaptation strategy. The damages inflicted upon the country by recurrent extreme climate events and the scale of human impact underline the critical need to emphasize climate adaptation. Despite this need, both the

[193] National Climate Change Policy 2021
[194] IUCN, Climate Change Gender Action Plan of the Government and People of Pakistan, 2022, Climate Change Gender Action Plan

National Climate Change Policy (NCCP) and the Nationally Determined Contributions (NDC) fall short of providing a coherent, step-by-step adaptation roadmap. Moreover, in addition to food, water, environmental, energy, and other types of security, climate change should also be framed as a national security concern. Neither document conceives the phenomena as such, thereby undermining the urgency of addressing it.

130. Given the scale of the problem facing Pakistan, it is critical to engage all stakeholders in the solution. The government's increasing lack of fiscal means to execute a comprehensive public sector response to recurring climate catastrophes underscores the critical need to engage the private sector. The current policy framework lacks guidance on favorable incentive structures for the private sector to engage in building climate-resilient businesses. As such, it is critical for Pakistan to not only divert public financing to adaptation measures but to also engage the private sector to contribute to the country's efforts to confront climate change.

131. Beyond private sector engagement, Pakistan needs to develop a multi-stakeholder response to the country's climate crisis. At this point, the NCCP and NDC do not envision a clear plan for engaging all relevant stakeholders in a national-level climate action plan. Relevant stakeholders are not only limited to the private sector, as discussed above, but include multiple players inside and outside Pakistan such as related ministries, DFIs, MDBs, Climate Investors, and NGOs.

ACAP Alignment with Country Level Priorities

132. By aiming to develop a climate-resilient agriculture sector with minimal crop losses and post-harvest food waste, ACAP's strategy attempts to align itself with Pakistan's food security targets (as outlined in Pakistan Vision 2025) ensuring adequate and stable provision of food supplies, especially amongst the country's most vulnerable population. Similarly, by investing in agribusinesses that employ smart farming techniques in the agri-production process, (such as precision irrigation), ACAP hopes to enable more effective utilization of water, minimize water wastage, and promote a practical and robust method for water conservation - bolstering the adaptation capacity of climate-vulnerable smallholders. This is aligned with the priorities identified in Pakistan's National Adaptation Plan; incentizing farmers to transition to climate-smart water and land management practices and modernizing surface and groundwater irrigation services to support transitions to climate-smart agriculture[195].

133. ACAP's investment in innovative processing and value-addition methods is also aligned with Pakistan's plan to reestablish its competitive advantage in the agriculture sector and transition towards producing high-value-added products with a capacity to generate substantial returns, both domestically and internationally. A key area identified in Pakistan's revised NDC is a need to develop and transition towards advanced technologies in the agriculture sector, specifically through the adoption of high-quality inputs such as drought-resistant seed varieties. One of ACAP's core strategic areas will address this need by connecting smallholder farmers across the value chain, through a digital platform, allowing them to have access to a variety of reliable, high-quality inputs amongst other services.

134. ACAP will function as one of the first private investment vehicles in the country to precisely map out an adaptation-focused approach to solving Pakistan's climate crisis. The Program will make use of rigorous impact metrics in an attempt to make a sizable impact on the livelihoods of climate-vulnerable farmers as well as on the economic outcomes of the country. From the beginning, ACAP's mandate has been to engage multiple stakeholders in the development of the Program, regularly convening all entities on a platform that builds on cross-sector synergies. The fund's success will also serve as a guide and provide valuable insights to the Government of Pakistan on how to develop an effective adaptation roadmap for building resilience in the agricultural sector.

Capacity of Accredited Entities and Executing Entities

135. Acumen, as AE, is committed to local country operations and expertise. Acumen has local offices in each of the regions where it invests (East Africa, West Africa, India, Pakistan, the United States, and Latin America), staffed by local employees that understand consumer needs and cultural context, and bring extensive local networks. Acumen's teams across the world work with relevant local government, NGO, and private sector partners to execute on its investment and impact goals and ensure alignment with local policies.

[195] Ministry of Climate Change and Economic Coordination, National Adaptation Plan, 2023, https://www.mocc.gov.pk/PublicationDetail/ZDg5NjgwMjMtMWVkZC00ODMyLTg1NzctNmQzODFhNTVmMWEz

136. ACAP Fund is also well aligned with the Strategic Plan for the Green Climate Fund 2024–2027.[196] ACAP as the first-of-its-kind in Pakistan scaling up adaptation capacity and building resilience, is well aligned with GCF priorities on addressing the adaptation finance gap and helping developing countries meet their NDC targets as well as "promoting innovation and catalysing green financing while increasing the share of funding allocated through the Private Sector Facility".

137. Acumen will play a strong role in engaging with the NDA.

NDA Relationship

138. Acumen has received a No Objection Letter from Pakistan's Ministry of Climate Change for ACAP. The team intends to continue engaging the Ministry of Climate Change for their expertise, guidance, monitoring, accountability, and engagement for this project. Acumen has outlined several of our commitments to the NDA and seeks to strengthen the relationship throughout the project lifecycle.

139. Acumen believes that the NDA will be an important partner in gathering stakeholders like CSOs, academic institutions, and public sector partners. We seek their guidance in disseminating and amplifying the insights we learn from this unique project.

Alignment with Climate Finance Mechanisms/Instruments

140. ACAP Fund's alignment with other climate finance funders is strong.

D.6. Efficiency and effectiveness (max`. 500 words, approximately 1 page)

Describe how the financial structure is adequate and reasonable in order to achieve the proposal's objectives, including addressing existing bottlenecks and/or barriers, and providing the minimum concessionality to ensure the project is viable without crowding out private and other public investments. Refer to section B.5 on the justification of GCF funding requested as necessary.

Please describe the efficiency and effectiveness of the proposed project/programme, taking into account the total financing and mitigation/ adaptation outcome the project/programme aims to achieve, and explain how this compares to an appropriate benchmark.

Please specify the expected economic rate of return based on a comparison of the scenarios with and without the project/programme.

IPlease specify the expected financial rate of return with and without the Fund's support to illustrate the need for GCF funding to illustrate overall cost effectiveness.

Please explain how best available technologies and practices have been considered and applied. If applicable, specify the innovations/modifications/adjustments that are made based on industry best practices.

141. Acumen has a strong track record of deploying and recycling capital as well as appropriate internal controls to monitor its investments and report to investors periodically. Through ACAP, Acumen plans to invest in businesses with a clear path to financial sustainability and expects to generate positive returns for investors as well as generate impact across the value chain.

142. As a blended finance fund, ACAP is targeting a 1:2 junior equity to senior equity structure (in order to de-risk investment into a critical sector). In doing so, Acumen aims to bring further investment into the country in both the climate change and agriculture sectors. Given investor risk perceptions on Pakistan, reflected in the 2023 downgrade by credit rating agencies (Moody's and Fitch) to reflect worsening liquidity and policy risks[197], the blended finance structure becomes even more critical in terms of unlocking capital for the Fund and enabling investments into climate focused agribusinesses.

[196] Green Climate Fund, Strategic Plan for the Green Climate Fund 2024–2027, 2023, https://www.greenclimate.fund/sites/default/files/document/strategic-plan-gcf-2024-2027.pdf
[197] https://www.fitchratings.com/research/sovereigns/fitch-downgrades-pakistan-to-ccc-14-02-2023

143. In terms of fund investors, ACAP intends to target a variety of investors ranging from DFIs, Multilateral Development Banks (MDBs), foundations to local and international commercial investors.

144. The blended finance structure ensures that senior equity is protected to attract prospective investors as it provides first loss cover via first loss equity/junior equity.

E. LOGICAL FRAMEWORK

*This section refers to the project/programme's logical framework in accordance with the **GCF's Integrated Results Management Framework** to which the project/programme contributes as a whole, including in respect of any co-financing.*

E.1. Project/Programme Focus

Please indicate whether this proposal is for a mitigation or adaptation project/programme. For cross-cutting proposals, select both.

☐ Reduced emissions (mitigation)

☒ Increased resilience (adaptation)

ACAP's theory of change provides a clear description of the intended impact that it hopes to achieve. The core impact theme is aligned with increasing adaptive capacity of farmers and the agricultural sector in Pakistan. Due to this, ACAP is categorized as a Adaptation project according to the GCF project description definitions.

For reporting purposes, where applicable, reduction in emissions or carbons sequestered will be measured. However, mitigation will not be a driver of the investment criteria, nor is the intention to invest into solutions that have a focus on emission reduction. Any emission reduction-based solutions will be co-benefits of the primary impact that is focused on adaptation.

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

This section of the logical framework is meant to help a project/programme monitor and assess how it contributes to the paradigm shift described in section D.2 above by applying three assessment dimensions - scale, replicability, and sustainability.

Accordingly, for each assessment dimension (see the definition per assessment in the accompanying guidance note), describe the current state (baseline) and the potential scenario (target) and rate the current state (baseline) by using the three-point-scale rating (low, medium, and high) provided in the guidance note. Also describe how the project/programme will contribute to that shift/ transformation under respective assessment dimensions (scale, replicability and sustainability). In doing so, please refer to section B.2(a) (theory of change).

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Agriculture is a key sector in Pakistan contributing 20% to GDP and accounting for	Low	By successfully implementing ACAP, scale will be achieved at three levels.	ACAP aims to mobilize USD 90M of catalytic risk tolerant capital over the next 10 years by providing financing solutions to innovative companies in the agriculture

	the livelihoods of 144M people living in rural areas of Pakistan. However, due to the complex macroeconomic conditions it has not attracted adequate capital to catalyze the sector towards climate adaptation. Risk tolerant patient capital invested into the sector is key to transforming the livelihoods of vulnerable businesses and smallholder farmers		1) The companies that receive the investment capital and Technical Assistance will be able to expand and grow both inclusively and sustainably, providing climate adaptation solutions across the sector. 2) The expanding companies will enable an increased number of farmers and supply chain workers to benefit from improved livelihoods - contributing to increased climate resilience. 3) Through the demonstration effect and effective stakeholder collaboration, ACAP will build awareness and collaboration towards climate adaptation within the agriculture sector in Pakistan	sector that contribute to the climate adaptation of smallholder farmers and vulnerable agri-dependent lives. The programme will be investing in 15-17 companies, impacting more than 13,055,893 direct and indirect beneficiaries. Finally, ACAP will be developing key insights and convening with key industry stakeholders to further amplify ecosystem level impact.
Replicability	Companies lack the necessary capital and support to effectively scale, and demonstrate a commercially viable, replicable model.	<u>Low</u>	Based on the blueprint demonstrated through the successful scaling of companies and the effective convening of stakeholders, there is an increased flow of critical capital into the sector from multiple providers, deploying much-needed funding at scale for climate adaptation across Pakistan.	ACAP will demonstrate how to develop effective private sector solutions for climate adaptation in a vulnerable frontier market. The Fund will be developing and deploying climate finance solutions for agribusinesses delivering adaptation impact in a highly vulnerable agrarian context while catalyzing significant follow on private capital. The process of successfully developing the blended finance fund model, identifying and supporting best-in-class investments with a gender lens, attracting private capital and focusing on generating insights and stakeholder engagement will be a powerful example for the wider sector to

				learn from and replicate within Pakistan and beyond in similarly challenging environments.
Sustainability	The lack of collaboration, knowledge sharing and availability of capital prevents companies operating in agriculture from being sustainable.	Low	The demonstration effect of scaling companies, while being intentional about knowledge sharing and convening with industry stakeholders, will ensure that ACAP supports the sustainability of the sector. ACAP will also deliver Technical Assistance and leverage its networks to ensure that companies are effectively supported such that their potential for growth and impact is optimized.	ACAP will be supporting companies through its investment and technical assistance to be sustainable post the intervention. Additionally, ACAP will implement best in class governance, institutional, and fiduciary frameworks to provide financing solutions to the sector. ACAP will also, where appropriate, execute on its exit strategies via trade sales such that the company can continue operating at an increasing scale via existing business models. Finally, ACAP will convene stakeholders in the agriculture sector to contribute to, and be aligned with, the national adaptation strategies of Pakistan to ensure the interventions contribute to a sustainable solution after the fund's lifetime.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

Select appropriate IRMF core and supplementary indicators to monitor project/programme progress. More than one IRMF (core and or supplementary) indicators may be selected as applicable for each GCF results area and project/programme outcome (as defined in the table in section B.2(b)). If IRMF indicators are unable to measure any given project/programme outcomes, project/programme-specific indicators should be developed under section E.5 (project/programme specific indicators).

GCF Result Area	IRMF		Baseline	Target	Assumptions / Note

	Indicator	Means of Verification (MoV)		Mid-term	Final[198]	
Total Programme Beneficiaries (Direct/Indirect)	Core 2: Direct and indirect beneficiaries reached	Internal: - Monitoring through annual ESG, impact reporting and reporting from ACAP Fund's portfolio companies. - ACAP Fund reporting consolidated figures annually. External: - Audit and review of portfolio companies reporting and data by independent consultants - User surveys	0	**Total Beneficiaries: 6,527,946** -Direct Beneficiaries: 996,633 -Indirect Beneficiaries: 5,531,313 -Male Beneficiaries: 3,329,456 -Female Beneficiaries: 3,198,490	**Total Beneficiaries: 13,055,893** -Direct Beneficiaries: 1,993,266 -Indirect Beneficiaries: 11,062,627 -Male Beneficiaries: 6,658,912 -Female Beneficiaries: 6,396,980	- Indirect beneficiaries are estimated using an average household size of 6.55. - Female representation is based on national demographics. - Assumptions: macroeconomic stability, no global financial crisis, no major pandemics, no distorting subsidies or market barriers are in place in recipient companies.
ARA1 Most vulnerable people and communities	Core 2: Direct and indirect beneficiaries reached	Internal: - Monitoring through annual ESG, impact reporting and reporting from ACAP Fund's portfolio companies.	0	**Total Beneficiaries: 6,527,946** -Direct Beneficiaries: 996,633 -Indirect Beneficiaries:	**Total Beneficiaries: 13,055,893** -Direct Beneficiaries: 1,993,266 -Indirect Beneficiaries:	- The direct and indirect beneficiaries for ARA1 and ARA2 are the same. - It is assumed that the climate resilience of beneficiaries will be enhanced through their use

[198] The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

		- ACAP Fund reporting consolidated figures annually. External: - Audit and review of portfolio companies reporting and data by independent consultants. - User surveys		5,531,313 -Male Beneficiaries: 3,329,456 -Female Beneficiaries: 3,198,490	1,1062,627 -Male Beneficiaries: 6,658,912 -Female Beneficiaries: 6,396,980	of the products and services provided by portcos. In addition beneficiaries will have access to improved climate resilient livelihoods. - Female representation is based on national demographics. - Assumptions: macroeconomic stability, no global financial crisis, no major pandemics, no distorting subsidies or market barriers are in place in recipient companies.
	Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options	Internal: - Monitoring through annual ESG, impact reporting and reporting from ACAP Fund's portfolio companies. - ACAP Fund reporting consolidated figures annually. External: - User surveys - Audit and review of portfolio companies	0	**Total Beneficiaries: 6,495,631** -Direct Beneficiaries: 991,699 -Indirect Beneficiaries: 5,503,931 -Male Beneficiaries: 3,313,989	**Total Beneficiaries: 12,991,261** -Direct Beneficiaries: 1,983,399 -Indirect Beneficiaries: 11,007,863 -Male Beneficiaries: 6,627,978	- It is assumed that beneficiaries will be able to enhance their climate resilience as a result of improved climate resilient livelihood options. - Female representation is based on national demographics.

		reporting and data by independent consultants		-Female Beneficiaries: 3,181,642	-Female Beneficiaries: 6,363,283	
	Supplementary 2.5: Beneficiaries (female/male) adopting innovations that strengthen climate change resilience	Internal: - Monitoring through annual ESG, impact reporting and reporting from ACAP Fund's portfolio companies. - ACAP Fund reporting consolidated figures annually. External: - User surveys	0	**Total Beneficiaries: 32,752** -Direct Beneficiaries: 5,000 -Indirect Beneficiaries: 27,751 -Male Beneficiaries: 16,842 -Female Beneficiaries: 15,910	**Total Beneficiaries: 65,503** -Direct Beneficiaries: 10,000 -Indirect Beneficiaries: 55,503 -Male Beneficiaries: 33,684 -Female Beneficiaries: 31,819	- It is assumed that all direct beneficiaries have adopted innovations that strengthen climate change resilience which will be climate adaptive as required through the investment criteria - Female representation is based on national demographics.
ARA2 Health, well-being, food and water security	Core 2: Direct and indirect beneficiaries reached	Internal: - Monitoring through annual ESG, impact reporting and reporting from ACAP Fund's portfolio companies. - ACAP Fund reporting consolidated figures annually.	0	**Total Beneficiaries: 6,527,946** -Direct Beneficiaries: 996,633 -Indirect Beneficiaries: 5,531,313	**Total Beneficiaries: 13,055,893** -Direct Beneficiaries: 1,993,266 -Indirect Beneficiaries: 1,1062,627	- Fund beneficiaries are calculated using assumptions within the Impact Model (Annex 22a), pipeline companies as representative for company breadth of impact and growth forecast. - The direct and indirect beneficiaries for ARA1 and ARA2 are the same.

		External: - User surveys - Audit and review of portfolio companies reporting and data by independent consultants		-Male Beneficiaries: 3,329,456 -Female Beneficiaries: 3,198,490	-Male Beneficiaries: 6,658,912 -Female Beneficiaries: 6,396,980	- It is assumed that the well being (increased income, food security etc) of beneficiaries and their households will improve as a result of their access to the products and services provided by portfolio companies. - Female representation is based on national demographics. - Assumptions: macroeconomic stability, no global financial crisis, no major pandemics, no distorting subsidies or market barriers are in place in recipient companies.
	<u>Supplementary 2.2: Beneficiaries (female/male) with improved food security</u>	Internal: - Monitoring through annual ESG, impact reporting and reporting from ACAP Fund's portfolio companies. - ACAP Fund reporting consolidated figures annually.	0	**Total Beneficiaries: 1,628,723** -Direct Beneficiaries: 248,660 -Indirect Beneficiaries: 1,380,063	**Total Beneficiaries: 3,257,445** -Direct Beneficiaries: 497,320 -Indirect Beneficiaries: 2,760,125	- Female representation is based on research conducted in the gender analysis and national demographics. - Responses could be impacted by

		External: - User surveys - Insights project(s) to determine food security benefits, either directly or through proxy of income improvement		-Male Beneficiaries: 830,699 -Female Beneficiaries: 798,023	-Male Beneficiaries: 1,661,399 -Female Beneficiaries: 1,596,047	1. Higher cost of living due to non-food inflation; 2. Conflicts and/or other external disruptions leading to food insecurity. 1) Perceptions of food security from types of food consumed at the time of the survey. This could include seasonal foods available, among others
	Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice	Internal: - Monitoring through annual ESG, impact reporting and reporting from ACAP Fund's portfolio companies will indicate the number of hectares of natural resources brought under improved practices - ACAP Fund reporting consolidated figures annually. External: - Independent surveys to measure the hectares of land brought under	0	203 Hectares	405 Hectares	- It is assumed that the quality of life of smallholder farmers who are supplying to these archetypes and rural workers being employed will improve as a result of increased incomes. - No perverse incentives (policies, prices, monoculture industries that affect natural capital) are introduced in the area. - The area is not seriously disrupted by a major extreme climate event, affecting restored areas, before resilience measures have been fully adopted

		climate resilient practices				

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)					
Select at least two relevant IRMF core (enabling environment) indicators to monitor and elaborate the baseline context and project/programme's targeted outcome against the respective indicators. Rate the current state (baseline) vis-à-vis the target scenario and select the geographical scope of the outcome to be assessed. Describe how the project/programme will contribute towards the target scenario. Refer to a case example in the accompanying guidance to complete this section.					
Core Indicator	**Baseline context (description)**	**Rating for current state (baseline)**	**Target scenario (description)**	**How the project will contribute**	**Coverage**
Core Indicator 5: Degree to which GCF projects/programmes contribute to strengthening institutional and regulatory frameworks for low-emission climate-resilient development pathways in a country-driven manner	The institutional and regulatory framework for climate resilient development is currently very limited in Pakistan. Private sector players have limited awareness and access to multi-stakeholder platforms addressing climate change impacts, and do not have the structures or skills to respond in a timely manner. Therefore, it is currently challenging to create, develop and support adaptation financing	Low	Institutional and regulatory frameworks in place that incentivize and facilitate the channeling of domestic and international climate finance for adaptation that targets poor and vulnerable people in Pakistan. These frameworks should cover key sectors such as agriculture, social protection, urban planning, and natural resources and environment.	By engaging with climate and agriculture sector stakeholders (public, private, CSOs, etc.), ACAP aims to create pathways for scaling climate resilient development to improve adaptation outcomes in Pakistan. Through demonstrating successful investments in climate resilient businesses, ACAP will be showcasing how funding initiatives and strategies that directly identify and respond to climate change	National

	measures that build national capacity to respond to climate induced shocks. Currently, the gap in Pakistan's climate financing needs stands at 8% of GDP and will need a holistic, multisectoral approach to address Pakistan's growing adaptation needs.			challenges can be structured in Pakistan. Additionally, ACAP's knowledge sharing and convening activities will seek to disseminate learnings and contribute to private sector players fully understanding their role in addressing climate change.	
Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level	Pakistan's agriculture sector is of major significance to its economic position. Due to the current frequency of climate shocks and the macroeconomic conditions, the sector is not seeing sufficient capital to adapt and scale appropriately. ACAP, through its demonstration effect and its convening power, will strengthen the climate resilience of the agriculture sector.	Low	Strong financial markets that are able to provide sufficient capital, appropriately risk tolerant, to enable companies to be established and grow to support the climate resilience of smallholder farmers. Increased capacity and willingness of financial institutions to provide customized financing solutions to companies/customers in the space is developed.	ACAP seeks to provide innovative financing solutions to best-in-class agribusinesses building climate resilience across Pakistan. The funding will not only support the growth of these critical business models directly but also has the potential to be replicated by other financiers in order to further amplify market impact.	National
Core indicator 8: Degree to which GCF investments contribute to effective knowledge	Despite significant climate vulnerability and the central importance of Agriculture in Pakistan, there is little	Low	Knowledge of financing products, technical assistance and best practices within each	ACAP aims to provide innovative financing solutions not currently available in the market	National

generation and learning processes, and use of good practices, methodologies and standards	collaboration and knowledge sharing between stakeholders that builds the sector towards more climate resilience. ACAP aims to make investments and create and disseminate insights to ensure knowledge is distributed and best practice is applied throughout the sector. ACAP will also regularly convene sector stakeholders to accelerate impact, amplify learnings and create opportunities for engagement.		sub sector are developed. Sector-wide improved collaboration and alignment on climate resilience. Collaboration between different stakeholders towards developing scalable solutions for climate adaptation.	and help develop new products that have the potential to be replicated. Investee companies will also be exposed to the implementation of good standards in environmental, social (including gender) and fiduciary elements. The Fund seeks to have monitoring and evaluation systems in place to identify lessons learned. ACAP's management team intends to also participate in forums of knowledge sharing to support regulations and policy in the space	

E.5. Project/programme specific indicators (project outcomes and outputs)

*This section should list out project/programme-specific performance indicators (outcomes and outputs) that are not covered in sections above (E.1-E.4). List down tailored indicators to monitor /track progress against relevant project/programme results (outcomes/outputs). AEs have the freedom to decide against which outcomes they would like to set project/programme specific indicators. If any co-benefits are identified in sections B.2(a)(b), and D.3, AEs are encouraged to add and monitor co-benefit indicators under the "**Project/programme co-benefit indicators**" section in table below. Add rows as needed.*

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	

Outcome 1 Improved Farmer and Vulnerable, Agri-dependent Livelihoods	*Percentage reporting income improvement*	*Portfolio data (as provided in Annual Reports), informed by company sales data* *User Surveys*	*0*	*70%*	*74%*	*Farmers are accessible for surveys* *Income improvement could include several indicators such as income increase, reduction in time spent at work, improved access, reliability of income, consistency of income, etc.* *We expect that ACAP investments will enable, across portfolio company customers, employee or supplier base an improvement in incomes by 70%. We will further refine portfolio impact and target to reach 74% by the end of the fund.* *The target of 74% is derived from a benchmark provided by 60 Decibels from their analysis of agriculture companies in Asia. Actual numbers for Pakistan may vary.*
	Number of Direct Jobs created	*Portfolio data (as provided in Annual Reports), informed by company sales data*	*0*	***Total Direct Jobs Created*:*** *792*	***Total Direct Jobs Created*:*** *1,583*	*As companies grow, due to the investment made by ACAP, additional human resources are required to support this*

				Male: 607 *Female: 184*	*Male: 1,140* *Female: 443*	*growth, which will result in new jobs being created.* *Female and male targets have been rationalized based on their labor force participation rate for Pakistan*[199]*.* **Indicative impact target. To be further iterated upon with guidance from local agribusinesses and experts in the field.*
Outcome 2 Improved Climate Adaptive Capacity of Farmers	*Percentage of farmers reporting increased access to credit, improved inputs, weather information or extension services*	*Portfolio data (as provided in Annual Reports), informed by company sales data* *User Surveys*	*0*	*75%*	*80%*	*The total number of farmers reached have their climate adaptive capacity improved due to improved access to services, increased incomes, and improvement of knowledge through the delivery of the products and services from portfolio companies.*
	Percentage of farmers that perceive themselves as being more equipped to absorb a climatic shock due to the	*User Surveys*	*0*	*55%*	*60%*	*Farmers are accessible for surveys* *Sampling process provides a representative*

[199] World Bank Group, World Development Indicators Database, last accessed January 19, 2024, https://data.worldbank.org/indicator/SL.TLF.TOTL.FE.ZS?locations=PK

	company's product/service					*group of farmers that can absorb climate shocks.* *The assumption is made that farmers with improved incomes, will at a minimum, be more equipped to absorb climate shocks.* *The target of 60% is derived from a benchmark provided by 60 Decibels from their analysis of agriculture companies in South Asia. Actual numbers for Pakistan may vary.*
Outcome 3 Improved Farmer Health & Food Security	*Percentage of farmers reporting increased food production (yield)*	*User Surveys*	*0*	*65%*	*72%*	*The assumption is that farmers availing the products and services provided by portcos will see an improvement in their yields which will strengthen their food security.* *The target is 72% (a benchmark provided by 60 Decibels from their analysis of agriculture companies in Asia). Actual numbers for Pakistan may vary.*

Outcome 4 Demonstrated Commercial Viability of Business Models Attracting More Private Capital	*Number of portfolio companies receiving co investment*	*Portfolio data (as provided in Annual Reports)*	*0*	*2*	*6*	*Regulatory environment and macro-economic conditions remain enabling for additional capital to be crowded in.*
	Number of portfolio companies receiving follow on capital investment.	*Portfolio data (as provided in Annual Reports)*	*0*	*3*	*7*	*macro-economic conditions remain enabling for additional capital to be crowded in.*
	Multiple for Follow-on capital	*Portfolio data (as provided in Annual Reports)*	*0*	*TBD*	*TBD*	*Assumptions:* *Regulatory environment and macro-economic conditions remain enabling for additional capital to be crowded in.* *Follow on will not be determined within the first year of investment. This will be determined during the course of the fund.*
Outcome 5 **Scalable, Replicable Fund Models Developed for Climate Adaptation**	*Number of climate adaptation focused investment programmes developed for Pakistan or other agri dependant climate vulnerable frontier countries*	*Insights Report*	*0*	*1*	*3*	*Assumptions:* *Global macroeconomic and policy environment remain enabling and conducive to investing* *Assumes that climate change and adaptation efforts remain critical to institutional investors.*

Outcome 6 **Enabling environment for adaptation solutions supporting climate resilient agriculture**	*Number of new climate adaptation partnerships developed between stakeholders in Pakistan or regionally*	*Press releases, industry analysis and insights reports*	*0*	*TBD*	*TBD*	*Local stakeholders are open and collaborative to work together towards sector level adaptation outcomes.* *ACAP's demonstration of successful adaptation investments are catalytic in aligning stakeholders.*
Output 1.1 ACAP Fund established	*Establishment of ACAP Fund* *Fully Fundraised*	*Registration documentation* *USD 90M fundraised*	*-*	*Registration documentation* *USD 90M fundraised*	*Registration documentation* *USD 90M fundraised*	*Legal and regulatory environment will enable and facilitate establishment and fundraising*
Output 1.2 Fund capital disbursed to venture, early-growth growth-stage agribusinesses	*Funds disbursed to portfolio companies (In USD)*	*Disbursement note (bank transfer statement)*	*0*	*USD 66,770,000 Disbursed to 15-17 portfolio companies across ACAP's three strategic investment themes*	*USD 66,770,000 Disbursed to 15-17 portfolio companies across ACAP's three strategic investment themes*	*Legal and regulatory environment will enable and facilitate investments Fund model assumptions hold true There are sufficient investable companies*
	Number of pipeline companies sourced	*Internal documentation and pipeline tracker*	*0*	*Total: 250*	*Total: 250*	*The investment pipeline is a dynamic list of companies that meet the ACAP investment criteria. This list will continue to change as new companies are identified.* *Based on the pipeline companies, the most promising companies are*

						selected for full due diligence, according to the investment process. The companies that are approved by the Investment Committee, will receive investment capital. *Pipeline sourcing activities will be concentrated during the investment period up to Year 5.*
Output 1.3 Strengthened business performance of portfolio companies	*Number of companies reporting increased revenue growth*	*Portfolio data (as provided in Annual Reports), informed by company sales data Portfolio company audited financials*	*Current value of company revenue before ACAP investment*	*70%*	*80%*	*Capital from ACAP's investment will unlock the company's growth and enable increased revenues Assumption: Regulatory environment and macro-economic conditions remain enabling for additional capital to be crowded in.*
Output 2.1 TA deployed across strategic themes to maximize portfolio viability and fund impact	*Number of TA transactions*	*Portfolio data (as provided in Annual Reports),* *Disbursement notes; TA reports*	*0*	*7*	*17*	*Companies that are identified for TA have the capacity to accept TA and deploy it effectively* *This assumes that ACAP invests in 17 portfolio companies.*
	Amount of TAF disbursed (USD)	*Disbursement notes; TA reports*	*0*	*USD 5M*	*USD 10M*	*Companies that are identified for TA have the*

						capacity to accept TA and deploy it effectively *This assumes that ACAP invests in 17 portfolio companies.*
Output 2.2 Monitoring and evaluation of the financial performance & climate impact of investments	*Number of Fund APRs*	*ACAP Fund APRs (Annual Progress Reports)*	*0*	*5*	*10*	*Investees adhere to investment contracts, including the requirement to report on key performance indicators* *These are fund level progress reports and will be carried out across the entire portfolio capturing financial and impact performance*
	Number of annual financial audits of portfolio companies	*Third party audits of company financials*	*0*	*42*	*85*	*As per the fund model we assume a holding period of 5 years per investment.* *This assumes that ACAP invests in 17 portfolio companies.*
Output 2.3 Increase in inclusivity and climate impact of portfolio companies	*Number of Gender Action Plans created for portfolio companies*	*TA reports*	*0*	*17*	*17*	*Companies have the capacity to accept TA and deploy it effectively* *Investees adhere to investment contracts, including the requirement*

						to report on key performance indicators
	Number of ESG action plans developed and (successfully) implemented	*TA reports ACAP Fund ESG Reporting*	*0*	*17*	*17*	*Investees adhere to investment contracts, including the requirement to report on key performance indicators* *This assumes that ACAP invests in 17 portfolio companies.*
Output 3 Establish platforms to catalyze stakeholder engagement, collaboration on & awareness of climate adaptation	*Number of events held to convene stakeholders*	*TA Reports*	*0*	*2*	*5*	*Regulatory environment and macro-economic conditions remain enabling for additional capital to be crowded in*
	Number of insights reports/academic articles published and disseminated	*TA Reports*	*0*	*5*	*13*	*Local stakeholders are open and collaborative to work together towards sector level adaptation outcomes*
Project/programme co-benefit indicators						
Co-benefit 1 Enhanced Gender Inclusivity	*TAF deployed to support agri dependent rural women and female run agribusinesses*	*Insights Report*	*0*	*1*	*2*	*It is assumed that ACAP's gender-focused TA activities will enhance the skills development and entrepreneurial participation of women in the agriculture sector.*
Co-benefit 2 Reduced GHG emissions	*Tons CO2eq avoided or sequestered*	*Portfolio data (as provided in Annual Reports), informed by company sales data*	*0*	*47,000 tCO2eq*	*95,608 tCO2eq*	*Pakistan's grid emission factor is assumed at 0.515.* *Food loss mitigation is assumed at 3.4kg CO2*

						per kg of food for rice and 0.77kg of CO2 per kg of food for maize. Farming practices result in 0.3 tons CO2eq per ha. Sequestered over the life of the investment

E.6. Project/programme activities and deliverables

All project activities should be listed here with a description and sub-activities. Please number the activities as shown below to indicate association of activities to the related outputs provided above in section E.5. Similarly, please number sub-activities as shown below to associate to the related activity.

Activities	Description	Sub-activities	Deliverables
Activity 1.1: Establish ACAP legal & operational structure	Target fund raise of USD 80M in investable capital toward establishing ACAP Fund.	Sub-activity 1.1.1: Raise USD 80M in Senior and Junior equity Sub-activity 1.1.2: Develop ACAP Operational Manual	Total of USD 80M raised successfully Operational Manual submitted
Activity 1.2: Identify pipeline companies across prioritized themes	Fund portfolio team to identify companies that operate in Pakistan under ACAP's investment themes.	Sub-activity 1.2.1: Identify Pipeline Companies	Reviewed up to 250 opportunities
Activity 1.3: Deploy capital to investee companies	Fund team to diligence and define appropriate investment terms per investee company	Sub-activity 1.3.1: Deploy investment capital to portfolio companies Sub-activity 1.3.2: Manage exits and closures of portfolio companies	Fund capital deployed to 15-17 portfolio companies Exits and legal closures of 15-17 portfolio companies
Activity 1.4: Monitoring, reporting & verification of portfolio company performance and impact	Post-investment verification of impact and company performance will be rigorously monitored. The Fund will also submit a mid-term and a final report on the same.	Sub-activity 1.4.1: Report on Fund Level financial performance and impact	10 Fund APRs (Annual Performance Reports): Including 1 mid-term and 1 final report
Activity 2.1: Establish TA Facility	ACAP aims to provide a USD 10M grant-funded Technical Assistance Facility (TAF) to further support portfolio companies, smallholder	Sub-activity 2.1.1: Establishment and Management of ACAP TAF	USD 10M TAF raised Members of TAF Committee finalized TA Team hired TA Operational Manual submitted

	farmers and vulnerable agri-dependent livelihoods.	Sub-activity 2.1.2: TAF Reporting	10 TAF APRs: Including 1 mid-term and 1 final-term evaluation
Activity 2.2: Provide targeted TA across five components: i. Climate Adaptation ii. Gender Initiatives iii. Business Development iv. ESG Reporting v. Impact Evaluation	ACAP Fund will provide TAF support across 5 critical dimensions.	**2.2.1 Climate Adaptation** Sub-Activity 2.2.1.1: Support activities increasing farmer/rural communities awareness of climate resilient agriculture practices and products Sub-Activity 2.2.1.2: Support activities increasing agribusiness awareness of climate resilient practices and risks Sub-Activity 2.2.1.3: Execute programmes to accelerate early-stage agribusinesses building climate resilience **2.2.2. Gender Initiatives** Sub-activity 2.2.2.1: Support activities to promote skills development and income enhancement of women in the agriculture sector Sub-activity 2.2.2.2: Support initiatives to increase entrepreneurial participation of women in the agriculture sector	**2.2.1. Climate Adaptation** Up to 20 trainings/grants Up to 17 trainings/grants 2 agri business accelerators **2.2.2 Gender Initiatives** 20 trainings/workshops conducted 5 grants to female agri-entrepreneurs deployed 15-17 gender action plans submitted (1 per portfolio company); gender trainings provided as needed Full time Gender expert and analyst hired **2.2.3. Business Development**

		Sub-Activity 2.2.2.3: Support companies to set, implement, and evaluate gender action plans and provide company level gender trainings Sub-activity 2.2.2.4: Recruit a full-time gender expert to ensure that gender is embedded across ACAP investment processes and generate gender insights **2.2.3. Business Development** Sub-Activity 2.2.3.1: Support companies with financial management and governance Sub-Activity 2.2.3.2: Support companies on improving investment readiness including marketing, data room management, and valuation support Sub-Activity 2.2.3.3: Support companies with the development of human resources processes and talent development and acquisition	TA funded support on financial management and governance provided to each of the 15 - 17 portfolio companies through the life of the Fund TA funded support on enhancing investor readiness provided to each of the 15 -17 portfolio companies through the life of the Fund TA funded support on HR/Talent provided to each of the 15 -17 portfolio companies through the life of the Fund 5 training sessions targeting distressed companies provided, as needed during the life of the Fund **2.2.4. ESG Reporting** TA funded support on ESG action plans provided to each of the 15 -17 portfolio companies 9 Environmental and Social Impact Assessments supported, as needed when companies engage in higher risk activities ESG associate hired

		Sub-Activity 2.2.3.4: Support distressed companies with access to recovery best practices **2.2.4. ESG Reporting** Sub-Activity 2.2.4.1: Support companies to set, implement and monitor their climate and ESG action plans Sub-Activity 2.2.4.2: Support companies with Environmental and Social Impact Assessment when companies engage in higher risk activities Sub-Activity 2.2.4.3: Hire ESG associate for monitoring ESG outcomes for portfolio companies **2.2.5. Impact Evaluation** Sub-Activity 2.2.5.1: Pre-investment impact assessment of pipeline companies Sub-Activity 2.2.5.2: Support companies with post-investment customer surveys to enhance impact awareness and solutions	**2.2.5. Impact Evaluation** 30 impact evaluations of high potential pipeline companies 17 post investment customer surveys conducted 5 deep dives into portfolio companies conducted 17 grants deployed/trainings conducted

		Sub-Activity 2.2.5.3: Support portfolio companies with post-investment detailed customer surveys to enhance impact awareness and solutions Sub-Activity 2.2.5.4: Support companies with development of impact evaluation processes and systems	
Activity 3.1: Generating sector insights & knowledge products	ACAP will generate insights and knowledge products incorporating all investment related learnings for stakeholders to learn from and replicate within Pakistan	Sub-Activity 3.1.1: Support academic research and insights into climate adaptation in Pakistan Sub-Activity 3.1.2: Hire an insights expert to develop sector insights and knowledge products Sub-Activity 3.1.3: Generate comprehensive insights reports to disseminate fund level learnings from investing in climate resilience in Pakistan Sub-Activity 3.1.4: Develop industry research on climate adaptation, environmental safeguarding, and specific impact areas to develop sector best practices for Pakistan	5 academic grants deployed Insights expert hired 10 insights reports generated 3 research reports generated
Activity 3.2: Convening & engaging stakeholders & co-investors to amplify impact	ACAP will convene stakeholders to enable meaningful dialogue with relevant stakeholders to build adaptation solutions for Pakistan at scale	Sub-Activity 3.2.1: Convene stakeholders including public and private sector leaders, agricultural leaders, academics, CSOs, and entrepreneurs to share best practices and develop knowledge networks	5 stakeholder events convened

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

Besides the arrangements (e.g. annual performance reports) laid out in Accreditation Master Agreement (AMA), please give a summary of the project/programme specific arrangements for monitoring, reporting and evaluation including a description of the monitoring and reporting system that will be used to assess the climate results of the proposed activity. Please also summarize the types of interim and final evaluations. Describe Accredited Entity (AE) project reporting relationships, including to the National Designated Authority (NDA)/Focal Point and between AE and Executing Entity (EE) as relevant, identifying reporting obligations from the EE to the AE. This should relate to the frequency of reporting on project indicators, implementation challenges and financial status. Please note that interim and final evaluations are expected to embed an assessment of project/programme's contributions to a paradigm shift and enabling environment using a three-point scale rating. Refer to the guidance note for the summary requirements and factor in additional evaluation /assessment activities under this section accordingly.

145. ACAP seeks to create eligibility criteria and methodologies, right-sized for the scale and resources available, pending any further iterations of the logic framework indicators after further engagement with the Green Climate Fund, relevant stakeholders, or other investors.

146. The following outlines the reporting schedule that ACAP will follow for GCF:

1. Annual Fund impact reporting
2. ESG reporting
3. Mid-project evaluation
4. Final evaluation

1. Annual Fund Report: Impact Reporting

The team will seek to ensure that investments are made into Companies that are aligned with ACAP's impact thesis and whose goals are aligned directly with building the climate resilience of their beneficiaries. This will be done through the due diligence process which will determine the impact, finance and values of the company. The activities during diligence will serve as the baseline for the subsequent monitoring of the company's financial and impact performance.

To track that the Fund is achieving its climate and impact objectives, the ACAP team will use a combination of company data, external data and market related data through company reports, annual review and impact studies. The ACAP team will conduct periodic surveys and audits to ensure the data received from annual reporting from the company is accurate and up to date. Importantly, ACAP ensures that it invests in companies that are impact aligned in terms of climate resilience and adaptation impacts. In the investment legal documents, ACAP requires its companies to report on impact data and the companies also enter into binding covenants to act in accordance with ACAP's impact principles.

As part of the external data collection, the Fund will also initiate various Lean Data surveys (or similar projects) over the course of the Fund's life to evaluate improvements in climate resilience among beneficiaries, including improved income levels, income stability, ability to bounce back from climate shocks, and other metrics. The Fund will consolidate data for analysis and form insights at Fund level on a regular basis, reporting to GCF as appropriate. This will help track progress against set objectives and enable the Fund to provide timely feedback to investee companies where realignment in their business strategy is required.

2. ESG Reporting

ACAP's investors will receive an annual report on the ESG performance of its investee companies. ACAP will actively monitor ESG matters in portfolio companies post-investment. This will typically include (i) monitoring development of ESG risk areas, (ii) assessing progress made on the ESG Action Plan and (iii) evaluating any changes in the business, which might create additional ESG risks.

Additionally, if any project is deemed a Category B investment, then the Fund will disclose a public version of an Environmental and Social Impact Assessment. The Fund will also disclose major ESG incidents and follow-on activity with investors.

On an annual basis, the Fund will conduct an independently run ESG review of each portfolio company. The purpose of the annual review is to assess ESG performance of the portfolio companies, identify those companies facing challenges implementing ESG action plans (as well as those with superior performance), and identify new ESG risks related to the business models. Insights gathered will foster detailed discussion amongst the investment team and the IC to determine how to best support the portfolio companies going forward. Further, the Fund will conduct both a mid-point and completion date review of the action plans to assess progress with the adoption of ESG gaps identified during the diligence stage and note additional gaps/inform additional initiatives to close on action items.

3. Mid-term evaluation

At the time of the Mid-term evaluation, ACAP expects some development results may have been achieved, since the portfolio will not have achieved full maturity. Therefore, the implementation of the investment strategy will be the focus of the evaluation and will also cover aspects like the number of portfolio companies invested by the Fund, Technical Assistance delivered, private sector capital catalyzed and total number of people served/impacted. Mid-term evaluation will be conducted by an independent evaluator.

4. Final evaluation

The Final evaluation will measure ACAP's ability to achieve the paradigm shift, outcomes, climate resilience benefits, country ownership, gender goals, and other factors outlined in the funding proposal both with the equity and grant components of GCF's contribution. The evaluation will likely include data analysis, focus group interviews, portfolio company interviews, and other evaluation activities. The final evaluation will be conducted by an independent evaluator.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Please describe financial, technical, operational, macroeconomic/political, money laundering/terrorist financing (ML/TF), sanctions, prohibited practices, and other risks that might prevent the project/programme objectives from being achieved. Also describe the proposed risk mitigation measures. Insert additional rows if necessary.

For probability: High has significant probability, Medium has moderate probability, Low has negligible probability

For impact: High has significant impact, Medium has moderate impact, Low has negligible impact

Prohibited practices include abuse, conflict of interest, corruption, retaliation against whistleblowers or witnesses, as well as fraudulent, coercive, collusive, and obstructive practices

Selected Risk Factor 1

Category	Probability	Impact
Macroeconomic	Medium	Medium

Description

Currency depreciation, Inflation, and Economic Downturns

Devaluation of the Pakistani rupee (PKR) can impact the dollar value of fund investments and compromise returns. As a single country climate focused fund, currency devaluation can have significant and multifaceted implications on expected returns of portfolio companies. The devaluation of the Pakistani rupee (PKR) can directly trigger a significant increase in the cost of doing business for companies that are import-dependent in their supply chains or for meeting capital expenditure needs. The devaluation impact can also worsen due to the impact on operational expenses (such as energy costs) for portfolio companies. Given that the Fund returns are denominated in dollar terms, it is important for ACAP to factor in and mitigate this risk (to the extent possible).

High inflation can substantially reduce buying power, affecting demand and profitability of portfolio companies. High levels of inflation will also likely trigger an increase in interest rates by the State Bank of Pakistan, that can further reduce economic activity and reduce the growth of portfolio companies while making it harder to raise capital locally. In FY 23, inflation rose to an all-time high in Pakistan, averaging 29.2% YoY compared to 12.2% in FY 22.[200] This surge was attributed to several key factors, including disruption to agriculture production caused by the 2022 floods (a once in a generation event[201]), adjustments in energy tariff and petroleum prices, as well as depreciation of the rupee. As a result, the State Bank of Pakistan responded to inflationary pressures by progressively raising the policy rate to reach 22%, a total increment of 825 basis points (bps).[202] These high rates imply a high cost of borrowing, which coupled with the government's increasing reliance on financing from commercial banks due to limited external financing inflow, can further restrict the private sector's access to credit – further reducing growth expectations for portfolio companies.

A *low growth* economic context can impact the valuation and profitability of portfolio companies and make it harder for them to raise growth capital. A market slowdown can also delay any upcoming exits of later stage companies through public markets/ IPOs while dragging down fund level IRRs.

Mitigation Measure(s)

While the Pakistani economy has been through a challenging year, it is likely that these trends will moderate and that devaluation risk (while not insignificant) will remain moderate. In July 2023, the International Monetary Fund (IMF) approved a 9-month Stand-By Arrangement (SBA) for Pakistan, and since that time, the country has made progress toward achieving its targets. This new SBA serves as a policy anchor and framework for financial support from both multilateral and bilateral partners in the upcoming period. It is intended to support the authorities' immediate efforts to stabilize the economy in the face of recent external shocks, maintain macroeconomic stability, and establish a framework for financial assistance from various partners. The IMF program has reduced the risk of a sovereign default, and in combination with several policy measures (including a crackdown on illegal dollar trade), has had a very positive impact on devaluation – not only arresting the decline, but in September 2023, causing the Pakistani

[200] The World Bank, Pakistan Development Update, 2023, https://thedocs.worldbank.org/en/doc/cfd9113f24c548efdc86dba482a5e2cf-0310062023/original/Pakistan-Development-Update-October-2023.pdf
[201] UNOPS, A difficult lesson from the Pakistan floods, 2023, https://reliefweb.int/report/pakistan/difficult-lesson-pakistan-floods
[202] ibid

Rupee to gain over 8% against the dollar and be considered one of the top performing currencies globally. While the continued imbalance of payments makes the currency appreciation trend unlikely to continue, the ACAP team expects that the Fund model will adequately account for expected devaluation during the Fund time horizon. Further, the Fund investment strategy is designed to ensure that investments remain profitable at a Fund level.

The ACAP investment strategy has an strong focus on export oriented/import substituting value chains, which are robust to (and in fact often benefit from) PKR devaluation

The ACAP Fund investment strategy factors in various scenarios for devaluation. The Fund strategy of prioritizing investments in import substituting and/or export oriented agribusinesses will enable fund level gains through devaluation – and ACAP will benefit substantially from an increasing demand for local agriculture products, based on a growing local population, accelerating demand for local production, and the strong impetus to localize agriculture supply chains and reduce import dependence.

Even under a high inflation scenario, the average price elasticity of food remains lower in absolute terms than the typical assumption of a unitary price elasticity (as food is a necessary good). Thus, demand is likely to remain resilient to price increases/fall in real incomes. We also expect that while interest rates are expected to go down in the near term, a high rate environment will further increase the demand for ACAPs equity-based financing and increase the scope and profitability of its investments.

Further, given that ACAP Fund has a longer investment horizon, we expect that investee company valuations will not drop sharply during periods of market volatility. Venture-backed companies are typically valued based on their last financing round, even if it occurred multiple quarters ago. While these valuations may be adjusted based on operating performance or market conditions, the adjustments are not typically made immediately, especially for companies that recently raised capital. This approach leads to a smoothing effect in portfolio valuations, and ultimately returns, where changes in value are much more muted.

The Fund is also looking at multiple exit options in addition to an exit via public markets. A slower IPO environment can be mitigated by a focus on alternative exit routes companies, via strategic or financial acquisitions.

Selected Risk Factor 2

Category	Probability	Impact
Macroeconomic	Low	Low

Description

Regulatory and Compliance Risks

Pakistan's agricultural sector is significantly influenced by government policies and regulations. Changing regulations and policies can create uncertainty and compliance challenges for future investee companies, potentially affecting performance. As such, unanticipated policy changes, price controls, and new taxes/increases in tax rates can negatively impact the the Program's profitability.

ACAP is also exposed to operational and compliance risks stemming from the processes, people & systems of the Program's external service providers. When engaging with third party contractors, the ACAP team will screen the identification information of each associated individual and/or entity against international sanctions lists, enforcement sources (i.e., civil or criminal offenses, indictments, and convictions), and relevant financial history (i.e., foreclosures, bankruptcies, etc.). Furthermore, to avoid any incidences or appearances of impropriety, the ACAP team ensures that each of its procurement decisions will be based on the following principles:

a) Best value for money: the optimum combination of costs and benefits based on the Fund's needs. Third party contractors are not necessarily selected for the lowest price but based on a balanced judgment of financial (i.e., cost) and non-financial factors (i.e., scope and quality, references, etc.).
b) Transparency and integrity: clearly defined processes for fairness to internal and external parties with consistent application.
c) Competitive and objective selection: where applicable, standardized evaluation criteria and procedures for impartiality and opportunity for all potential contractors and vendors.

Mitigation Measure(s)

The future ACAP LP will operate via a local team of experts who can advise on local political, regulatory, social, economic and environmental conditions. The Program will also regularly assess regulatory risk at a fund level and consult with local regulatory agencies to understand their planning processes and potential for changes. The Program will carry out regular regulatory and legislative reviews across its verticals to ensure that any potential compliance risks are mitigated. The future ACAP LP will use all reasonable measures to operate in compliance with all applicable local and international laws and regulations.		
Selected Risk Factor 3		
Category	**Probability**	**Impact**
Political and Operational	Medium	Low
Description		
Political and Operational Risks Companies operating in Pakistan may face logistical, infrastructure, and supply chain challenges that can impact their operations and profitability. Political instability and changes in government can impact business operations, sector growth and profitability.		
Mitigation Measure(s)		
The ACAP Program has been developed by Acumen, after nearly two decades of investment experience in Pakistan. Acumen is confident that its on-the-ground knowledge, coupled with a local team, and strong stakeholder relationships with public and private sector entities will enable the future LP to navigate any operational challenges effectively. Further, the ACAP team will conduct rigorous due diligence (DD) of all potential investee companies, to ensure that the future LP will only invest in companies that are more resilient to political/operational volatility. If and where necessary, the Program will work closely with investees to develop mitigation plans for responding to market shocks and operational disruptions.		
Selected Risk Factor 4		
Category	**Probability**	**Impact**
Technical and Operational	Medium	High
Description		
Liquidity/Exit Risk The Fund will primarily be making long term equity investments in venture, early growth, and growth stage companies. These investments are generally illiquid, and investors will not be able to realize returns until a specific exit event. Further, potential profits may not always be immediately realizable and can be lost prior to exit, while potential losses may get amplified by the time an exit is secured. The ability to exit investments is critical for ACAP to meet its targets. These exits may be challenging due to limited exit options, illiquid markets, or potential difficulties finding suitable buyers if market conditions are generally unfavorable or if portfolio companies do not generate attractive returns.		
Mitigation Measure(s)		
The ACAP Program has been developed from the ground up, based on extensive local knowledge and after a targeted 2-year process of fund diligence to source both investment opportunities and exit options. All investment decisions will be made after identifying realistic and profitable exit options within the time horizon of the future LP. As such, the ACAP team will focus on preparing investee companies for public markets, while maintaining strong relationships with local and offshore Agriculture sector corporations and investors to identify effective exit options via a strategic sale to later stage investors.		

Furthermore, the future ACAP LP may utilize alternative investment structures reflective of market conditions, and in addition to investing equity, ACAP may structure debt instruments and equity-like instruments that have a predetermined liquidation date to help improve the likelihood of financial exits. The team may also consider using vehicles such as dividends and warrants to help offset the unpredictability of markets and reduce exit risk.		
Selected Risk Factor 5		
Category	**Probability**	**Impact**
Technical and Operational	High	Low
Description		
Single Sector Exposure The future ACAP LP will invest in climate adaptation through the agriculture sector and will thus be exposed to several sector level risks that can potentially affect the entire investment portfolio. Agriculture remains highly vulnerable to weather conditions and water availability. Extreme climate events (such as floods, droughts, outbreaks of pests/diseases and extreme temperatures) can lead to large scale crop failures, causing significant losses for both farmers and investors. Pakistan also faces serious water scarcity issues, and the agriculture sector remains heavily dependent on the availability of water. The unpredictability of water access (for both irrigated and rain fed farms), combined with overall water scarcity can be a significant risk affecting crop yields and overall profitability for the entire sector. In addition, agricultural commodity prices can be highly unpredictable, as they are influenced by global markets and local factors. Fluctuations in prices can in turn affect revenues and profitability of portfolio companies.		
Mitigation Measure(s)		
Future ACAP LP investments will be diversified across the agriculture value chain to reduce the risk of sector exposure. The ACAP team has a concentration limit of USD 10M on a single company. Furthermore, the future ACAP LP will also restrict its investment mandate to limit politically dependent commodities (like wheat or sugar). Given the concentration limit detailed above, the Fund will exclude investments in large companies engaged in said value chains.		
Selected Risk Factor 6		
Category	**Probability**	**Impact**
Technical and Operational	Low	High
Description		
Concentration Risk The Fund will be investing in a limited number of companies, leading to a concentration of risk. A decline in the value of a significant investment can have a substantial impact on the overall performance of the fund. Political instability and changes in government can impact business operations, sector growth and profitability.		
Mitigation Measure(s)		
Along with a rigorous post investment deal management process, ACAP Fund may deploy its investments in tranches based on conditions precedent. This phased approach will help to manage certain company risks to reduce the exposure to any single company. The Fund also has a USD 10M cap on a single company, and all follow on investment decisions will require the same diligence process as that required for an initial investment. Follow on capital will only be deployed where positive prospects for performance are high and fund management has a clear line of sight on exit options.		
Selected Risk Factor 7		
Category	**Probability**	**Impact**
Other	Low	High
Description		
Reputational Risk		

Fund investments may have negative impacts and cause reputational damage.

Reputation risk may arise from potential involvement with companies facing ethical issues. Further, agricultural practices can have significant environmental implications, such as deforestation, excessive water usage, and the use of harmful pesticides and fertilizers. If portfolio companies are found to be contributing to environmental degradation, it can lead to negative publicity, damaging the reputation of a company's investors.

The agriculture sector in Pakistan is also mostly informal, employing a large labor force susceptible to reputational risks associated with exploitative labor practices, child labor, and the violation of workers' rights. Furthermore, poorly managed agriculture companies may negatively impact local communities' access to resources, livelihoods, or cultural heritage, which would further lead to reputational risks for investors.

Mitigation Measure(s)

When considering potential investees, on site due diligence is mandatory, as is regular reporting of company performance via structured KPIs/scorecards. Given the relatively small number of expected portfolio companies and the extensive on-the-ground experience of the ACAP's local team, portfolio companies will be actively managed and regularly monitored.

The Fund will also appoint a member to the board of directors of all portfolio companies to ensure strategic oversight and effective corporate governance. Furthermore, the Fund will ensure that necessary covenants are included in the shareholder agreements.

Lastly, the Fund will also ensure high levels of transparency and public accountability with a Grievance Redress Mechanism (via phone, email, post or on the website.)

Selected Risk Factor 8

Category	**Probability**	**Impact**
Investment Capital	Low	High

Description

Inability to catalyze private sector financing

If donors and funders do not engage, ACAP will not be able to raise enough commercial capital or grant funding to launch the future LP.

Mitigation Measure(s)

The ACAP team has already engaged various potential investors and donors on the future LP. We expect that catalytic GCF funding will overcome the high (perceived) risk and lower the entry barrier for international private sector investors and DFIs. The financial return and catalytic impact ACAP can offer its investors will be very attractive for the amount of risk protection provided. As the first of its kind fund, we are also confident that there are no other investment vehicles that allow international private sector investors and DFIs to get this type exposure to building high-impact climate adaptation solutions in a frontier market. As such, we are confident that we will be able to raise the required level of co-financing to operationalize the future LP.

This confidence was further bolstered following the USD 3B short-term financial package secured by the Pakistani government from the International Monetary Fund on June 30, 2023. The influx of capital has eased international concern regarding the long-term viability of the Pakistan economy and the deal, larger than previously expected, is expected to unlock other bilateral and multilateral financing. Given the recent positive development, the Fund is confident that investor anxiety around the macro-economic climate has been assuaged.

Selected Risk Factor 9

Category	**Probability**	**Impact**
Technical and Operational	Low	Medium

Description

Fund Execution Risk

Pre-Investment Due Diligence Risk: The largely informal and undocumented nature of the sector also creates a due diligence risk as it may be difficult to obtain accurate and reliable financial and operational information on portfolio companies. Cultural and language barriers can further complicate due diligence processes – as the sector is primarily rural.

Capital deployment Risk

The ACAP LP is unable to fully deploy investment capital due to slow project pipeline development, stringent deal screening criteria or a lack of investable opportunities in target sectors.

Venture, Early Growth, and Growth Stage Investment Risk: ACAP LP will invest in venture, early growth, and growth stage agribusinesses that have transformative potential. By definition, these business models are still evolving as venture, early growth, and growth stage companies are still developing the internal capabilities and capacities to operate successfully in a challenging frontier market. As such, investee companies may fail or may take longer than anticipated to achieve financially viable business models and there is a risk that the the future LP will not achieve its investment objectives.

Fund Management Risk

There is also a risk that the ACAP LP is not managed properly. Weak governance and talent issues may exist within the LP and inexperienced or ineffective managers may make poor investment decisions, leading to suboptimal returns.

Mitigation Measure(s)

The future ACAP LP will be the first fund of its type in Pakistan and we expect to realize a significant first mover advantage by investing in high impact, best-in-class companies, with the highest likelihood of generating attractive returns. The ACAP team already has a pipeline of over 200 investable opportunities through Acumen's work with earlier stage agribusinesses and continues to track future opportunities in the sector. Given the quality of ACAP's potential pipeline and the growth potential of the sector, the team fully expects to deploy all investable capital during the investment horizon of the fund.

The LP management team will set up rigorous due diligence (DD) processes which include an in-depth analysis of financial, operational, market, and ESG performance. All investments will require mandatory on-site DD to ensure that the LP can accurately assess high quality investment opportunities and select enterprises with strong management teams, high quality products, and sound reputations – even in cases where formal, documented sources of information are not fully available.

ACAP investment processes (that have been developed after two decades of in-country experience) will significantly reduce the risk of making sub-optimal investments and enable the team to support portfolio companies in a way that amplifies their climate impact while optimizing financial returns and commercial viability.

To further reduce the risk of failure, the LP will not only provide investment capital and targeted TA but expects to provide strategic advice and additional resources holding a seat on the Board of Directors of each of the future LP's portfolio companies, identifying, and working with co-investors, providing access to local networks of experts and value-add partners, and offering specific technical assistance programs.

In addition to hiring a team of investment and sector experts, the Fund will set up a robust post investment monitoring and reporting system to monitor and track fund performance. Annual audits will be conducted in accordance with the Program's audit protocol. The LP's governance structure will focus on high transparency, annual public reports, and direct investment oversight by an Investment Committee following international best practices.

ACAP's Code of Conduct will provide a clear ethical framework in which the LP will operate. The code of conduct will contain an overview of the values, commitments, responsibilities and integrity that the Program stands for, including the 'zero tolerance' principle when it comes to bribery, fraud and corruption.

Selected Risk Factor 10

Category	**Probability**	**Impact**
Social and Cultural Risks	Medium	High

Description

Limited additionality of grants funding for gender equity

The agriculture sector is largely informal and based in rural areas with more conservative social/cultural values. There is a risk that TA investments will not meet objectives on gender due to social/cultural resistance to gender focused interventions.		
Mitigation Measure(s)		
The LP's gender assessment and action plan has been developed from the ground up, after extensive field research and consultation with sector experts. As such, it incorporates local social and cultural barriers that may impact outcomes. The Program will also hire a dedicated Gender Expert who will be responsible for making sure that gender equity is a focal area for all the Fund activities - from due diligence (DD) to investment and TA assistance. All investee companies will be tracked on gender equity metrics and post investment support will include a bespoke gender action plan for each investee company. In addition, the LP will provide trainings to companies to prioritize more gender inclusive growth.		
Selected Risk Factor 11		
Category	**Probability**	**Impact**
ML/TF	Low	High
Description		
ML/TF Risk that the Fund is misused for money laundering ("ML") and terrorism financing ("TF") ML/TF or any other financial misconduct or crime.		
Mitigation Measure(s)		
Both the LP and Acumen have AML/CFT policies that aim to ensure that ML/TF risks in the processes of fundraising for the LP, and deploying future investments are properly identified, monitored, mitigated and reported in order to prevent the the LP, governing bodies, and service providers from being misused for ML/TF or any other financial misconduct or crime. The use of any materials and technology procured by the LP is subject to AE's Code of Ethics which is applicable to everyone associated with the AE, "whether they are employees, consultants, fellows, volunteers, or members of our Board or committees."		

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

Provide the environmental and social risk category assigned to the proposal as a result of screening and the rationale for assigning such category. Present also the environmental and social assessment and management instruments developed for the proposal (for example, Environment and Social Impact Assessment (ESIA), Environment and Social Management Framework (ESMF), Environmental and Social Management Plan (ESMP), Environmental and Social Management System (ESMS), environmental and social audits, etc.). Provide a summary of the main outcomes of these instruments. Present the key environmental and social risks and impacts and the measures on how the project/programme will avoid, minimize and mitigate negative impacts at each stage (e.g. preparation, implementation and operation), in accordance with GCF's Environmental and Social Safeguard (ESS) standards. If the proposed project or programme involves investments through financial intermediations, describe the due diligence and management plans by the Executing Entities (EEs) and the oversight and supervision arrangements. Describe the capacity of the EEs to implement the ESMP and ESMF and arrangements for compliance monitoring, supervision and reporting. Include a description of the project/programme-level grievance redress mechanism, a summary of the extent of multi-stakeholder consultations undertaken for the project/programme, the plan of the Accredited Entity (AE) and EEs to continue to engage the stakeholders throughout project implementation, and the manner and timing of disclosure of the applicable safeguards reports following the requirements of the GCF Information Disclosure Policy and Environmental and Social Policy.

Describe any potential impacts on indigenous peoples and the measures to address these impacts including the development of an Indigenous Peoples Plan and the process for meaningful consultation leading to free, prior and informed consent, pursuant to the GCF Indigenous Peoples Policy.

Attach the appropriate assessment and management instruments or other applicable studies, depending on the environmental and social risk category as annex 6.

147. After an environmental and social impact assessment, assessing project fundamentals, reviewing relevant policies and procedures including Acumen's and GCF's environmental and social policies, assessing executing entity capacity, and developing the ACAP Environmental and Social Management System, the ACAP team has determined the project to be a Category I-2 with mostly environmental and social Category C investments and activities. The Acumen Climate Action Fund team will lead the project's ESG efforts with the support of the AE.

148. Acumen rated the Acumen Resilient Agriculture Fund to be a Category I-3 project making Category C investments. Similar projects in Southeast Asia have evaluated themselves as Category I-2/B due to specific investing opportunities and risks to the region. While ACAP intends to invest primarily in Category C investees, there may be some Category B investing opportunities.

149. While ACAP assumes that a majority of investments in the agriculture sector will be E&S Category C, the team acknowledges the potential for some amount of investments to have Category B risks contained within the scope of the investment. ACAP expects to have a more rigorous investment process that may include an environmental and social impact assessment if the team perceives an investment may be Category B. Category B investments may be engaged in purchasing land, land tenancy issues, or have a demonstrated pattern of site-specific and reversible negative environmental and social impacts.

150. Acumen has developed an Environmental and Social Management System (ESMS) to identify, manage, and mitigate environmental and social risks and impacts. The ESMS aligns with both Acumen's ESG Policy for GCF-Funded Projects and GCF's revised Environmental and Social Policy.

(i) Scope of E&S assessment

Acumen seeks to do no harm with the ACAP activities. Acumen used industry best practices, tools, metrics, methodologies, and engagements to develop a broad and detailed assessment of the project environmental and social risks and impacts. ACAP also extensively engaged with stakeholders to get their input into project design, market research, project management and implementation, climate, gender, and poverty strategies, assessing the viability of the initiative, and measuring interest in continued engagement. ACAP engaged with potential investors and donors, pipeline companies, potential partners, industry leaders and experts, industry associations, government entities, regulators, expert consultants, and beneficiaries in designing and developing the project. The team has held one on one meetings, group discussions, town halls, webinars, and surveys. ACAP intends to continue this engagement with stakeholders throughout project development and implementation. The ACAP project team also utilized extensive desktop research to understand specific risks for the project, investees, the country context, and relevant mitigants.

(ii) Key Project Level Environmental and Social Risks

The project has seven primary environmental and social risk concerns:

-Biodiversity Loss and Pesticide Use

-Deforestation via livestock and agricultural farming expansion

-Water usage

-Community Health and Safety

-Informal Labor Practices

-Land Tenancy Issues

-ESG readiness

(iii) Biodiversity Loss

When overused or misused, pesticides have done serious damage to biodiversity and soil quality. Researchers have found toxic levels of DDT, and especially DDE, in bird egg samples in Haleji Lake. Vultures have had an unusually high mortality rate and population decline. Researchers also found that freshwater fish they sampled contained above the permissible amount of DDT contents, and that 100% of the fish they sampled had metabolites and DDT/HCH in their systems. Pesticide use is down from its peak in Pakistan, but remains a critical part of controlling pests and disease in agriculture.

(iv) Deforestation

According to the World Bank, Pakistan loses nearly 27,000 hectares of natural forest area every year. 68% of Pakistan's population uses firewood as their primary source of energy in households. Agriculture has been a cause of deforestation as farmers have needed more space for livestock and other farming practices.

(v) Water Usage

Pakistan uses 90% of its water for agriculture. Mostly, farmers use water to flood their fields to irrigate their crops. This water waste has led to a scarcity of usable water for farming practices. The amount of available water per person has dropped from 1,500 cubic meters in 2009 to about 1,017 cubic meters in 2021 according to the IMF[203].

(vi) Community Health and Safety

Communities face some vulnerabilities as a result of this project including livelihood loss, exposure to physical hazards, social conflicts, and some potential impacts on indigenous peoples.

(vii) Informal Labor Practices

Pakistan's agricultural economy and both industrial and smallholder farmers rely on informal labor practices. One risk we note is the inclusion of children in informal and unpaid labor. While family farming is not uncommon around the globe, it is important to note the distinction. Some children are being exposed to harsh conditions, extreme physical labor, long hours, and unpaid work.

(viii) Land Tenancy Issues

Land tenancy in Pakistan has created challenges for smallholder and vulnerable farmers. Smallholder farmers in several regions in Pakistan are tenants of large landowners where landowners provide land, inputs, and technology in

[203] Ahmad et al, Impact of water insecurity amidst endemic and pandemic in Pakistan: Two tales unsolved, 2022, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9464874/#:~:text=As%20per%20IMF%2C%20Pakistan's%20per,approximately%2083%20MAF%20%5B21%5D

exchange for a portion of the farmer's yield. When farmers do not achieve their yield, they remain indebted to landowners. Some of these practices have led to the exploitation of smallholder farmers.

(ix) ESG Readiness

Early-stage companies in Pakistan have limited experience with managing environmental and social risks and opportunities. Some companies will have weak and insufficient environmental and social practices that can harm communities and create risks for the Fund.

(x) Environmental and Social Risk Mitigants

To ensure the ACAP Fund achieves its mission and impact, the team has designed a thorough ESMS aligned with GCF Policy, Acumen Policy, and industry best practices.

The ESMS shares the project development, management, and monitoring and reporting policies and procedures. All investments will be screened against the project Exclusion List before being accepted for due diligence.

The project team will diligence companies with a variety of ESG diligence tools including base-line questionnaires, site visits, company evidence in the form of policies and procedures, desktop research, a climate resilience assessment tool, and supplementary questionnaires.

The ESG findings will be memorialized in an ESG due diligence report and the investment memo. The Investment Committee is expected to consider ESG factors in their deliberations. The ESG due diligence report will also provide the information needed for developing the term sheet binding, time bound, ESG Action Plan.

ACAP will offer Portfolio Companies technical assistance tailored to supporting company-specific ESG, consumer protection, gender, and business readiness needs. This will include: needs diagnostics, webinars, direct support, and policy and procedure guidance and resources.

The team will monitor and report on ESG strategy across the portfolio and the project to relevant teams.

ACAP will also ensure that the team has the organizational capacity to identify and mitigate risks across the project and investments. The team will hire a Climate, ESG, and stakeholder engagement officer who will be responsible for ensuring the implementation of the ESMS.

The team will also ensure that they listen to stakeholder voices so that communities affected by the project can continue to shape and improve the project design and implementation.

(xi) ESG Technical Assistance

ESG support will be provided to portfolio companies to improve their environmental and social impact, mitigate environmental and social risks, and build organizational capacity. The TA will provide consulting support to help companies create and execute on environmental and social action plans.

Additionally, technical assistance will be used to provide in depth analysis of projects with E&S category B characteristics.

(xii) Indigenous Peoples

The program intends to have a landing page on the AE website that will share information about the program including the ESMS. We seek to provide guidance on how indigenous communities can request support from the GCF independent Redress Mechanism and/or the GCF Secretariat's indigenous peoples focal point at any stage, including before a claim has been made. Additionally, the team seeks for ACAP investees to have indigenous peoples' policies that will communicate how GCF can be reached. This is especially important for investees that directly engage with indigenous communities.

Acumen, as the AE, and the ACAP team have reached out to indigenous organizations to learn from them, share the project design, and ask for their input. We seek meaningful engagement with indigenous populations who may be impacted by the project prior to first close. We intend to report to indigenous organizations about project activity and direct them to climate resilient agricultural practices expanding in areas that have indigenous populations so that they can benefit from access to affordable, clean electricity opportunities.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

Provide a summary of the gender assessment and project/programme-level gender action plan that is aligned with the objectives of GCF's Gender Policy. Confirm a gender assessment and action plan exists describing the process used to develop both documents. Provide information on the key findings (who is vulnerable and why) and key recommendations (how to address the vulnerability identified) of the gender assessment. Indicate if stakeholder consultations have taken place and describe the key inputs integrated into the action plan, including: how addressing the vulnerability will ensure equal participation and benefits from funds investment; key gender-related results to be expected from the project/programme with targets; implementation arrangements that the AE has put in place to ensure activities are implemented and expected outcomes will be achieved, monitored and evaluated.

Provide the full gender assessment and project-level gender action plan as annex 8.

151. Executive Summary

Acumen has conducted a thorough and robust gender assessment for ACAP aligned with the objectives of GCF's Gender Policy and Acumen's Gender Sensitivity Policy for GCF-funded projects. The gender assessment informed the development of the ACAP Gender Action Plan. The ACAP gender assessment and GAP are detailed in Annex 8.

The team is conducting extensive stakeholder engagement across Pakistan including speaking with companies, government entities, industry associations, NGOs, and civil society organizations. The ACAP team continues engaging with local gender experts and civil society organizations on local gender challenges and opportunities that informed our gender lens investing strategy and gender work.

The team relied on desktop research, industry experts, gender experts, agriculture companies, and survey data to inform the gender assessments, GLI strategy, and Gender Action Plan.

(i) Findings

Barriers to gender equality in Pakistan's agriculture sector

The link between agriculture and gender equality is well recognized. Poverty and limited access to financial opportunity has a strong gender dimension as the lack of access to agricultural employment and services inhibits women's well-being, and their access to economic opportunities, and it strongly affects women's living conditions and time-use.

One of the key barriers in the market relates to lack of sex-disaggregated data within Pakistan both at the consumer and workforce levels.

Gender and social norms limit women's opportunities to generate income and wealth and, as a result, access to technologies and their abilities to enter the workforce. Women in rural communities may also have limited access to legal protections, healthcare, and educational opportunities. All of these constraints limit their opportunities for employment, entrepreneurship, and benefitting from agribusinesses in Pakistan.

Agribusinesses in Pakistan may have a limited understanding of the business case for gender inclusion in their businesses, and of how to make their practices more inclusive. Several companies we spoke with shared that they are beginning to build institutional capacity on gender and want to build inclusive policies and procedures into their operations.

Serving women with climate resilient agricultural products and services

Overall, evidence shows that farms managed by women are less productive than farms managed by men. Women have less access to wealth and incomes, and therefore have limited access to agricultural products and services. Many women do not manage farms and cannot make decisions on farm products and services. Moreover, as women participate in agriculture on a primarily informal basis, their labor often does not include agricultural products and services.

Women's participation in agricultural sector

Women in Pakistan have limited opportunities to participate in the formal labor markets.

The gender assessment indicates that several barriers limit women's participation in the workforce and in the economy more generally, with prevailing gender norms, gaps in education levels and discriminatory policies and practices affecting women's productive activities and economic empowerment in several countries.

(ii) ACAP Gender Work

ACAP seeks to support the Pakistan agricultural sector and agribusinesses with a gender lens, and strong commitment to gender equity. ACAP is also strongly committed to supporting companies to build their gender capacity to support their operations and customers. Additionally, ACAP is committed to mitigating risks of sexual harassment, abuse, and exploitation across the project, investees, and beneficiaries.

All companies will undergo a gender assessment to determine their gender strengths and weaknesses. Opportunities for growth will be captured in a binding Gender Action Plan. All companies in the ACAP portfolio will have to create a GAP that will be monitored over the company engagement with ACAP. Companies must commit to strong anti-sexual harassment policies, non-discrimination policies, and their best effort to collect sex-disaggregated data.

ACAP intends to provide companies with technical assistance to support building and executing GAPs, creating strategies to hire women, and to retain female customers. ACAP will also have technical assistance to build climate awareness for female farmers and train women for jobs in the agriculture sector. Additionally, ACAP intends to build women's awareness of climate resilient products and services and local climate hazards by partnering with gender experts on improving inclusive marketing and customer retention activities.

ACAP will monitor and report on the gender assessments, GAPs, sex-disaggregated data, and other KPIs in the GAP and logic framework.

Lastly, ACAP will continue to engage female stakeholders and stakeholders from marginalized or unheard communities to ensure that their insights inform the project design, implementation, and management.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

Describe the project/programme's financial management including the financial monitoring systems, financial accounting, auditing, and disbursement structure and methods. Refer to section B.4 on implementation arrangements as necessary.

Articulate any procurement issues that may require attention, e.g. procurement implementation arrangements and the role of the AE under the respective proposal, articulation of procurement risk assessment undertaken and how that will be managed by the AE or the implementing agency.

As the AE for the project, Acumen will have overall responsibility and oversight for the project, including project preparation and implementation, financial management and procurement. As such, the financial management and procurement procedures detailed below were developed in concert with the AE and based of the organizational policies adopted by the AE in 2021.

1. Accounts Payable / Purchases / Expenses

a. ***Initiating and Authorizing***:

i. Procurement decisions will be made by the Program EE (the future ACAP LP), in consultation with the AE (Acumen Fund, Inc.). When purchasing supplies, property, equipment, and services from an external source, the Program EE (the future ACAP LP):

1. Will alert the AE's General Counsel for potential conflict of interests.

2. Shall neither solicit nor accept gratuities, favors, or anything of monetary value from contractors, potential contractors, or similar service providers.
3. Will follow the Procurement Policy including consulting the AE (Acumen) for any goods or services ≥ USD100K, awarding the winning bidder, documenting the decision justification, and the evaluation of each vendor following the contract term.
   ** The Program EE will seek to engage service providers with track records of delivery in alignment with the above policies and the AE's procurement standards. Service providers must demonstrate capacity and execute around the scope of work developed by the Program EE team and shared with the AE (Acumen) for review, where relevant. All procurement decisions will be reviewed by the Program Director (the ACAP Managing Director).
4. When required by investors, the Program EE may publish an RFP for a specific project.

ii. Vendor selection is dependent on:
1. The results of a standardized background check. Each vendor and its related entities and owners are subject to Patriot Act searches.
2. Best value for money - the optimum combination of costs and benefits based on the Program's needs. Goods and services are selected not necessarily for the lowest price but based on a balanced judgment of both financial (i.e., cost) and non-financial factors (i.e., scope/quality, diversity and inclusion, references, etc.).

b. ***Recording and Processing****: Both the AE and the EE will adhere to acceptable International Accounting Standards. As such, a*ccounts payable will be reviewed and processed at the end of each week. Each invoice is submitted to the Program's accounting platform for review and approval and must have an associated vendor record in the system. Before approving, the appropriate finance employee will review all invoices, wires, checks and account coding.

c. ***Reconciling****:* The Program reconciles all bank, petty cash, and investment accounts monthly, as follows:
   i. Once available, the Finance Associate (or designate) reviews all relevant bank statements.
   ii. The Finance Associate (or designate) will complete the reconciliation within seven days.
   iii. The following is the general outline to be used to reconcile accounts:
   1. Comparison of bank statements and cash receipts, verifying dates and totals deposited.
   2. Comparison of bank and/or wire transfer dates received against dates sent.
   3. Investigations of items rejected by the bank (i.e., returned checks or deposits).
   4. Comparison of canceled checks and the disbursement journal (check numbers, payee, and amount).
   5. Examination of canceled checks for authorized signatories, irregular endorsements, and alterations.
   6. Accounting for the sequence of checks from month-to-month.
   7. A review and proper mutilation of void checks.
   8. Investigate, write-off or reissue/stop-pay checks outstanding for an extended period.

   [Furthermore, the Program (future ACAP LP) is subject to an annual financial audit by an independent external auditor in accordance with internationally recognized financial reporting and auditing standards. The AE's Audit and Finance Committee may be consulted to discuss potential issues and/or irregularities discovered during the external audit process as needed.]

2. Expense Reporting Procedures

a. ***Principles and Procedures****:*

i. The Finance Associate Director (or designate) will be responsible for compiling monthly and year-to-date expense reports. Expense reports are reconciled to the general ledger and accounting records.
ii. Quarterly financial reports, analyzing the Program's financial position and the effectiveness of its management, will be presented to its Limited Partners as required.
iii. Records to be maintained include copies of contracts, invoices, changes to contracts or budgets, proof of payments, and allocation tracking.

3. **Expected Contractors**
The Program may contract consultants for several purposes. Technical Assistance consultants will be hired for expertise in ESG; Consumer Protection; Gender Equity; Business development (as it relates to fundraising and financial systems management) for companies; Legal and Accountants/auditors.

The Program will focus on early-stage investments, employing equity, equity-like, impact index loans, grants, and joint venture financing options. As such, there may be other types of contracts with other service providers that the Program continues to explore including legal services and fund administration [**see capacity assessment in B.4**].

4. **Disbursements:**
The conditions associated with the **disbursements** of funds from GCF to the AE and from the AE to the EE are detailed in the ACAP LP Term Sheet.

G.4. Disclosure of funding proposal

Note: The Information Disclosure Policy (IDP) provides that the GCF will apply a presumption in favour of disclosure for all information and documents relating to the GCF and its funding activities. Under the IDP, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Information provided in confidence is one of the exceptions, but this exception should not be applied broadly to an entire document if the document contains specific, segregable portions that can be disclosed without prejudice or harm.

Indicate below whether or not the funding proposal includes confidential information.

☐ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

☒	Annex 1	NDA no-objection letter(s) **(template provided)**
☒	Annex 6	E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3): **(ESS disclosure form provided)** ☐ Environmental and Social Impact Assessment (ESIA) or ☐ Environmental and Social Management Plan (ESMP) or ☒ Environmental and Social Management System (ESMS) ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
☒	Annex 8	Gender assessment and project/programme-level action plan **(template provided)**

H.2. Other annexes as applicable

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

No-objection letter(s) issued by the national designated authority(ies) or focal point(s)

F.No. CFU/GCF/002/2015
Government of Pakistan
Ministry of Climate Change
(LG&RD Complex, 3rd Floor, Sector G-5/2, Islamabad)
(Ph: +92-51-9245585)

******

To: The Green Climate Fund ("GCF") Secretariat
Song Do,
South Korea.

Islamabad, the 27th December, 2022

Re: Funding proposal for the GCF by Acumen regarding Acumen Climate Action Pakistan Fund

Dear Madam/ Sir,

We refer to the programme titled *Acumen Climate Action Pakistan Fund* in Pakistan as included in the funding proposal submitted by Acumen to us on 20 August 2022.

The undersigned is the duly authorized representative of Ministry of Climate Change, the National Designated Authority of Pakistan.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Pakistan has no-objection to the programme as included in the funding proposal;
(b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Pakistan;
(c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Hussain

Syed Mujtaba Hussain
Sr. Joint Secretary (IC)/GCF Focal Point
Ministry of Climate Change
Pakistan

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Acumen Climate Action Pakistan Fund
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Private
Accredited entity	Acumen Fund Inc.
Environmental and social safeguards (ESS) category	Category I-2
Location – specific location(s) of project or target country or location(s) of programme	Pakistan
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	Thursday, January 25, 2024
Language(s) of disclosure	English and Urdu
Explanation on language	These are the official languages in Pakistan.
Link to disclosure	English: https://acumencapitalpartners.com/wp-content/uploads/2024/02/Acumen-Climate-Action-Pakistan-Environmental-and-Social-Management-System-English.pdf Urdu: https://acumencapitalpartners.com/wp-content/uploads/2024/02/Acumen-Climate-Action-Pakistan-Environmental-and-Social-Management-System-Urdu.pdf
Other link(s)	N/A
Remarks	An ESMS consistent with the requirements for a Category I-2 programme is contained in the "Environmental and Social Management System".

Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	
Description of report/disclosure on accredited entity's website	Thursday, January 25, 2024
Language(s) of disclosure	English and Urdu
Explanation on language	These are the official languages in Pakistan.
Link to disclosure	English: https://acumencapitalpartners.com/wp-content/uploads/2024/02/Acumen-Climate-Action-Pakistan-Environmental-and-Social-Management-System-English.pdf Urdu: https://acumencapitalpartners.com/wp-content/uploads/2024/02/Acumen-Climate-Action-Pakistan-Environmental-and-Social-Management-System-Urdu.pdf
Other link(s)	N/A
Remarks	The Land Acquisition and Resettlement Policy Framework and the Indigenous People Policy Framework are included in the Acumen Climate Action Pakistan "Environmental and Social Management System".
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Thursday, January 25, 2024
Place	The Acumen Climate Action Pakistan Environmental and Social Management System is available at the Acumen Pakistan office located at: Office 912-913, Islamabad Stock Exchange Building, Jinnah Avenue, F7/1, Islamabad, Pakistan The Acumen Climate Action Pakistan Environmental and Social Management System can also be found at the Nationally Designated Authority office in Pakistan. The address is listed below: Pakistan Ministry of Climate Change Additional Secretary, Ministry of Climate Change and Environmental Coordination 3rd Floor, Ministry of Climate Change, LG&RD Complex, G-5/2, Islamabad, Pakistan
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, March 4, 2024

Note: This form was prepared by the accredited entity stated above.

Independent Technical Advisory Panel's assessment of FP229

Proposal name:	Acumen Climate Action Pakistan Fund
Accredited entity:	Acumen Fund, Inc.
Country/(ies):	Pakistan
Project/programme size:	Medium

I. Assessment of the independent Technical Advisory Panel

1.1 Overview

1. Pakistan is extremely vulnerable to climate change. By the end of the twenty-first century, average daily temperature increases of 1.3–5.3 °C are anticipated.[1] An expected rise in precipitation will increase the frequency and intensity of extreme weather events such as floods, droughts and heatwaves. Due to climate change, Pakistan faces average annual gross domestic product (GDP) losses of 5.5–9.0 per cent.[2]

2. Agriculture is the largest sector of the Pakistani economy, with the majority of the population, directly or indirectly, dependent on this sector. It contributes to about 24 per cent of GDP.[3] Climate change will significantly and disproportionally impact this sector, reducing staple crop yields and productivity. This may affect food prices and increase food insecurity and malnutrition risk. Approximately 90 per cent of farmers in Pakistan are smallholders, which are the most vulnerable group to climate change as they cannot adapt to its effects.

3. The Acumen Climate Action Pakistan (ACAP) Fund will invest in 15–17 climate-focused agribusinesses to increase the climate resilience of smallholder farmers (SHFs). The investment areas are: (1) farmer platforms and financial solutions to remove informational asymmetries and improve access to finance, such as platforms connecting retailers to end users or solutions involved in the spatial tagging of livestock to enable insurance; (2) smart farming to increase yields such as climate-resilient production practices like farm mechanisation services at farm-gate or the propagation of high-yielding value chains through tissue culture technology; and (3) post-harvest technologies to reduce on-farm losses such as controlled atmosphere logistics and storage facility services.

4. The ACAP Fund is a private equity fund providing concessional private capital and technical assistance to agribusinesses with growth potential, improving SHF resilience. It will be Pakistan's first adaptation-focused fund to support the venture, early growth and growth stage agribusinesses building climate resilience. It is a USD 90 million blended finance facility with USD 80 million of tiered investable capital and a USD 10 million grant-funded technical assistance facility. The blended capital structure will de-risk other investors' engagement,

[1] See *Climate Risk Country Profile: Pakistan* (World Bank and Asian Development Bank. Available at: https://climateknowledgeportal.worldbank.org/sites/default/files/2021-05/15078-WB_Pakistan%20Country%20Profile-WEB.pdf), table 2; low end of the representative concentration pathway (RCP) 2.6 scenario; high end of the RCP8.5 scenario.
[2] See *Asia-Pacific Riskscape @ 1.5°C: Subregional Pathways for Adaptation and Resilience* (The Economic and Social Commission for Asia and the Pacific. Available at: www.unescap.org/kp/2022/asia-pacific-disaster-report-2022-escap-subregions-summary-policymakers), range from current to RCP8.5 scenario.
[3] www.pbs.gov.pk/content/agriculture-statistics

catalysing further investment in climate-resilient agriculture. The ACAP Fund's investment strategy involves using equity and debt financing, and technical assistance. It fills a market gap as other entities either target larger farmers and entities or offer loans with small ticket sizes, high interest rates and high collateral requirements. Development finance institutions (DFIs) only offer larger ticket sizes and are generally difficult to access for growth entities. Technical assistance is critical to address the capacity challenges that growth-stage agribusinesses face due to the informal nature of the agriculture sector. Technical assistance will be deployed across climate adaptation, impact measurement and capacity-building, business development, gender, and environmental, social and corporate governance support.

5. Acumen has been investing in the agriculture sector since 2008, funding over 40 entities in 11 countries that impact SHFs through financial instruments such as debt, equity and hybrid instruments. Entities in which Acumen has invested previously include those which (1) provide access to finance for SHFs; (2) provide critical goods and services to SHFs, such as low-cost greenhouses and drip irrigation systems; (3) provide supply chain solutions to SHFs to allow them to access high-value markets; and (4) unlock alternative income streams for SHFs (e.g. by training them on aquaculture practices and by providing an end-to-end aquaculture platform).

6. The ACAP Fund builds on the experiences of Acumen with other funds and agribusiness, including in Pakistan, where Acumen has implemented an accelerator programme over the past two years, which resulted in the following key takeaways: (1) agribusinesses lack linkages with each other or with larger buyers/suppliers; (2) business models are not tailored to build adaptive capacity of smallholder farmers; and (3) highly experienced teams are critical.

7. The USD 90 million project will be funded by: (1) USD 28 million from GCF composed of USD 25 million junior equity and a grant of USD 3 million; (2) USD 55 million of equity (USD 2 million junior and USD 53 million senior) from DFIs, private institutions, family offices and foundations; (3) USD 7 million grant from DFIs, private institutions, family offices and foundations. The to-be-formed ACAP Fund Manager will be indirectly wholly owned by the accredited entity (AE). The ACAP Fund will have an advisory committee to serve as an adviser to the Fund Manager and will consult with the Fund Manager concerning the ACAP Fund's activities and operations, particularly identifying and resolving conflicts of interest.

1.2 Impact Potential *Scale: High*

8. The programme has a potential adaptation impact and, as a co-benefit, a potential mitigation impact. The adaptation impact is estimated at 2 million direct (2 per cent of the population) and 13 million total beneficiaries. Around 50 per cent of these are expected to be women.

9. The agriculture sector and within this industry, especially SHFs, will be significantly impacted by climate change. Each of three ACAP Fund investment themes has a specific climate rationale and climate-resilience impact as follows:

(a) The investment theme "Farmer platforms and financial solutions" has as climate-resilience impact of increased farmer incomes and reduced food insecurity by providing high-yielding inputs appropriate for increased weather variability and extreme weather events, information access and extension services, and access to financial services to support the adoption of climate-resilient practices.

(b) The investment theme "Smart farming" aims to increase and diversify farmer incomes through increased yields by providing small farm-focused mechanisation, through farm optimization, and by providing alternatives to traditional farming and regenerative agriculture, which improves farmer resilience against the weather variability arising due to climate change, reducing farmers' exposure to downside risks and crop failures and diversifying income streams.

(c) The investment theme "Post-harvest technologies" aims to increase farmer income by removing middlemen, reducing food waste, and adding higher value to the product, enabling farmers to withstand climate shocks better.

10. To ensure that funded companies work with SHFs and climate-vulnerable rural populations, Acumen will:

(a) Realize a pre-investment due-diligence process vetting the entity's historical work with SHFs and assessing the alignment of products and services provided to the target group. On-the-ground teams and on-site due diligence implement this;

(b) Realize impact-focused shareholder agreements including a "mission drift" clause which can trigger an exit of Acumen's investment and also a board representation and observer seat to oversee strategic decision-making and corporate governance; and

(c) Make strategic use of financial instruments to incentivize companies, including tranched capital deployment, follow-on capital availability, technical assistance to provide grant-funded support to amplify the investee company's focus and post-investment monitoring and reporting.

11. Direct beneficiaries are SHFs as customers or suppliers and agriculture-dependent rural employees. SHFs are defined as having less than 12.5 acres of land. The number of beneficiaries is calculated per investment area. Indirect beneficiaries are estimated by multiplying the average household size in rural Pakistan and deducting the direct beneficiary number to avoid double counting. The investment theme of farmer platforms and financial solutions is dominant for the number of expected beneficiaries, accounting for 99 per cent of direct and indirect expected beneficiaries Expected female representation is based on national demographics. The direct beneficiaries estimate is based on interviews with selected entities per investment theme to determine the number of SHFs reached and growth rates per investment theme based on weighted average returns from deal-level internal rates of return, including failure-rate assumptions. Estimates are based on indicative pipeline entities and are thus subject to change. The calculation approach is considered to be appropriate As with all programme and funding proposals, it is subject to the investment theme and investment archetypes with assumptions of how funds will be used and can thus be considered a high-level initial approximation.

12. While the funding proposal targets all vulnerable "farmers", the farmers, in reality, are split into two groups: bankable/less vulnerable and non-bankable/most vulnerable. More opportunities are available to the bankable farmers. Understandably, financial service providers will want to deal with creditworthy farmers, but that might exclude other farmers.

13. A mitigation co-benefit of 95,000 tonnes of carbon dioxide (t CO_2) reduced cumulatively over the project lifetime (investment period of 5 years is assumed). The greenhouse gas reduction is based on (1) reduced food waste (94 per cent of impact), (2) climate-smart agricultural practices (1 per cent of impact), and (3) farm optimization (5 per cent of impact). Reduced food waste is due to various measures such as providing storage solutions that reduce post-harvest losses Its calculation is based on a food waste footprint report 2013[4] produced by the United Nations Food and Agriculture Organization, which calculates a top-down CO_2 reduction of 3.4 kg CO_2 per kg of food waste reduced. This is based on a mixture of food types. The actual impact of food waste will depend on the investments in storage solutions and the value chain in which the invested company operates. The Verra methodology for food loss and food waste (VM0046)[5] is not followed for initial estimates due to a lack of data but shall be used to monitor this parameter. Climate-smart agricultural practices result in carbon sequestering (or avoidance) with a default value of 0.3 t CO_2 per hectare applied to manage good practices. Farm optimization is based on solar-powered devices using Pakistan's carbon grid factor

[4] FAO, (2013), Food wastage footprint
[5] https://verra.org/methodologies/vm0046-methodology-for-reducing-food-loss-and-waste-v1-0/

published by the United Nations Framework Convention on Climate Change secretariat (harmonized international financial institutions grid factor)[6]. It is noted that the programme only claims the mitigation impact as a co-benefit. The calculation procedure for determining the climate mitigation co-benefit is considered appropriate. It is also pointed out that the AE will establish a mitigation methodology per investment area aligned with the best practices and existing acknowledged procedures during the due-diligence process of investment.

14. The project has the potential to be scaled and replicated based on its finance approach, which is based on supporting profitable ventures and not a purely grant-financed support project. Therefore, the indirect impact is potentially very high.

15. Overall, the independent Technical Advisory Panel (iTAP) considers that the project has the potential to have a high impact.

1.3 Paradigm shift potential *Scale: Medium*

16. The ACAP Fund will be Pakistan's first private investment vehicle focused on climate adaptation. The ACAP Fund model can be replicated within Pakistan depending on its commercial success and a conducive enabling environment. The success of the project should be able to demonstrate the commercial viability of investing in companies providing services in the area of climate-resilient agriculture with smallholder farmers. In turn, this could further attract private sector capital to Pakistan in the agriculture sector, depending also, however, on having a conducive enabling environment. The project is linked with public and private stakeholders and profits from the on-ground experience of Acumen in this sector, thus facilitating the creation of an enabling environment for investments.

17. The ACAP Fund team plans to collaborate with the AE to publish a report for the international community that details the potential impact of creating an adaptation-focused private investment fund. This could foster the replication of similar ventures in other countries and regions.

18. The capital to be raised for the fund is primarily from DFIs/multilateral development banks and foundations. No firm commitments currently exist, however, which constitutes an impact risk.

19. The investee companies are intended to be financially sustainable enterprises that continue operating after project termination. Companies that can attract later-stage additional investments are targeted, thus expanding their impact. Successful exits from the fund could provide an essential signal to private investors concerning the profitability and risk of investing in this area, thus supporting a paradigm shift. Acumen will also engage with local investors (private equity/venture capital funds and corporates) to develop a syndicate of co-investors for each deal, opening exit options via share buyouts.

20. Pakistan's agriculture sector is more fragmented compared to its neighbours India and Bangladesh, with some structural differences related to land distribution, allocation and use of irrigation water, government intervention in markets (e.g. for seeds), investment in research and development, access to farm credit and an enabling environment to encourage investment in the sector. Policy reforms to enable Pakistan to overcome these barriers are critical. Therefore, active and successful engagement with other relevant public and private-sector players in the agriculture sector is crucial to allow for more enabling framework conditions. This is included as an activity within the project's technical assistance.

21. At USD 80 million, the fund size is minimal considering the Pakistani market and the scope of the investment opportunity presented by the AE. A partnership with other investment firms is not deemed a viable option to enlarge the fund due to the lack of impact-focused

[6] Harmonized IFI Default Grid Factors 2021 v3.2 | UNFCCC

investment funds targeting SHFs and climate-vulnerable rural populations The approach to delivering initial positive results with a limited fund size is acknowledged. This can then serve as a blueprint for a larger fund. This will be required as the initial fund size is considered insufficient to create the momentum and critical mass required for a transformational change towards the influx of private capital into the sector regarding the sector size, complexities and investment risks in Pakistan. Further investments, including that from GCF, are essential for achieving transformational change in this area. The approach of starting with a small initial fund to showcase a successful intervention strategy is deemed appropriate; however, this fund will not be sufficient to achieve a significant transformational impact, although it will be a critical step towards such change.

22. Overall, the iTAP considers that the paradigm shift potential of the project is medium.

1.4 Sustainable Development Potential *Scale: Medium to High*

23. The project is aligned with several Sustainable Development Goals: SDG 1 (No poverty), SDG 2 (Zero hunger), SDG 3 (Good health and well-being), SDG 5 (Gender equality), SDG 6 (Clean water and sanitation), SDG 7 (Affordable and clean energy), SDG 8 (Decent work and economic growth), SDG 11 (Sustainable cities and communities), SDG 13 (Climate action), and SDG 15 (Life on land).

24. Environmental co-benefits are related to reduced usage of fertilizers and organic farming, reduced nitrogen and chemical run-off, efficient water usage, sustainable land use with less post-harvest losses, and decreased pressure on land.

25. Economic benefits will be achieved through investing in businesses with services that will improve the incomes of smallholder farmers in Pakistan. Farmers can improve their incomes through crop diversification, improved efficiencies using less input materials and increased productivity and yields, and improved sales prices of agricultural products through enhanced quality of products and market information.

26. Social benefits will be realized through improving the livelihood of farmers by increasing crop productivity (through the adoption of climate-smart agriculture) and incomes, resulting in enhanced quality of life, time savings (through efficient processing, improved logistics and better access to information), and improved nutrition through increased quantities of food and a healthier diet Through its technical assistance, the project will help farmers adapt to and mitigate the impact of climate hazards on their lives. The project also expects to create around 1,600 jobs over the fund's life.

27. The project seeks to invest and support agribusinesses in Pakistan with a gender lens. The ACAP Fund aims to invest in companies that have gender-smart business practices. It also works through technical assistance with the invested companies in enhancing their gender expertise.

28. Overall, the project's sustainable development potential is considered medium. It can make contributions in the area of gender and have a positive social, economic and environmental impact.

1.5 Needs of the Recipient *Scale: High*

29. The country's vulnerability to climate change is well established. The important agriculture sector suffers from systemic problems that have exacerbated its vulnerability to climate change.

30. Pakistan has large and persistent fiscal deficits, which have led to public debt accumulation, crowding out private sector borrowing and posing macroeconomic risks. As of

2022, personal sector credit was only 15 per cent of GDP.[7] Limited access to credit results in low levels of private investment. High-level government borrowing from the financial sector crowds out private-sector financing and has kept the financial system from extending finance to riskier and vulnerable sectors like agriculture.

31. International private investors have little interest in climate-financing opportunities due to high return on investment barriers in Pakistan, high currency risks and perceived political instability. The most significant challenges for local investors are credit risk and the long-term tenure of climate projects.

32. The iTAP considers that the need of the recipient is high.

1.6 Country ownership *Scale: High*

33. The nationally determined contribution (NDC) of Pakistan includes nature-based solutions, the development of land-use change initiatives, and resilient community infrastructure prominently. The updated NDC, the national climate change policy and the national adaptation plan (NAP) of July 2023 support initiatives in the agricultural field to increase climate resilience. The updated NDC has, for example, been a key area of the transition towards advanced technologies in the agriculture sector, specifically through the adoption of drought-resistant seed varieties. The NAP highlights the critical nature of the agriculture-water nexus and the need for climate-smart agriculture practices. The NAP lists green jobs and livelihoods as one of its six pillars and acknowledges the importance of sustainable agriculture and land-use practices. The project also aligns with Pakistan's food security targets as outlined in *Pakistan Vision 2025*.[8]

34. The no-objection letter of the Government of Pakistan has been provided.

35. The project has made a stakeholder assessment and responded satisfactorily to the issues raised during the assessment. Acumen has almost two decades of experience in Pakistan and has engaged with various stakeholders, including government bodies, the private sector, finance institutions (national and international), and academic and research institutions.

36. Acumen has local country operations and expertise. It intends to collaborate with the executing entity to monitor and report ACAP Fund activities and impacts with the national designated authority. The accredited entity will also monitor ACAP Fund activities and impacts to ensure alignment with country priorities and climate ambitions.

37. The iTAP assesses the country ownership as high.

1.7 Efficiency and Effectiveness *Scale: Medium*

38. The ACAP Fund is a blended finance fund targeting a 1:2 junior to senior equity structure. The blended finance structure is important given the risk perspectives of investors and enables an influx of private capital to the agriculture sector and into the resilience field. The accredited entity intends to establish a USD 80 million, early growth and growth stage agriculture investment fund with a 10-year term (subject to two one-year extensions). The target capital structure is USD 27 million junior and USD 53 million senior equity tranches.

39. ACAP targets DFIs, multilateral banks, foundations, family offices, and commercial investors. It remains open to how much private capital can be attracted. Annex 7 to the funding

[7] World Bank. 2022. *Pakistan Development Update: Financing the Real Economy*. Available at: https://thedocs.worldbank.org/en/doc/410d0506bba8afc6fd9d9541148bfe4d-0310062022/original/PDU-April-2022-April18-ForWEB-Final.pdf

[8] Available at: www.pc.gov.pk/uploads/vision2025/Pakistan-Vision-2025.pdf

proposal (Summary of consultations and stakeholder engagement plan) lists the multilateral banks, governments, United Nations organizations and international non-governmental organizations consulted.

40. The accredited entity requests USD 25 million junior equity capital from GCF. This vehicle and the anchoring investment of GCF should enable Acumen to raise USD 2 million additional junior equity and USD 53 million senior equity from investors with the same goals but a lower risk appetite. Given the fund's risk profile, and despite the junior equity enhancement provided by GCF, this return might be insufficient to attract private-sector investors. GCF participation is considered material to provide de-risking in attracting additional capital. GCF capital can thus be leveraged by a factor of 2.2.

41. Acumen also expects to raise USD 10 million in grant finance for the technical assistance fund, USD 3 million of which will come from GCF and USD 7 million from DFIs, foundations and family offices. GCF grants could thus be leveraged by a factor of 2.3. The technical assistance component is 12.5 percent of the total target investment fund[9].

42. The total investment amount is USD 90 million, USD 28 million from GCF, with an expected co-financing ratio of 79 per cent and a leverage factor of 2.2.

43. The fund's investment approach is to make equity, mezzanine and debt investments in venture, early growth and growth stage agribusinesses, which cater their services to farmers and aim to improve farmers' income and resilience to climate change. The ACAP Fund is a for-profit fund. This can potentially increase the impact, replicability and sustainability, and bring efficiency and effectiveness to investment decisions.

44. The overall iTAP assessment concerning efficiency and effectiveness is considered medium.

II. Overall remarks from the independent Technical Advisory Panel

45. The iTAP endorses this funding proposal based on the assessment above.

[9] This is a higher percentage share than in some other comparable programs but justified based on the focus on resilience and the targeted sector.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP229)

Proposal name:	Acumen Climate Action Pakistan Fund
Accredited entity:	Acumen Fund Inc.
Country/(ies):	Pakistan
Project/Programme size:	Medium

Impact potential

Acumen is pleased to note that the note that the ACAP Fund impact potential has been given a "high" rating. As noted by the Panel, each investment theme of the fund has a specific climate rationale and will have a direct impact on the climate resilience of smallholder farmers and climate vulnerable rural populations. In addition, the fund has been designed as a blended finance instrument that mitigates the real and perceived risks of investing in frontier markets like Pakistan and can unlock considerable indirect impact through its potential to be replicated and scaled.

We recognize that while a subset of program investments (such as those focused on financial services) may be targeting climate vulnerable farmers that are more bankable, at a program level Acumen remains focused on maximizing impact on the most vulnerable. To that end, in addition to investing in companies that can impact highly climate vulnerable, low-income communities, the program intends to also strategically deploy its TA facility to ensure that investments extend and amplify their impact on the most vulnerable – including unbankable farmers, women and landless rural populations.

Paradigm shift potential

Acumen notes iTAP's balanced and helpful discussion of the program's paradigm shift potential. As highlighted in the commentary, ACAP will be Pakistan's first private investment vehicle focused on climate adaptation. The program aims to build on Acumen's on-ground experience, engaging a diverse set of public and private stakeholders towards developing a collaborative and enabling environment for climate focused investments.

We note iTAP's encouraging comment about the scope of investment opportunities and agree with the need for transformational change in this sector. In designing ACAP, Acumen has leveraged its learnings from over two decades of impact focused investing in challenging regions around the world and approached fund development through a stage wise scaling strategy as a pathway to sustainable transformation. ACAP is a first-time fund that can serve as an opportunity to jumpstart activity in a market, with strong potential for scaling and replication. If the fund is successful, the model can then lead to larger follow-on funds that can further scale the impact and transform the sector.

Sustainable development potential

The iTAP's "medium to high" rating of the program's sustainable development potential is noted. As the commentary suggests, along with important environmental co-benefits, the program has a range of economic and social benefits that include improved farmer incomes, livelihoods, food security, and overall quality of life.

The comments regarding the program's emphasis on gender impact are well noted. Given the importance of advancing gender equity for the region and the sector, Acumen considers this an important development indicator for the fund. Social impact remains at the core of fund development and will be a key part of program implementation.

Needs of the recipient

Acumen appreciates the thorough iTAP review and is pleased to note the positive alignment between ACAP's interventions and the needs of the recipient. As the commentary notes, Pakistan faces significant structural and macroeconomic challenges in responding to climate change, even as its extreme vulnerability is well documented. This makes a program like ACAP, that can de-risk and catalyze climate finance towards where it is most needed even more important.

The ITAP's conclusion that the recipient has a high need for the ACAP is a welcome validation of the program's relevance and its potential impact on creating a climate resilient agriculture sector in Pakistan.

Country ownership

Acumen welcomes iTAP's positive review of country ownership with a "high" rating. The program aligns with Pakistan's recently updated NDCs and national adaptation plans, as well as Pakistan's food security targets. Acumen's on-ground presence and extensive engagement with key stakeholders has been a key aspect of fund development and design. Throughout the program development phase, the fund team has consulted extensively with various ministries and public sector entities to ensure engagement and national alignment. As such, we remain confident of consistent country level support for the program and its impact.

Efficiency and effectiveness

Acumen notes the iTAP's positive review of efficiency and effectiveness of its blended finance structure, targeting a 1:2 junior to senior equity ratio to mitigate investor risks and attract private capital to the agriculture sector. The funding proposal underscores the catalytic role of GCF's funding in attracting additional capital from private sector investors.

The ACAP Fund aims to improve impact, replicability, and sustainability in the agriculture sector. Therefore, the positive iTAP assessment efficiency and effectiveness rating of "medium" is well noted.

Overall remarks from the independent Technical Advisory Panel:

Acumen notes the positive remarks from the iTAP on this program, along with recognition of Acumen's track record, and on the ground experience. Acumen also notes, with thanks, the positive endorsement of the funding proposal from iTAP to the GCF Board.

Acumen Climate Action Pakistan Fund Gender Assessment

November 2023

Executive Summary

The Acumen Climate Action Pakistan Fund (ACAP) seeks to build climate resilience for the agricultural sector and smallholder farmers in Pakistan by investing in impactful agribusinesses. Smallholder farmers are increasingly vulnerable to the impacts of climate change in Pakistan ranging from extreme heat to floods that have devastated over 1/3 of Pakistan's farmers. The impacts of climate change have been damaging as Pakistan has suffered from soil degradation, disease outbreaks, crop damage, and reduced crop diversity. Farmers have experienced decreased yields, less income, and more vulnerability to extreme weather events. By investing in agribusinesses selling products and services to smallholder farmers, we hope to bolster farmer resilience with access to farmer platforms and digital marketplaces, smart farming, and post-harvest technologies.

The ACAP Fund is based on the premise that if investment capital and technical assistance are provided to small and growing agribusinesses in combination with the development of strategic private-public partnerships then agribusinesses and Pakistan's agricultural sector at large will have demonstrated a viable adaptation pathway to creating a more resilient agri-ecosystem and food system. This will in turn support vulnerable farmers and catalyze private capital because the investment and partnerships will bolster adoption of climate resilient business models in Pakistan's increasingly vulnerable agri-sector, enhancing its ability to adapt to climate threats impacting food security.

To ensure the effectiveness of the ACAP activities, it is important to understand the differences between men and women in Pakistan, the role of women in this project, gender differences on the impacts of climate change, and vulnerabilities and barriers that are important to the stakeholders of this project.

The Acumen Climate Action Pakistan (ACAP) Fund has developed this gender assessment to understand the important role of women in agriculture in Pakistan, and how ACAP can play a positive role for gender equity in agriculture in Pakistan. To that end, ACAP has engaged in extensive stakeholder engagement and desktop research to assess the gender dynamics within agriculture in Pakistan.

The assessment includes an overview of the evidence on women's access to agricultural inputs and services, women's participation in the agribusiness sector workforce, and the key barriers to advancing equality in the sector. This document outlines key entry points for the ACAP initiative, followed by detailed in-country assessments focusing on health, education, legal rights, human rights, SEAH, vulnerable populations, sector specific gender outcomes, and project barriers and opportunities.

The ACAP gender assessment has informed the Gender Action Plan and the intended activities of the fund.

Objective of the Assessment

Rationale

ACAP seeks to build the climate resilience of farmers and agribusiness with a gender lens. The project's impacts, outputs, and activities are enhanced when women are included. Women are an important part of the social, cultural, and economic fabric of Pakistan. Gender dynamics, cultural norms, opportunities, and barriers play a significant role in Pakistan's history and economy. In order to achieve ACAP's goals, the project must investigate the impact of gender on this project and how ACAP can impact gender.

Ensuring strong gender-based outcomes aligns with Acumen and the Green Climate Fund's gender initiatives. Acumen and GCF both mandate that GCF-funded projects include a gender assessment to ensure best practices and strong gender-based outcomes. This assessment aligns with Acumen's Gender Sensitivity Policy for GCF-Funded Projects and GCF's Gender Policy.

Objective

ACAP aims to incorporate Pakistan's gender context, barriers, and opportunities into the project using the findings of this gender assessment. The gender assessment examines Pakistan's gender-specific roles in agriculture and society, constraints and needs, and will impact the activities and goals shared in the ACAP funding proposal and Gender Action Plan

Methodology

The project's gender research is based on both desktop research and stakeholder engagement. Our project benefits both from academic and official research and the voice of stakeholders to ensure that we have a broad and deep understanding of how gender dynamics may impact project activities and outcomes.

Desktop research included reviewing relevant Pakistani policies and laws, public commitments and goals, government track record and demographics, budgets, and other public policy initiatives. Our research also included research from the Pakistani government and outside entities like the United Nations, the Food and Agricultural Organization, the World Bank, and other reputable organizations. Finally, our research was bolstered by academic research on gender, agriculture, human rights, health, education, and other relevant topics.

ACAP also engaged a diverse range of stakeholders to learn about gender dynamics within Pakistan. ACAP met with civil society organizations, women's organizations, public leaders, agribusiness management and staff, and both male and female farmers. Their insights and experience guided much of the ACAP gender assessment and Gender Action Plan.

The ACAP team held stakeholder consultations with Aurat Foundation, which is a Pakistani NGO working extensively on advocacy and action, with extensive grassroots links, and strong relations with women parliament members) in order to improve our understanding of climate change, rural women and applicable labor laws.

Pakistan Gender Assessment

Overview of gender equality and gender gaps in the country

Pakistan has a young and growing population. At 227 million people (49.2% female; 50.8% male), it is the fifth largest country in the world. While women in Pakistan have made measurable progress across several fronts, they still face significant challenges compared to the rest of the world.

According to the Global Gender Gap Index Report 2022, Pakistan ranks higher on political empowerment at 95/156, but near the bottom on economic participation and opportunity (145/156), educational attainment (135/156), and health and survival (143/156).[1] With the uptake in female participation in politics, Pakistan has improved it's ranking in the World Economic Forum's Global Gender Gap political participation ranking at 95/146. With that context, there are still very few female politicians and government leaders. Moreover, Pakistan's election data shows that there is a 12.5 million person voting gap based on gender.[2]

In this assessment, we will review legal status and rights for women in Pakistan, women's health in Pakistan, women's economic participation and entrepreneurship in Pakistan, gender and farming, and opportunities for gender equity activities for ACAP.

[1] UN Women (2023), https://asiapacific.unwomen.org/en/countries/pakistan

[2] World Economic Forum (2021), https://www3.weforum.org/docs/WEF_GGGR_2021.pdf

Legal Status of Women

Legal protections for women were enshrined in the Constitution of Pakistan in 1973. The Constitution has provisions for the equality of women in several articles. Article 3 contains a commitment to eliminate the exploitation of women. Article 25 (1) and (2) states that "all citizens are equal under the law and are entitled to equal protection of law" and "there shall be no discrimination on the basis of sex alone," respectively.

Women in Pakistan have earned increasingly strong legal protections and rights in recent history. Last year, the government of Pakistan developed the National Gender Policy Framework[3], and the Anti-Rape (Investigation and Trial) Ordinance and the Domestic Violence against Women (Prevention and Protection) Act in all four provinces of Pakistan in 2020.

Women are allowed to vote, run for elected office, and hold government positions. Women are also allowed to own property.

Rural women in Pakistan have high illiteracy rates and have very limited civic education on their protections and rights, which has limited their understanding of laws and protections. Beyond lacking awareness of laws and legal protections, many women in Pakistan are unaware of resources available to them like shelters. Additionally, the systems in place seem remote, expensive, and inaccessible for many women in Pakistan.

Women also lack financial resources that may enable their legal protections. One study found that 78% of women and girls reporting abuse did not have any income.[4]

Moreover, there is limited female representation in the legal sector including in the police, prosecution, and the courts. There are no women in the Pakistani Supreme Court and, as of 2018, only 14.54% of judges are women.[5] The limited representation of women in the Pakistani legal system has impacted the perception of inaccessibility of the legal system.

Beyond perceived inaccessibility, there are also other issues with inaccessibility including a dearth of courts and police stations in rural and remote areas. Both police officials and judges alike encourage female victims of rape and sexual assault to mediate their solutions rather than use formal channels for resolving SEAH activities. Pakistani people have used Dispute Resolution Councils comprised of community elders and respected individuals to resolve civil disputes. DRCS have been quick and effective but there are far too few for the number of cases outstanding. Women in remote parts of Pakistan have to travel far to access DRCS. Adding to the challenges for women, court rooms and police stations have limited infrastructure for women like women's rest rooms, sitting rooms, and private rooms.

Some researchers have noted that police in Pakistan have exhibited bias towards women when reporting and investigating SEAH cases. They also have limited resources for forensics investigations of women and researchers have reported inefficient and dysfunctional staff making evidence gathering more difficult.

Lawyers and prosecutors have also had limited success prosecuting rapists or defending women. If women are painted as promiscuous or of low moral character, they are far less likely to succeed in court. Moreover, the previously mentioned challenge with evidence gathering has made prosecuting SEAH crimes far less successful.

[3] Ministry of Planning Development & Special Initiatives (2023) https://www.pc.gov.pk/web/gender

[4] Corridors of Knowledge for Peace and Development, Jan. 1, 2020, pp. 304-322

[5] Livia Holden (2019) Women judges in Pakistan, International Journal of the Legal Profession, 26:1, 89-104, DOI: 10.1080/09695958.2018.1490296, https://doi.org/10.1080/09695958.2018.1490296

Courts have been dominated by male judges who have used harsh sentencing techniques and strict interpretations of laws that have disproportionately harmed women. A 2003 study found that 80% of female prisoners were unable to provide evidence of rape and were convicted of adultery instead.

While these are explicit rights that are constitutionally or legally mandated, implementation of these rights and the reality on the ground have different outcomes. Lack of knowledge of legal rights and protections remains to be a significant barrier for women.

Gender and Policies

The Pakistan government developed and now executes the Agriculture and Food Security Policy to detail their role as regulator and facilitator of the agriculture sector in Pakistan. The policy seeks to build an inclusive sector that supports smallholder farmers and reduces sharecropping. The policy also aims to improve resource scarcity and degradation issues relating to land and water. As shared in the funding proposal, misuse of fertilizer has significantly impacted biodiversity and water. Lastly, the policy also has a goal of reducing hunger and malnutrition. The policy does not explicitly have a section dedicated to gender, but the focus on vulnerable people will disproportionately help women.

Commonly Held Perceptions on Gender

Some researchers have noted that large parts of Pakistan’s society share and uphold patriarchal views. Men are expected to be the primary income earners, decision makers, and the head of the household. Men are also expected to have leadership positions in both the public and private sector. Many in Pakistan expect men to have stereotypically masculine behaviors.

Women are expected to be traditionally feminine in Pakistan. Many in Pakistan’s society expect women to be caretakers of children and parents and to maintain the home. They are expected to support men’s activities but to not lead the household or the farm.

Researchers have shared that some of these deeply held patriarchal beliefs come from local and conservative interpretations of religion. Cultural norms also play an important role in the commonly held beliefs around gender.

These norms are played out in segregated spaces in the home for men and women, differences in attire including veiling for some women, and access to public and private spaces.

Additionally, research has shown that commonly held beliefs and perceptions on gender that resulted in stark gender inequality have also contributed to violence against women.

Perceptions on gender equity do change when people have more access to education and have more wealth.[6]

Division of Labor

Women have far fewer opportunities to participate in the formal economy than men. According to Nature, only 22% of women participate in the formal labor force compared to 84% of men. On a positive note, female labor force participation as a percentage of male labor force participation jumped from 13.32% in 1990 to 27.18% in 2018. However, this number is the lowest in the region.[7] As mentioned in the agricultural section, women primarily work in the agricultural sector. There is virtually no female participation in the mining and quarrying industry. Most women work as skilled agriculture, elementary,

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[6] Khalid, M. (2021) Assessment of Gender-Role Attitudes among People of Pakistan. Open Journal of Social Sciences, 9, 338-350. doi: 10.4236/jss.2021.912023.

[7] Khan, M.Z., Said, R., Mazlan, N.S. et al. Measuring the occupational segregation of males and females in Pakistan in a multigroup context. Humanit Soc Sci Commun 10, 5 (2023). https://doi.org/10.1057/s41599-022-01498-6

craft and related trade workers. The study in Nature concluded that both economic and sociological theories partially explain the pattern of occupational segregation in Pakistan.

61% of women working in the formal labor market are in urban areas compared to 39% in rural Pakistan.[8] Women account for 65% of the $2.8 billion USD informal sector which is largely based in rural areas.[9]

In Pakistan, men are considered to be the primary earners while income from women are considered to be a secondary benefit. Due to some gendered social norms, women are primarily tasked with informal labor including running the household, raising children, and taking care of elders.

Education

The Pakistan Constitution guarantees the right to free and compulsory education to all children between the ages of 5-16 years old. Additionally, the Constitution codifies that religion cannot be forcefully taught to youth. Pakistan's government has made improvements to their education system a key policy goal. The government has funneled tremendous sums of money into education in the last few years to mixed results.

Approximately 22.5 million children are out of school right now in Pakistan.[10] Almost a third of primary aged girls are out of school compared to just one fifth of boys.[11] By ninth grade, only 13% of girls are still participating in school.[12] Girls only average 3.9 years of schooling compared to 6.4 for boys. Punjab is the only province in Pakistan that just over half of girls completed at least primary school while only 19% of girls completed at least primary education in Balochistan.[13]

This is partially driven by cultural and familial perceptions of the value of girl's education to their economic opportunities. The gender gap and the overall lack of educational participation is also driven by difficulties accessing schools in rural communities. At a systemic level, limited educational financing, challenges with implementing policies and national education goals, and inequitable distribution of resources has hampered the country's educational growth, particularly for the adolescent girl demographic.

On the positive side, the promulgation of Article 25-A of the Constitution after the Devolution makes education a right and obligates the State to provide free and compulsory education to all children of the age 5 to 16. Article 37-B of the Constitution also reiterates that 'the State of Pakistan shall remove illiteracy and provide free and compulsory secondary education within minimum possible time.[14]

Health and Gender

[8] Khan, Tasnim, and Rana Ejaz Ali Khan. "Urban Informal Sector: How Much Women Are Struggling for Family Survival." *The Pakistan Development Review*, vol. 48, no. 1, 2009, pp. 67–95. *JSTOR*, http://www.jstor.org/stable/41260908. Accessed 18 Aug. 2023.

[9] UN Women (2016), https://asiapacific.unwomen.org/-/media/field%20office%20eseasia/docs/publications/2016/05/pk-wee-status-report-lowres.pdf?la=en&vs=5731

[10] UNICEF (2023), https://www.unicef.org/pakistan/education#:~:text=An%20estimated%2022.8%20million%20children,population%20in%20this%20age%20group.

[11] https://edufinance.org/latest/blog/2021/empowering-girls-through-education-in-pakistan#:~:text=In%20Pakistan%2C%20an%20estimated%2022.5,compared%20with%2021%25%20of%20boys.

[12] Global Citizen (2018), https://www.globalcitizen.org/en/content/girls-pakistan-education-report-2018/

[13] Georgetown WPS (2021) https://giwps.georgetown.edu/index-story/consistently-low-rates-of-womens-inclusion-across-pakistans-provinces/

[14] Pakistan Constitution (2023), https://www.infopakistan.pk/constitution-of-pakistan/article/37-Promotion-of-social-justice-and-eradication-of-social-evils#:~:text=b.,education%20within%20minimum%20possible%20period.

Women's life expectancy in Pakistan is 69 years old as of 2020.[15] The Pakistani fertility rate has significantly dropped in recent decades and reached 3.6 births per woman.[16] Birth rates are higher in rural Pakistan. The maternal mortality rate has also been dropping over the decades and, in 2017, was at 140 women per 100,000 live births.[17] Pakistan's infant mortality rate also decreased from 185 in 1960 to 54 babies per 1,000 live births.[18] Researchers and statisticians have found it difficult to get accurate data on rural versus urban health statistics in Pakistan.

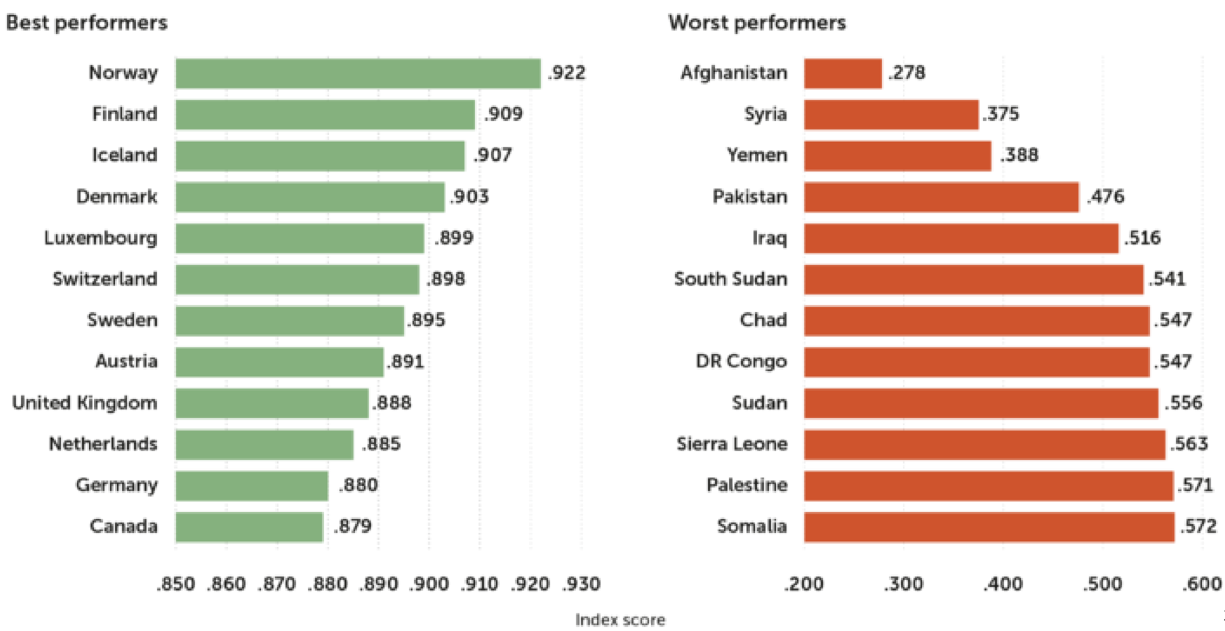
A large number of low-income women in Pakistan do not have access healthcare facilities. There is a limited number of healthcare facilities near rural communities. Women in rural communities will have to travel to urban areas for healthcare. Both travel and healthcare can be expensive, which is another limiting factor.

While the maternal mortality rate significantly dropped over the last half century, it is still one of the highest rates in Southeast Asia. Researchers have noted needs for improvements in maternal healthcare in Pakistan. Women reported having limited access to the right doctor and felt unprepared for birth. They were not appropriately engaged in the decision-making process nor given a birth plan or counseling.

Additionally, educational disparities, malnutrition, poverty, and violence against women have contributed to the poor maternal health outcomes and other health related issues.

Women's Economic Participation and Entrepreneurship

The dozen best and worst performers on the WPS Index 2021



[19]

The Georgetown Institute for Women, Peace and Security (WPS) ranked 167 out of 170 on their WPS index in 2021 after finding very low levels of inclusion for women across regions in Pakistan.

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15 World Bank (2023), https://data.worldbank.org/indicator/SP.DYN.LE00.FE.IN?locations=PK

16 World Bank (2023), https://data.worldbank.org/indicator/SP.DYN.TFRT.IN?locations=PK

17 World Bank (2023), https://data.worldbank.org/indicator/SH.STA.MMRT?locations=PK

18 World Bank (2023), https://data.worldbank.org/indicator/SP.DYN.IMRT.IN?locations=PK

19 Georgetown WPS (2021) https://giwps.georgetown.edu/index-story/consistently-low-rates-of-womens-inclusion-across-pakistans-provinces/

Women in Pakistan have faced significant barriers towards work, income, and wealth generating opportunities. Women only make up 22.63% of the Pakistani labor force. The average Pakistani woman only makes 16.3% of a man's income. Women make up 81 percent of the 5.26 million Pakistani people working in the informal sector. Even as women overwhelmingly contribute to the informal sector, accounting for 65% of the value of the informal economy, they still only make $15-20 USD per month.[20] Low pay is related to low-income security, few social protections, high economic vulnerability, and occupational health issues.

Pakistan has some of the worst female employment rates across Balochistan, KPK, and Sindh, which is around 10%. These would mirror the worst countries in the world.

According to the World Bank, women only account for 1% of entrepreneurs in Pakistan.[21] The lack of free movement and limited access to skills and technology needed for entrepreneurship and formal labor jobs restricts opportunities for women in Pakistan. Additionally, women lack support networks, business premises, training, and agency assistance. Researchers have noted that societal norms like patriarchy and deeply held beliefs about gender roles have limited women's opportunities for entrepreneurship.[22]

Women are also unlikely to receive encouragement from male family members further hampering mobility and fundraising. A World Bank Report found that less than 25% of female entrepreneurs are microfinance borrowers. Nearly 68% of women borrowers needed a male relative's permission in order to qualify for any kind of loan.[23]

Given the scarcity of female entrepreneurs, researchers have studied the characteristics of female entrepreneurs. Successful female entrepreneurs in Pakistan have demonstrated internal factors including self-confidence, risk taking, and need for achievement, and external factors including economic and socio-cultural privileges.[24]

Vulnerable Groups

Female Headed Households

In 2018, female headed households in Pakistan comprised 12.5% of all households.[25] This increased from 10.4% in 2007. These households primarily led by women face a heightened risk of poverty due to several factors, including the marital status of most household leaders, their limited access to productive resources, income, and essential services. Additionally, the weakening of traditional familial support systems, the household's size and composition, and other related factors contribute to their vulnerability. Furthermore, these women encounter the added challenge of balancing market-oriented activities with their domestic responsibilities. Given their available education and training, as well as their dual roles as mothers and workers, their employment options primarily lie within the informal sector. Unfortunately, within this sector, opportunities for impoverished women are severely limited, reflecting the cultural norms in Pakistan that restrict females to roles where gender segregation can be maintained.[26]

Gender and Farming

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[20] UN Women (2023) https://asiapacific.unwomen.org/en/countries/pakistan/economic-empowerment-and-sustainable-livelihood#:~:text=Women%20account%20for%20a%20mere,81%20per%20cent%20are%20women.
[21] Women Entrepreneurship Finance Initiative (2020) https://we-fi.org/enabling-women-entrepreneurs-in-pakistan-through-training-and-legal-reform/#:~:text=In%20Pakistan%2C%20only%20one%20percent,loans%20and%20financial%20know%2Dhow.
[22] Roomi, M. A., & Parrott, G. (2008). Barriers to Development and Progression of Women Entrepreneurs in Pakistan. The Journal of Entrepreneurship, 17(1), 59–72. https://doi.org/10.1177/097135570701700105
[23] World Bank (2012), https://www.worldbank.org/en/news/feature/2012/10/17/are-pakistans-women-entrepreneurs-being-served-by-the-microfinance-sector
[24] Khan, R.U., Salamzadeh, Y., Shah, S.Z.A. et al. Factors affecting women entrepreneurs' success: a study of small- and medium-sized enterprises in emerging market of Pakistan. J Innov Entrep 10, 11 (2021). https://doi.org/10.1186/s13731-021-00145-9
[25] World Bank (2018), https://data.worldbank.org/indicator/SP.HOU.FEMA.ZS?end=2018&locations=PK&start=1991&view=chart
[26] Mohiuddin, Y. (1992). Female-headed households and urban poverty in Pakistan. In Women's work in the world economy (pp. 61-81). London: Palgrave Macmillan UK.

Women play a pivotal role in Pakistan's agrarian economy. Agriculture employs 65% of the female labor force and unfortunately, almost all of it is unpaid.[27] Farming is the main labor activity of rural women with approximately 75 percent of women and girls employed in the sector.[28] While female roles vary across regions, their on-farm work is intensive and significant. Despite being essential to the entire process, gender inequity in this largely traditional sector restricts women's ability to reap the benefits of their work. They are predominantly engaged in labor intensive tasks such as manual harvesting, seed transportation, and livestock related activities.

Most roles for women in agriculture are unpaid, underpaid, unrecognized, and entirely informal. Moreover, many women employed in agriculture also conduct most of the unpaid household labor including child rearing and maintaining their homes. Most women providing informal labor in the agricultural sector are unprotected by local labor laws and are therefore more likely to be exploited in their work.

Only 4% of landowners in Pakistan are women. The empowerment of women in the agriculture sector is integral to its growth - bridging the gender gap could potentially increase agricultural output by 2.5 to 4% in developing countries.[29] Investing in this sector has a spillover effect on gender equality in agriculture and large-scale investment in agriculture has been known to have a positive economic impact on women's direct employment status, value chain participation, skills development, and land/capital access. This improved and more equitable access to productive resources for women can increase overall yields by 20-30%.[30]

Gender and Access to Agricultural Products and Services

Women-managed farms, as acknowledged earlier, are a very small subset of farms in Pakistan. Data has shown that women-managed farms are 30% less productive than farms managed by men.[31] In Pakistan, men tend to have control over finances, financial decision making, and access to agricultural products and services.

Overall, women have less access to assets, training opportunities, and services. Shared below are examples of gender differences in access to agricultural products and services including: land and livestock, agricultural extension services, and agricultural credit.

<u>Land and Livestock</u>

11% of married women ages 15 to 49 years old own homes and only 4% own land. In almost all cases, men are very likely to own agricultural land due to both wealth and socio-cultural norms.[32]

Rural homes are more likely to own livestock. There is limited information about women ownership of livestock. However, given their limited resources and income, women are less likely to acquire livestock.

USAID trained over 35,000 women in livestock management and horticulture, which they claim has helped create full-time equivalent jobs in agriculture and agribusinesses.

<u>Agricultural Extension Services</u>

[27] Trading Economics (2023) https://tradingeconomics.com/pakistan/employees-agriculture-female-percent-of-female-employment-wb-data.html#:~:text=Employment%20in%20agriculture%2C%20female%20(%25,compiled%20from%20officially%20recognized%20sources.
[28] FAO (2023), https://www.fao.org/3/cc5970en/cc5970en.pdf
[29] FAO (2010) https://www.fao.org/publications/sofa/2010-11/en/#:~:text=If%20women%20had%20the%20same,countries%20by%202.5%E2%80%934%20percent.
[30] FAO (2010), https://www.fao.org/publications/sofa/2010-11/en/
[31] CABI (2023) https://blog.plantwise.org/2023/01/17/women-farmers-in-pakistan-arent-realising-their-potential-heres-why/
[32] National Institute of Population Studies [Pakistan] and ICF International. 2013. Pakistan Demographic and Health Survey 2012-13. Calverton, Maryland, USA: National Institute of Statistics and ICF International.

Pakistan's various government entities are deeply engaged in agricultural extension services, transferring expertise and knowledge to smallholder farmers in rural regions of the country. Provincial departments have taken a lead on providing these services to smallholder farmers. There is limited knowledge on the gender aspect of extension services.

USAID has also claimed that over 30,000 women farmers have used new technologies and management practices that resulted in increased yields and productivity.[33]

Agricultural Credit

In Pakistan, the high cost of deploying credit to informal entities has resulted in limited credit opportunities for men and women smallholder farmers. Many agribusinesses operate in the informal economy and therefore have limited incentives to formalize their business practices. Given their informal nature, including informal structure, and remaining unregistered, commercial creditors like formal banking institutions have limited incentives to work with many agribusinesses and smallholder farmers. Additionally, the high cost of credit has limited formal agribusinesses from accessing credit.

As men tend to own agribusinesses, land, and products and services, they are more likely to have the collateral and necessary inputs to access credit. Women are much less likely to access credits through formal institutions. Both men and women, especially those in rural areas of Pakistan, have low levels of access to agricultural credits.

Some smallholder farmers have accessed credit via informal relationships with friends and family but that is not well documented.

The First Women Bank in Pakistan has encouraged female farmers to establish asset ownership and has worked with almost 3,000 women in rural areas working in the carpet industry.[34]

Gender and Climate Resilience

A recent ranking of 87 low- and middle-income countries on a climate-agriculture-gender inequality hotspot index, developed by CGIAR researchers, shows Pakistan ranking number one in Asia. These hotspots are defined as areas where climate hazards, women's exposure to climate hazards because of their involvement in agriculture, and women's vulnerability due to prevailing gender inequalities intersect.

What's more, a province-level analysis highlighted Balochistan, Punjab and Sindh as hotspots within the country. Women farmers in these provinces engage heavily in rice and oil seed production, and are often solely responsible for weeding, seed cleaning and storing of crops as well as for dairy farming. This puts them at risk of suffering under current floods and future climate disasters.

During a disaster, women farmers are not compensated for crop losses as they do not own the land they toil on. In the wake of the 2010 floods, the government of Pakistan devised a crop insurance scheme that could only be availed by farmers who owned up to 25 acres of land.[35] This meant that women stood to gain nothing from this relief measure.

Women farmers' ability to cope with floods and other climate shocks also hinges on their access to credit, knowledge and skills. Only 5.3 percent, 1.6 percent and 0.9 percent of rural women in Punjab, Sindh and Balochistan, respectively, own and use a bank account, and they hardly get loans from formal channels as they have barely any land to offer as collateral.

Finally, social norms dictate for Pakistani women to be homebound and fulfil household responsibilities. They have limited time and opportunities to acquire new knowledge and skills that could help them cope

[33] USAID (2022), https://www.usaid.gov/pakistan/our-work/gender-equality-and-female-empowerment
[34] First Women Bank of Pakistan, Ltd. (2023), https://fwbl.com.pk/about-us/#history
[35] CGIAR (2022) https://gender.cgiar.org/news/rural-women-pakistan-are-most-affected-apocalyptic-floods#:~:text=In%20the%20wake%20of%20the,nothing%20from%20this%20relief%20measure.

in crises. A 2018 report highlighted that a lack of training opportunities prevent the country's rural women from adding value to agricultural products and services, limiting their earning opportunities.

In sum, these factors – lack of access to land, loans and learning – result in women having limited resources to cope with and recover from climate disasters.

Pipeline Companies

ACAP has interviewed pipeline companies to better understand the gender dynamics within companies in Pakistan. As ACAP has only begun fundraising and cannot make any investing decisions, companies were willing to give limited information about their gender commitments and staff composition.

Generally, across the 10 entities the ACAP team interviewed, the staff consisted of between 5-30% female employees. Several staff had zero employees and several reached higher than 30%. This is similar to what academics and economists have found about traditional employment for females in Pakistan.

Several companies noted that they seek to serve and support female farmers. Particularly, several companies noted that their products and services can create efficiencies and reduce the physical labor demanded of women.One company shared that their products and services could improve nutrition for women and girls.

Additionally, the ACAP team asked companies about interest in gender equity initiatives and technical assistance to support their work. Several companies noted that they had barriers to achieving gender equity in their organizations and that they would benefit from additional expertise. Others noted that they wanted to reach more female farmers. While there are limits to the gender equity work that can be accomplished in Pakistan, we note that companies share our enthusiasm for engaging on this difficult and important topic.

ACAP Program Activities, Outcomes, and Engagement

ACAP and Gender

Given the significance of women in agriculture and the gender inequity within the sector, ACAP will be applying a gender lens across all investments. We will be using the 2X Challenge criteria to ensure that we will be making more gender smart and inclusive investments and moving portfolio companies toward gender equity and create sector examples. The 2X Challenge was launched at the G7 Summit 2018 as a bold commitment to inspire DFIs/IFIs and the broader private sector to invest in the world's women and to measure value created by women's contributions. Investments that use 2X criteria reflect the multiplier effect of investing in women.

ACAP will do due diligence with a gender lens on potential portfolio companies. Companies will be asked about their gender policies, SEAH (sexual exploitation, abuse and harassment), and their impact on female customers.

ACAP will also provide technical assistance to improve opportunities for women smallholder farmers, support more equitable agribusinesses, and strengthen gender equity in the agriculture sector in Pakistan. The team seeks to use the TAF to increase female farmer awareness of climate resilient agriculture practices and improve gender practices of businesses.

Companies will be required to report sex-disaggregated jobs and impact numbers when feasible. Moreover, companies are expected to have gender action plans that they report progress on to ACAP. ACAP will monitor portfolio companies for gender incidents, issues with SEAH, and improvement on gender-based policies, practices, and protections.

SEAH and ACAP

ACAP has created strong SEAH protections at the fund level and across investments to ensure that the fund, companies, staff, and beneficiaries have strong, accessible, and transparent SEAH protections and recourse if incidents occur.

ACAP will have fund level protections for staff incorporated in the HR policies and by adhering to local sexual harassment laws. The team will also have access to the ACAP grievance mechanism.

ACAP will due diligence companies for SEAH protections, monitoring, and investigative functions. Gaps are expected to be addressed in the portfolio company-level Gender Action Plan.

Some companies may have insufficient workplace or customer SEAH protections, monitoring tools, and investigative functions. Early stage companies are often formalizing policies and procedures.

As this is a risk to both the company and ACAP, the ACAP team may determine that the company is eligible for technical assistance to help improve SEAH policies, procedures, training, and activities.

ACAP portfolio companies are expected to have the following policies and procedures:

- SEAH policies
- SEAH trained staff
- SEAH monitoring
- SEAH survivor-centered approach grievance mechanism

Priorities of Women and Men in ACAP

ACAP intends to engage men and women throughout the project design process to best understand the different needs of people based on gender, geography, socio-economic status, and other factors.

This fund will help male and female investors, entrepreneurs, and other stakeholders better understand the gender dynamics of agriculture and agribusinesses in Pakistan through the collection of and reporting on sex-disaggregated data across ACAP portfolio company employees and customers.

ACAP seeks to invest in companies that will provide products and services that will improve agricultural efficiencies and outcomes thereby building the climate resilience of farming families throughout Pakistan.

Both men and women farmers in rural communities are prioritizing products and services that will increase efficiencies, income, and economic stability. Men will have greater access to the products and services of the ACAP portfolio companies as men have more access to wealth, formal sector jobs, formal local and international food markets, and credit. ACAP will utilize technical assistance to support businesses in expanding their customer outreach to include women farmers. This will include providing gender expertise and marketing support to portfolio companies.

Our research demonstrated that both male and female farmers have limited awareness of climate hazards in Pakistan and have limited knowledge of climate resilient agricultural practices, technologies, and services. Men and women farmers prioritized products and services that will help them manage the extreme weather events, droughts, and other weather conditions that have harmed their livelihoods. The ACAP team seeks to use the technical assistance facility to fund training for female farmers on climate hazards and technologies. The TAF will also fund training for male farmers when appropriate.

As established earlier, women are predominantly in the informal labor market. Women are exposed to vulnerabilities without access to the formal market including limited mobility, few social protections, and economic incertitude. More and more women across Pakistan need jobs in the formal sector. ACAP seeks to create more opportunities for women to enter the formal agricultural sector through a variety of methods:

- Funding training to improve the employability of women seeking agribusiness employment. Training will be in technologies, tools, and skills that improve their ability to gain employment.

- Improving safe and equitable workspaces of businesses in the ACAP portfolio. Gender experts will work with businesses to create gender action plans that will improve policies, procedures, and organizational capacity relating to gender. We believe that safer workspaces with a gender equity lens will create more opportunities for women to seek employment.

ACAP also intends to promote and incorporate female leadership in the program's design. We seek to learn the priorities of female entrepreneurs and leaders across the public and private sector.

ACAP seeks to continue learning about the priorities of men and women through extensive stakeholder engagement and impact measurement and management. Our team intends to work with surveying organizations to listen to and learn from customer voices about ACAP portfolio companies' products and services. With continued input from men and women farmers, entrepreneurs, and leaders across Pakistan, ACAP will be well-equipped to invest in and support a more gender equitable agriculture sector in Pakistan.

Gender and Anticipated outcomes of ACAP

From the above assessment, there is an unequal sharing of power between women and men, the gender gap in access to education and employment opportunities, the unpaid care burden, prevalence of gender-based violence, and all other forms of deep-rooted gender-based discrimination.In addition, involving women in decision-making can help drive the adoption of climate change policies and strengthen mitigation and adaptation efforts by ensuring they benefit the needs of women.

But often, women's participation in decision-making and their climate leadership potential is hindered by their unpaid care responsibilities. Across the world, women carry out more than 75 percent of unpaid care work, or 3.2 times more than men. When climate-induced disasters hit, this figure only increases as women take on additional burdens to help their households and communities recover and rebuild.

Climate-induced stressors can also impact access to education and the labour market for women and girls, increasing the time they must spend on household chores and therefore perpetuating a cycle of disempowerment. Moreover, in the aftermath of climate-induced disasters, women and girls can be more vulnerable to gender-based violence and therefore need access to quality services essential for their safety and recovery, as well as seats at the decision-making table.

These are some of the reasons why long-term climate solutions cannot be achieved without pushing for gender equality and women's empowerment and leadership. Now, more than ever, it is imperative to put gender equality at the heart of climate action.

<u>Diligence and Gender Action Plans</u>

ACAP expects to incorporate gender into the due diligence process for potential portfolio companies. Companies will be specifically diligenced on SEAH, safe and equitable workspaces, safe customer interactions, and sex-disaggregated data when available. Any deficiencies found in diligence are expected to be included in Gender Action Plans.

Gender Action Plans will be a fund-mandated plan of activities that are agreed upon between the Fund and the portfolio company. The activities will include improvements necessary to comply with Fund standards and ambitious goals to improve gender equity in the operations and activities of the portfolio company. The portfolio company will be supported on the gender action plan by ACAP's gender expert and other resources.

Companies are expected to have the following policies, procedures and commitments or commit to developing and implementing them as part of their gender action plan:

- HR policies with sufficient gender equity provisions
- Provisions, policies, and procedures for equal employment opportunities
- Commitment, procedures, and policies to retain and promote female employees
- Sexual harassment policy
- Grievance redress mechanism with SEAH survivor-centered mechanisms and activities

- Sex-disaggregated customer and employee data

Technical Assistance Facility

The TAF is intended to both support the ACAP portfolio companies and ecosystem. The TAF will support businesses with collecting sex-disaggregated data, marketing to women, and implementing gender action plans. The TAF will provide broader ecosystem level support by training female farmers, training women on agribusiness skills to improve employability, and supporting sectoral initiatives. The team will also invest in a gender expert to support the TAF and portfolio company activities.

Activities include:

- Activities promoting skills development and job development of women in agriculture sector
- Initiatives to increase participation of women in the agriculture sector
- Supporting companies to set, implement, and evaluate gender action plans and provide company level gender trainings
  - Ensuring policies and procedures promote and strengthen gender equity
  - Support and promote job
- Develop and ensure that gender is embedded across ACAP investment processes through a full-time gender expert

Monitoring and Reporting

The Fund will monitor gender activities and outcomes on an annual basis using the Fund's Gender Action Plan as a basis for reporting. The Fund will collect sex-disaggregated employment and customer data across the portfolio when feasible. It may not be possible to collect sex-disaggregated customer data for B2B businesses.

The Fund intends to report gender outcomes to the Fund investors and relevant stakeholders.

ACAP and Access to Services and Technologies

Women have limited access to agricultural services and technologies due to limited income, wealth, and being relegated to informal labor practices due to economic opportunity and socio-cultural attitudes to women working. Men primarily make agricultural spending decisions and are the primary user and beneficiary of agricultural services and technologies.

ACAP seeks to improve women in Pakistan's access to agricultural services and technologies through both company-level and ecosystem-level activities. ACAP intends to invest in agricultural businesses that produce and sell services and technologies that will benefit men and women smallholder farmers including farmer platforms, digital marketplaces, and financial solutions, smart farming, and post-harvest technologies. ACAP intends to invest in these businesses to improve the climate resilience of farmers, businesses, and Pakistan's agriculture sector. While we assume that men will more often benefit from these products and services directly as they are more likely to have the income and decision-making authority to purchase these products, ACAP intends to guide businesses to more gender-inclusive marketing and sales strategies.

Additionally, ACAP found both via interviews with companies and through research that businesses have difficulties collecting sex-disaggregated data. ACAP's gender expert will work with agri-businesses to improve collecting sex-disaggregated data. Collecting sex-disaggregated customer and employee data can empower companies to change marketing or hiring practices.

Too often men have more access to and awareness of the latest technologies to improve their farms and ease their work. ACAP intends to use the technical assistance facility to support initiatives that train women on climate resilient agricultural practices and technologies. ACAP wants to empower women to be more engaged in climate resilient agriculture.

ACAP and Leadership

ACAP intends to execute robust stakeholder engagement activities to encourage a more climate resilient and gender equitable agricultural sector in Pakistan. ACAP's gender expert and insights expert will work together to coalesce leaders throughout Pakistan to engage on issues ranging from gender in the workplace to public-private engagement to climate resilience and vulnerable populations.

ACAP intends to gather public and private leaders to 5 or more gatherings to learn and share insights from the Fund and leaders across Pakistan. There will be a particular focus on the gender learnings including on entrepreneurship, employment, climate hazards, and climate resilient impacts. We expect representatives from the Fund, the public sector, business leaders, civil society organizations, non-governmental organizations, and universities. We also seek to sensitize financial institutions in Pakistan to the challenges and opportunities of supporting women entrepreneurs. We believe that our sharing our work and coalescing partners like financial institutions can raise awareness of the barriers faced by female entrepreneurs and help them overcome them.

ACAP seeks to create gender inclusive events and insights. Invitees will be asked to be consider including both female and male representatives. We seek to uplift and champion female leaders when sharing insights and bringing leaders together.

ACAP and Vulnerable Sub-Groups

As agriculture is a largely female sector in Pakistan, the ACAP team will be dedicating a significant component of the fund TAF for gender initiatives. The team has dedicated significant time to understanding the climate impacts for women in Pakistan and particularly in the agriculture sector which is extremely climate vulnerable.

Globally, women are more vulnerable to climate change impacts compared to men. This is also evident in Pakistan, where women disproportionately bear the brunt of climate change effects. They confront heightened risks of economic instability, compromised well-being, limited educational achievements, and personal safety threats, including incidents of gender-based violence. Gender disparity amplifies this vulnerability as women are systematically excluded from decision-making processes concerning matters directly impacting their lives. Societal biases and cultural norms hinder women from engaging in and contributing to the formulation and execution of strategies addressing climate change. Consequently, their potential to serve as effective catalysts for change remains unrealized.

Diverse communities encounter climate change in distinct ways. Their ability to manage, react, and adjust varies based on geographical location, livelihoods, and socioeconomic status. Climate change particularly impacts impoverished individuals, those dependent on natural resources for sustenance, and those lacking the resources (including assets and access to services) to withstand natural calamities and extreme weather occurrences. Among the groups most susceptible to climate change's effects, women bear a disproportionate burden. However, women seldom participate in decision-making processes concerning matters directly affecting their lives, including climate change initiatives.

Throughout Pakistan, women are part of the millions of landless laborers and small-scale farmers engaged in combating rural poverty. They actively participate in activities such as farming, tending to livestock, and managing cottage industries. These women provide vital support to male household members in tasks like cultivating crops, harvesting, and processing, and they assume additional duties when men seek employment outside the farm or migrate in pursuit of work opportunities.

The specific roles and responsibilities of women vary, influenced by factors such as geographical location, farming methods, and local cultural traditions. Therefore, while developing the fund thesis, the ACAP team have:

- Conducted focus group discussions with rural women and female farmers to understand their challenges and the context within which they live their daily lives. The team will be conducting further research and interviewing rural women across multiple areas in Pakistan in order to better understand the needs of a diverse group of vulnerable women, children and the elderly.
- The team has undertaken research to understand differing cultural dynamics across provinces in Pakistan which we will augment through primary research/consultations

- We will also be engaging with gender experts to ensure that program design accounts for the differing needs of women and vulnerable groups. The fund will have a gender expert leading this work as we develop and refine interventions

ACAP and Stakeholder Engagement

The ACAP team will be engaging with stakeholders that work across the climate gender nexus in Pakistan. As part of this the team will prioritize working with both government and non-government stakeholders that work with vulnerable groups, particularly women and indigenous people. To that end the ACAP team has identified stakeholders we will consult in this process. Please see a partial list below:

- Development Finance Institutions:
  - International Union of Conservation of Nature (IUCN) who engage with indigenous people and gender stakeholders in Pakistan, and have lately assisted the Ministry of Climate Change in developing a Climate Change Gender Action Plan (ccGAP)
  - World Wildlife Fund (WWF) Pakistan who work on climate, water and gender and run programs in areas where indigenous communities reside.
  - United National Development Program (UNDP), who work on the issues of gender and climate particularly as they affect vulnerable groups.
- Civil Society Organisations (CSOs):
  - Civil Society Coalition for Climate Change (CSCCC) which works on research, knowledge-sharing, and advocacy for climate change. We aim to speak with CSOs working across the climate space on gender and with vulnerable groups.
- Vulnerable Groups:
  - In order to better understand potential impacts of ACAP and similar instruments on vulnerable groups, the team will be reaching out to vulnerable groups in areas where potential ACAP portfolio company companies may operate to undertake stakeholder consultations.

ACAP intends to engage with these stakeholders both through coalescing groups to share insights and spur further gender equity action (as shared earlier), and by reporting our activities in annual stakeholder engagement reports, meetings, and knowledge products.

Conclusion

ACAP seeks to support a more gender equitable agricultural sector through investing activities, technical assistance, and ecosystem support. In our gender assessment, we found that while Pakistan has made progress on gender equity in recent decades, there are significant gaps on rights, health, education, and economic participation between men and women. Our research has informed our gender activities and anticipated outcomes. We seek to continue learning from men and women farmers, entrepreneurs, leaders, and other relevant stakeholders. Their engagement will continue to drive our activities and passion for a more equitable agricultural sector.

ACAP Gender Action Plan:

Impact Statement: If the appropriate investment capital and TA is provided to agribusinesses (B2C)[1] focused on climate resilience, then they will be able to drive greater gender equity, promote participation and build adaptive capacity of women farmers in the value chain. Further, the fund can be leveraged to support companies with female led and owned businesses; promotion of senior female leadership and boost diversity at the workplace.

Outcome statement:

- Increased access of funding to female led and inclusive businesses in Pakistan
- Increased awareness and integration of gender forward practices within business operations through technical assistance and support of gender consultant resulting in 443 new direct jobs created for women
- Increased awareness and accessibility of women to products/services as per the business models under the fund
- Increased economic resilience and adaptive capacity of women farmers leading to better access to basic needs, farming inputs and assets by training at least 1,000 women on climate resilient agricultural practices

Output statement:
Funding and technical assistance lead to increased capacity of agribusinesses to serve low-income women customers, hire and retain women employees, promote women in senior leadership as well as creating gender inclusive workplaces.

ACAP's Gender Objectives:

1. Invest in women-led businesses and businesses that promote gender-forward business practices across climate adaptation and mitigation focused sectors in Pakistan.
2. Support businesses in adopting gender forward practices, build internal capacity on gender, and champion safe and equitable working environments.
3. Integrate female SHFs and women in the agriculture ecosystems as customers, employees, leaders and entrepreneurs
4. Provide vulnerable women farmers with sustainable livelihoods through climate resilient technology solutions

1 This GAP has been built for Business to Consumer companies part of the ACAP fund

5. Do no harm from a gender perspective.

Activities	Baseline	Targets	Indicators	Responsibility	Indicative budget
1. Internal Capacity development and training support					
1.1 Invest in a gender expert at ACAP	0	1	Hiring of Gender Expert	Implementation supported by project management team	Salary of gender expert over the life of the fund as defined in TA budget $864,000
1.2 Equip Acumen team and portfolio with GLI training and access to tools, templates and case studies to support the portfolio	0	n/a	# trainings # of How to documents/tools and templates created for private sector GLI incentive	Gender expert/External consultant	Staff time
2. Apply a gender lens to its investment decision-making					
2.1 PIPELINE					

2.1.1 Identify businesses with at least 1 female founder	n/a	Track number (annually)	# of women owned/led companies investees considered for pipeline (for sourcing and due diligence)	Gender expert at ACAP (Acumen)	Staff time
2.2 DILIGENCE SCREENING					
2.2.1 Perform assessments of how companies are currently operating (before investment) that serves as a baseline for monitoring process in gender equity	n/a	100% of companies	# of gender inclusion due diligence questionnaires Questionnaires include: -HR Policies -Commitments to equitable and safe work environments -Sexual Harassment Policy -Grievance mechanism with SEAH-survivor centered approach -Flexible work plan -Family leave policies -Organizational capacity on gender and SEAH	Implementation supported by gender expert and chosen screening platform	Staff time + Cost of platform
2.2.2 Identify and track 2x metrics for	n/a	100% of companies	# of companies	Implementation supported by project management team,	Staff time

companies during pre investment: Data collection pre-investment, and use information for decision making by the Investment committee.				with the oversight of Gender Expert at ACAP	
2.3 Deal Execution and Agreement					
2.3.1 Require companies to develop a Gender Action Plan during the investment period	n/a	100% of companies 15-17 companies	# of GAPs developed -GAPs must meet minimum criteria as outlined below including SEAH provisions and provisions to create safe and equitable workplaces	Deployed by investee companies and reviewed by Gender Expert	$340,000
2.3.2 Require companies to collect and report sex-	To be defined at due diligence phase	70% of B2C companies	# of companies that report sex-disaggregated sales data	Deployed by investee companies, Reviewed by Gender Expert at ACAP	

disaggregated customer data when feasible[2]					
3. Offer post-transaction gender-focused technical assistance to investees (21.75% of $10 MM)					
3.1 Monitoring: Tracking 2x via Acumen's annual review data and supporting progress through accompaniment, and TA to improve female jobs and customers	n/a	100% of companies 15-17 companies	# of companies 15% increase in female employment across baseline	Implementation supported by project management team, with the oversight of Gender Expert at ACAP	Staff time
3.2 Support the development and implementation of gender	n/a	All companies 15-17 companies	# of companies	Implementation supported by project team, with the oversight of Gender Expert at ACAP	ACAP gender champion

2 We will collect sex disaggregated data of customers only for B2C companies in the ACAP portfolio

inclusive business practices to meet minimum requirements and to implement their GAP					
3.3 Support companies in their GAP implementation and building gender inclusive internal strategies and practices from the indicator options depending on business context through TA support	n/a	All companies 15-17 companies	Depending on company's GAP, these can include: # HR policies initiated/implemented to promote gender equity # Female internship program setup in companies # of female leaders and managers coached on leadership # Training conducted for promoting leadership development of female employee and farmers across investee companies Companies must meet these minimum criteria	Implementation with the oversight of Gender Expert at ACAP for Internal strategies and consultant for external ecosystem work	Staff time + cost of trainings

			or they must be addressed in the GAP: -HR policy with sexual harassment protections -Sexual harassment policy -		
3.4 Promote strategies on increasing external efforts for gender responsive marketing or hiring practices from the indicator options depending on business context through TA support	n/a	All companies	Depending on company's GAP, these can include: # of market research surveys conducted to understand needs of women # of marketing campaigns conducted based on survey # of women hired/trained in decision making roles (leadership/staff/board)	Implementation with the oversight of Gender Expert at ACAP for Internal strategies and consultant for external ecosystem work	
4. Supporting women gain access to agriculture sector and awareness of adaptation opportunities					

4.1 Support activities to promote skills development, climate adaptation awareness, or job development of women in agriculture sector	0	20 trainings or workshops	# of trainings and workshops	ACAP technical assistance facility funding and staff support	TA budget $200,000
4.2 Support initiatives to increase participation of women in the agriculture sector	0	5 grants for trainings or similar types of initiatives	# of gender equity in agricultural sector initiatives supported	ACAP technical assistance facility funding and staff support	TA budget, $250,000
5. Insights, data aggregation and ESG/impact measurement support to amplify impact of companies					
5.1 Deep dive studies to share lessons with business-supporting organizations like	0	1-2 knowledge products	# knowledge outputs developed Knowledge outputs can include: -blogs -whitepapers -side events at	Gender Expert at ACAP along with Impact team	Staff time

financial institutions, ecosystem players, investors and companies filling a much-needed data gap and in informing future programming and investments.			stakeholder meetings		
5.2 Knowledge sharing events that will include gender components	0	5 events	1 session per event shall be on gender Invitees include: -government entities -financial institutions -CSOs and NGOs -Organizations representing indigenous communities and vulnerable populations -Organizations representing women in Pakistan -Local businesses -Academic Institutions	ACAP team; insights staff	$500,000

5.3 Sharing public redacted annual performance report with NDA, government entities,	0	10	Stakeholder report shared with the following groups: -CSOs and NGOs -Organizations representing indigenous communities and vulnerable populations -Organizations representing women in Pakistan -Local businesses -Academic Institutions	ACAP team, insights staff	Staff time
Total budget for GAP implementation					**21.5 % of USD 10,000,000**